

Graphs – Introduction (theory)

MATH 2740 – Mathematics of Data Science – Unit U08

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The University of Manitoba campuses are located on original lands of Anishinaabeg, Ininew, Anisininew, Dakota and Dene peoples, and on the National Homeland of the Red River Métis. We respect the Treaties that were made on these territories, we acknowledge the harms and mistakes of the past, and we dedicate ourselves to move forward in partnership with Indigenous communities in a spirit of Reconciliation and collaboration.

Outline

Why use graphs/networks?

Binary relations

Undirected graphs

Directed graphs

Matrices associated to a graph/digraph

Trees

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Binary relations

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Binary relations

Graphs versus networks

Mostly a terminology difference:

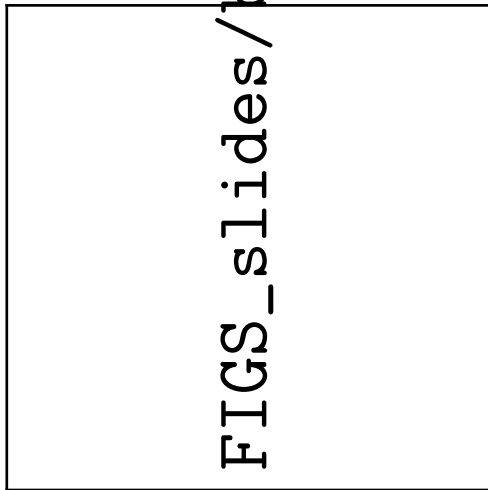
- ▶ graphs in the mathematical world
- ▶ networks elsewhere

I will mostly say *graphs* (this is a math course) but might oscillate

Beware: language is not consistent, so make sure you read the definitions at the start of whatever source you are using

The genesis of graphs – Euler's bridges of Königsberg

Cross the 7 bridges in a single walk without recrossing any of them?



The genesis of graphs – Euler's bridges of Königsberg

Cross the 7 bridges in a single walk without recrossing any of them?



FIGS_slides/bridge_color_with_graph

The genesis of graphs – Euler's bridges of Königsberg

Cross the 7 bridges in a single walk without recrossing any of them?

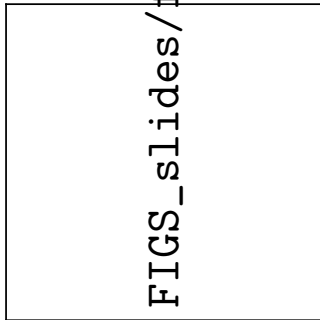


Mathematical problem

Is it possible to find a *trail* containing all *edges* of the graph?

Finding a cycle with all vertices

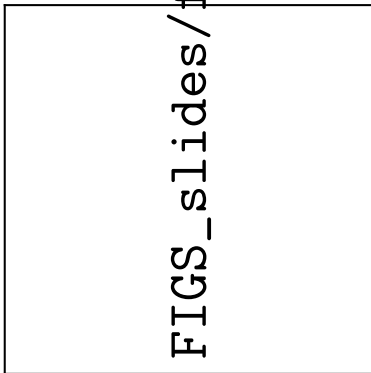
A salesperson must visit a couple of cities for their job. Is it possible for them to plan a round trip using highways enabling them to visit each specified city exactly once?



- ▶ vertices correspond to cities
- ▶ two vertices are connected iff a highway connects the corresponding cities and does not pass through any other city.

How far is it to drive through n cities?

What is the minimal length of driving needed to drive through n cities?



- ▶ vertices correspond to the cities
- ▶ all cities are connected; each edge has a value assigned to it

Mathematical problem

What is the minimal spanning tree associated to the graph?

Graphs/networks encode relations

Graphs are used in a variety of contexts because they encode *relations* between objects

Many objects in the world have relations... so graphs are quite easy to find

We will see many examples later, for now we cover the mathematical background

Graphs vs digraphs vs multigraphs vs multidigraphs vs ...

Name-wise and notation-wise, this domain is a bit of a mess

- ▶ The vertex set V is essentially the only constant
- ▶ *Undirected graph* $G = (V, E)$, where E are the *edges*
- ▶ *Undirected multigraph* $G_M = (V, E)$
- ▶ *Directed graph* (or *digraph*) $G = (V, A)$, where A are the *arcs*
- ▶ *Directed multigraph* (or *multidigraph*) $G_M = (V, A)$
- ▶ Any of the above is called a *graph* and is denoted $G = (V, X)$, when we seek generality

And just to confuse the whole thing more: we often say *graph* for *unoriented graph*

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Binary relations

Binary relation

Definition 1 (Binary relation)

- ▶ A **binary relation** is an arbitrary association of elements of one set with elements of another (maybe the same) set
- ▶ A binary relation over the sets X and Y is defined as a subset of the Cartesian product $X \times Y = \{(x, y) | x \in X, y \in Y\}$
- ▶ $(x, y) \in R$ is read “ x is R -related to y ” and is denoted xRy
- ▶ If $(x, y) \notin R$, we write “not xRy ” or $x \not R y$

Definition 2 (Properties of binary relations)

A binary relation R over a set X is

- ▶ **Reflexive** if $\forall x \in X, xRx$
- ▶ **Irreflexive** if there does not exist $x \in X$ such that xRx
- ▶ **Symmetric** if $xRy \Rightarrow yRx$
- ▶ **Asymmetric** if $xRy \Rightarrow y \not Rx$
- ▶ **Antisymmetric** if xRy and $yRx \Rightarrow x = y$
- ▶ **Transitive** if xRy and $yRz \Rightarrow xRz$
- ▶ **Total** (or **complete**) if $\forall x, y \in X, xRy$ or yRx

Definition 3 (Equivalence relation)

A relation that is reflexive ($\forall x \in X, xRx$), symmetric ($xRy \Rightarrow yRx$) and transitive (xRy and $yRz \Rightarrow xRz$) is an **equivalence relation**

Definition 4 (Partial order)

A relation that is reflexive ($\forall x \in X, xRx$), antisymmetric (xRy and $yRx \Rightarrow x = y$) and transitive (xRy and $yRz \Rightarrow xRz$) is a **partial order**

Definition 5 (Total order)

A partial order that is total ($\forall x, y \in X, xRy$ or yRx) is a **total order**

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The `igraph` library

Throughout these slides, we use the package `igraph`

I illustrate the functions that can be used to study some of the mathematical notions I introduce

I use mostly examples from the `igraph` documentation

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

Complete, bipartite and other notable graphs

Planar graphs

Graph colouring

Graph

Intuitively: a graph is a set of points, and a set of relations between the points

The points are called the *vertices* of the graph and the relations are the *edges* of the graph

We can also think of the relations as being one directional, in which case the relations are the *arcs* of the digraph (a contraction of “directed graph”)

Graph, vertex and edge

Definition 6 (Graph)

An **undirected graph** is a pair $G = (V, E)$ of sets such that

- ▶ V is a set of points: $V = \{v_1, \dots, v_p\}$
- ▶ E is a set of 2-element subsets of V : $E = \{\{v_i, v_j\}, \{v_i, v_k\}, \dots, \{v_n, v_p\}\}$ or $E = \{v_i v_j, v_i v_k, \dots, v_n v_p\}$

Definition 7 (Vertex)

The elements of V are the **vertices** (or nodes, or points) of the graph G . V (or $V(G)$) is the vertex set of the graph G

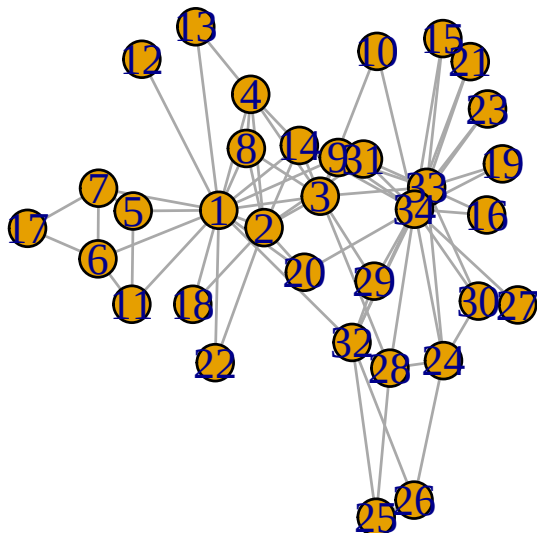
Definition 8 (Edge)

The elements of E are the **edges** (or lines) of the graph G . E (or $E(G)$) is the edge set of the graph G

Setting up graphs in igraph

```
G <- make_empty_graph()
G <- make_graph(edges = c(1, 2, 1, 5), n = 10,
                 directed = FALSE)
G <- make_graph("Zachary")
plot(G, main = "Data on a Karate club")
```

Data on a Karate club



Order and Size

Definition 9 (Order of a graph)

The number of vertices in G is the **order** of G . Using the notation $|V(G)|$ for the *cardinality* of $V(G)$,

$$|V(G)| = \text{order of } G$$

Definition 10 (Size of a graph)

The number of edges in G is the **size** of G ,

$$|E(G)| = \text{size of } G$$

- ▶ A graph having order p and size q is called a (p, q) –graph
- ▶ A graph is finite if $|V(G)| < \infty$

Some simple measures

```
V(G)
```

```
## + 34/34 vertices, from 1b6ce14:
```

```
## [1] 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
```

```
## [26] 26 27 28 29 30 31 32 33 34
```

```
head(E(G))
```

```
## + 6/78 edges from 1b6ce14:
```

```
## [1] 1--2 1--3 1--4 1--5 1--6 1--7
```

```
gorder(G)
```

```
## [1] 34
```

```
gsize(G)
```

```
## [1] 78
```


Incident – Adjacent

Definition 11 (Incident)

- ▶ A vertex v is **incident** with an edge e if $v \in e$; then e is an edge at v
- ▶ If $e = uv \in E(G)$, then u and v are each incident with e
- ▶ The two vertices incident with an edge are its ends
- ▶ An edge $e = uv$ is incident with both vertices u and v

Definition 12 (Adjacent)

- ▶ Two vertices u and v are **adjacent** in a graph G if $uv \in E(G)$
- ▶ If uv and uw are distinct edges (i.e. $v \neq w$) of a graph G , then uv and uw are adjacent edges

Incident vertices

```
incident(G, 1)
```

```
## + 16/78 edges from 1b6ce14:
```

```
## [1] 1-- 2 1-- 3 1-- 4 1-- 5 1-- 6 1-- 7 1-- 8 1-- 9 1--11 1--12 1--13 1--
```

```
## [13] 1--18 1--20 1--22 1--32
```

```
incident_edges(G, c(1, 2))
```

```
## [[1]]
```

```
## + 16/78 edges from 1b6ce14:
```

```
## [1] 1-- 2 1-- 3 1-- 4 1-- 5 1-- 6 1-- 7 1-- 8 1-- 9 1--11 1--12 1--13 1--
```

```
## [13] 1--18 1--20 1--22 1--32
```

```
##
```

```
## [[2]]
```

```
## + 9/78 edges from 1b6ce14:
```

```
## [1] 1-- 2 2-- 3 2-- 4 2-- 8 2--14 2--18 2--20 2--22 2--31
```

Adjacent vertices

```
adjacent_vertices(G, v = 1)

## [[1]]
## + 16/34 vertices, from 1b6ce14:
## [1]  2  3  4  5  6  7  8  9 11 12 13 14 18 20 22 32

adjacent_vertices(G, v = c(1, 2))

## [[1]]
## + 16/34 vertices, from 1b6ce14:
## [1]  2  3  4  5  6  7  8  9 11 12 13 14 18 20 22 32
##
## [[2]]
## + 9/34 vertices, from 1b6ce14:
## [1]  1  3  4  8 14 18 20 22 31
```

Definition 13 (Multiple edge)

Multiple edges are two or more edges connecting the same two vertices within a multigraph

Definition 14 (Loop)

A **loop** is an edge with both the same ends; *e.g.* $\{u, u\}$ is a loop

Definition 15 (Simple graph)

A **simple graph** is a graph which contains no loops or multiple edges

Definition 16 (Multigraph)

A **multigraph** is a graph which can contain multiple edges or loops

Testing for these properties

```
any_multiple(G)
```

```
## [1] FALSE
```

```
any_loop(G)
```

```
## [1] FALSE
```

```
is_simple(G)
```

```
## [1] TRUE
```

Graph and binary relations

A simple graph G can be defined in term of a vertex set V and a binary relation over V that is

- ▶ irreflexive ($\forall u \in V, u \not R u$)
- ▶ symmetric ($\forall u, v \in V, u R v \implies v R u$)

The set of edges $E(G)$ is the set of symmetric pairs in R

If R is not irreflexive, the graph is not simple

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

Complete, bipartite and other notable graphs

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Graph colouring

Definition 17 (Degree of a vertex)

Let v be a vertex of $G = (V, E)$.

- ▶ The number of edges of G incident with v is the **degree** of v in G
- ▶ The number of edges of G at v is the **degree** of v in G
- ▶ The degree of v in G is noted $d_G(v)$ or $\deg_G(v)$

Theorem 18

Let G be a (p, q) –graph with vertices $v_1, \dots, v_p$, then

$$\sum_{i=1}^p d_G(v_i) = 2q$$

Degree

```
degree(G)
```

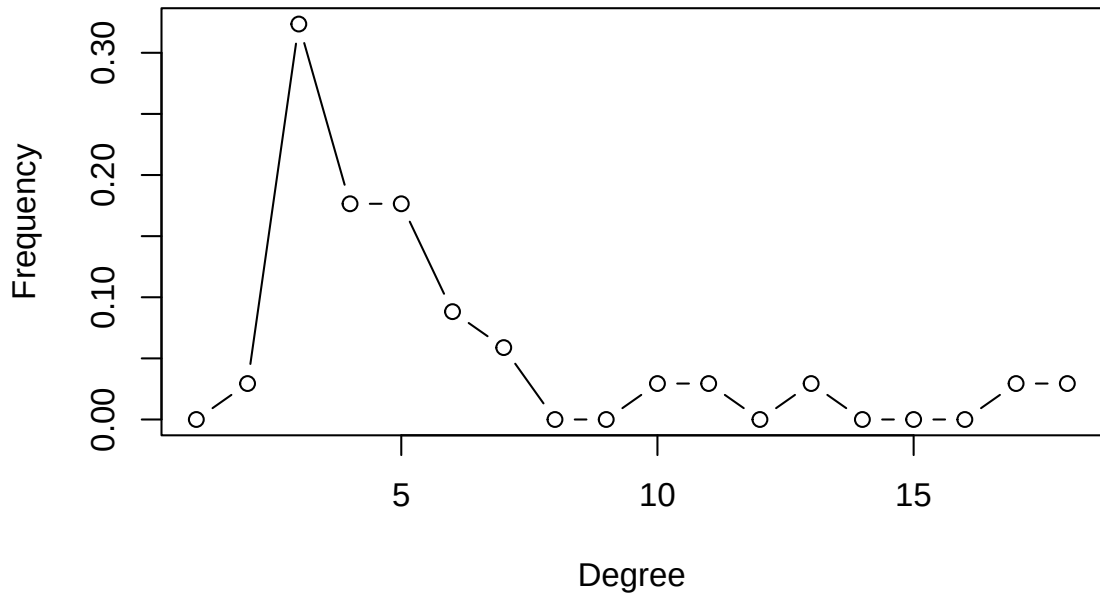
```
## [1] 16  9 10  6  3  4  4  4  5  2  3  1  2  5  2  2  2  2  2  3  2  2  2
## [26]  3  2  4  3  4  4  6 12 17
```

```
degree_distribution(G)
```

```
## [1] 0.00000000 0.02941176 0.32352941 0.17647059 0.17647059 0.08823529
## [7] 0.05882353 0.00000000 0.00000000 0.02941176 0.02941176 0.00000000
## [13] 0.02941176 0.00000000 0.00000000 0.00000000 0.02941176 0.02941176
```

```
plot(degree_distribution(G),
     type = "b",
     xlab = "Degree", ylab = "Frequency",
     main = "Degree distribution of the Karate graph")
```

Degree distribution of the Karate graph



Definition 19 (Odd vertex)

A vertex is an **odd vertex** if its degree is odd

Definition 20 (Even vertex)

A vertex is called **even vertex** if its degree is even

Theorem 21

Every graph contains an even number of odd vertices

Illustration of recent results

Theorem 394 states that a (p, q) -graph has $\sum_{i=1}^p d_G(v_i) = 2q$

```
sum(degree(G)) == 2*length(E(G))  
## [1] TRUE
```

Theorem 397 states that every graph contains an even number of odd vertices

```
sum(pracma::mod(degree(G), 2))  
## [1] 12
```

(mode(x, 2) returns 1 if x is odd so sum counts how many are odd)

Regular graph

Definition 22 (Regular graph)

If all the vertices of G have the same degree k , then the graph G is k -regular

```
length(unique(degree(G))) == 1
```

```
## [1] FALSE
```

Undirected graphs

Undirected graph

Degree of a vertex

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Isomorphic graphs

Definition 23 (Isomorphic graphs)

Let $G_1 = (V(G_1), E(G_1))$ and $G_2 = (V(G_2), E(G_2))$ be two graphs. G_1 and G_2 are **isomorphic** if there exists an isomorphism ϕ from G_1 to G_2 , that is defined as an injective mapping $\phi : V(G_1) \rightarrow V(G_2)$ such that two vertices u_1 and v_1 are adjacent in $G_1 \iff$ the vertices $\phi(u_1)$ and $\phi(v_1)$ are adjacent in G_2

If ϕ is an isomorphism from G_1 to G_2 , then the inverse mapping ϕ^{-1} from $V(G_2)$ to $V(G_1)$ also satisfies the definition of an isomorphism. As a consequence, if G_1 and G_2 are isomorphic graphs, then

- ▶ G_1 is isomorphic to G_2
- ▶ G_2 is isomorphic to G_1

Theorem 24

The relation “is isomorphic to” is an equivalence relation on the set of all graphs

Theorem 25

If G_1 and G_2 are isomorphic graphs, then the degrees of vertices of G_1 are exactly the degrees of vertices of G_2

Testing isomorphism

Create two isomorphic graphs by permuting the vertices of the first, then test if they are isomorphic

```
g1 <- sample_pa(30, m = 2, directed = FALSE)
g2 <- permute(g1, sample(vcount(g1)))
# should be TRUE
isomorphic(g1, g2)

## [1] TRUE
```

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

Complete, bipartite and other notable graphs

Planar graphs

Graph colouring

Subgraph

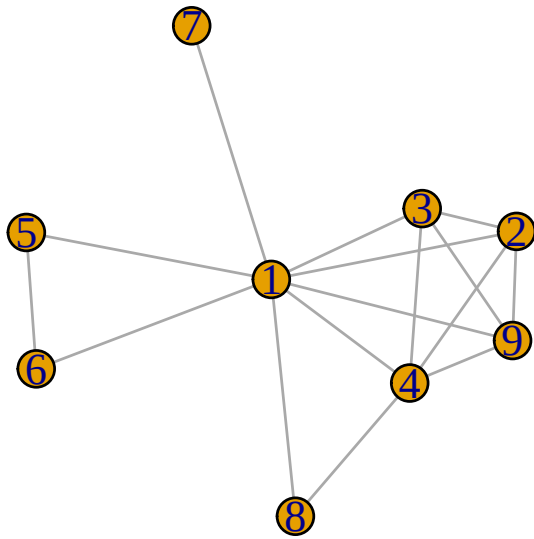
Definition 26 (Subgraph)

Let $G = (V, E)$ be a graph. A graph $H = (V(H), E(H))$ is a **subgraph** of G if $V(H) \subseteq V$ and $E(H) \subseteq E$

Extracting a subgraph

```
G_sub = induced_subgraph(G, c(1:5, 11:14))  
plot(G_sub, main = "A subgraph of the Karate graph")
```

A subgraph of the Karate graph



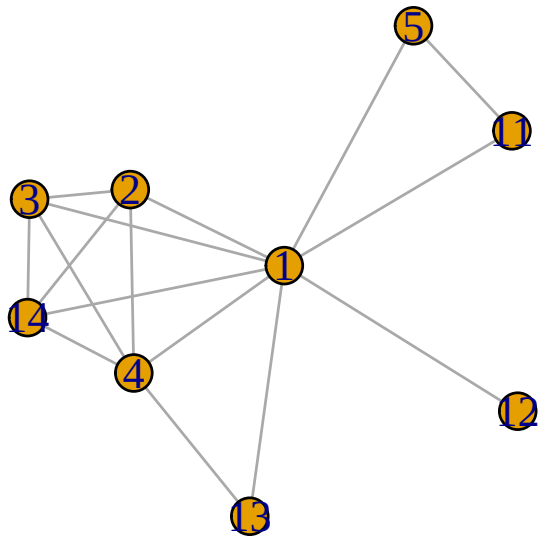
Naming vertices

Note that vertices (and as a consequence, edges) will be relabelled, so G_{sub} has vertices labelled 1 to 9

You can name vertices if you want to avoid this

```
V(G)$name = 1:length(V(G))
G_sub = induced_subgraph(G, c(1:5, 11:14))
plot(G_sub, labels = V(G)$name,
      main = "Subgraph of the Karate graph preserving names")
```

Subgraph of the Karate graph preserving names



Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two graphs

Definition 27 (Union of G_1 and G_2)

$$G_1 \cup G_2 = (V_1 \cup V_2, E_1 \cup E_2)$$

Definition 28 (Intersection of G_1 and G_2)

$$G_1 \cap G_2 = (V_1 \cap V_2, E_1 \cap E_2)$$

Definition 29 (Disjoint graphs)

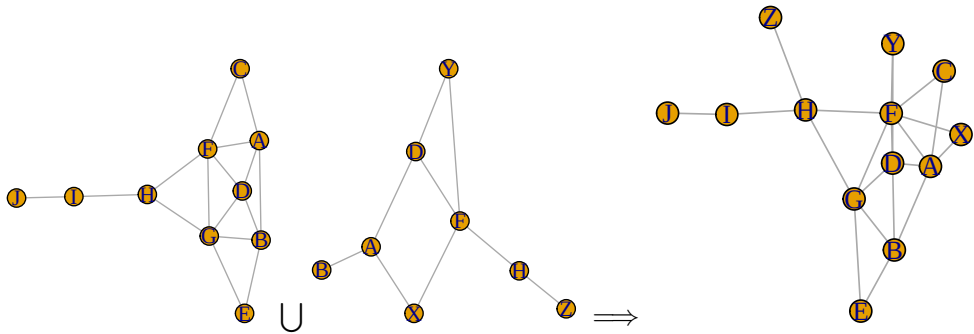
If $G_1 \cap G_2 = (\emptyset, \emptyset) = \emptyset$ (empty graph) then G_1 and G_2 are **disjoint**

Definition 30 (Complement of G_1)

The **complement** $\bar{G}_1$ of G_1 is the graph on V_1 , with the edge set
 $E(\bar{G}_1) = [V_1]^2 \setminus E_1$ ($e \in E(\bar{G}_1) \iff e \notin E_1$)

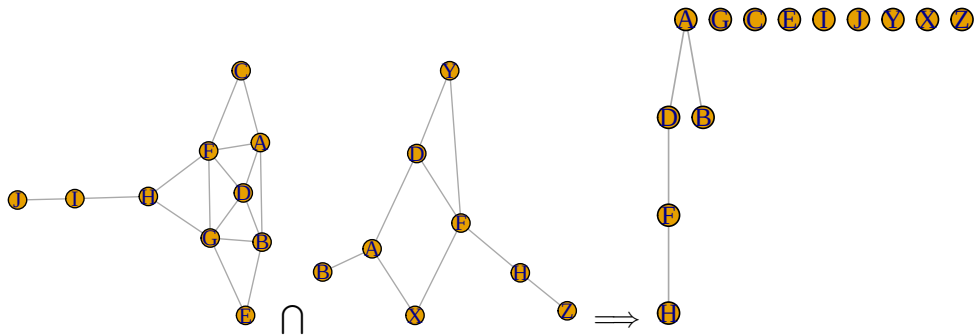
Union

```
net1 <- graph_from_literal(  
  D - A:B:F:G, A - C - F - A, B - E - G - B, A - B, F - G,  
  H - F:G, H - I - J  
)  
net2 <- graph_from_literal(D - A:F:Y, B - A - X - F - H - Z, F - Y)  
print_all(net1 %u% net2)  
  
## IGRAPH 4f186d3 UN-- 13 21 --  
## + attr: name (v/c)  
## + edges from 4f186d3 (vertex names):  
## [1] I--J H--Z H--I G--H G--E F--X F--Y F--H F--C F--G B--E B--G A--X A--  
## [16] A--B D--Y D--G D--F D--B D--A
```



Intersection of two graphs

```
G_inter = net1 %s% net2  
plot(G_inter)
```



Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

Complete, bipartite and other notable graphs

Planar graphs

Graph colouring

Connected vertices and graph, components

Definition 31 (Connected vertices)

Two vertices u and v in a graph G are **connected** if $u = v$, or if $u \neq v$ and there exists a path in G that links u and v

(For *path*, see Definition 420 later)

Definition 32 (Connected graph)

A graph is **connected** if every two vertices of G are connected; otherwise, G is **disconnected**

A necessary condition for connectedness

Theorem 33

A connected graph on p vertices has at least $p - 1$ edges

In other words, a connected graph G of order p has $\text{size}(G) \geq p - 1$

Connectedness is an equivalence relation

Denote $x \equiv y$ the relation “ $x = y$, or $x \neq y$ and there exists a path in G connecting x and y ”. $\equiv$ is an equivalence relation since

1. $x \equiv x$ [reflexivity]
2. $x \equiv y \implies y \equiv x$ [symmetry]
3. $x \equiv y, y \equiv z \implies x \equiv z$ [transitivity]

Definition 34 (Connected component of a graph)

The classes of the equivalence relation $\equiv$ partition V into connected sub-graphs of G called **connected components** (or **components** for short) of G

A connected subgraph H of a graph G is a component of G if H is not contained in any connected subgraph of G having more vertices or edges than H

Vertex deletion & cut vertices

Definition 35 (Vertex deletion)

If $v \in V(G)$ is a vertex of G , the graph $G - v$ is the graph formed from G by removing v and all edges incident with v

Definition 36 (Cut-vertices)

Let G be a connected graph. Then v is a **cut-vertex** G if $G - v$ is disconnected

Edge deletion & bridges

Definition 37 (Edge deletion)

If e is an edge of G , the graph $G - e$ is the graph formed from G by removing e from G

Definition 38 (Bridge)

An edge e in a connected graph G is a **bridge** if $G - e$ is disconnected

Theorem 39

Let G be a connected graph. An edge e of G is a bridge of $G \iff e$ does not lie on any cycle of G

(For *cycle*, see Definition 423 later)

Undirected graphs

Undirected graph

Degree of a vertex

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Walk

Definition 40 (Walk)

A **walk** in a graph $G = (V, E)$ is a non-empty alternating sequence $v_0 e_0 v_1 e_1 v_2 \dots e_{k-1} v_k$ of vertices and edges in G such that $e_i = \{v_i, v_{i+1}\}$ for all $i < k$. This walk begins with v_0 and ends with v_k

Definition 41 (Length of a walk)

The **length** of a walk is equal to the number of edges in the walk

Definition 42 (Closed walk)

If $v_0 = v_k$, the walk is **closed**

Trail and path

Definition 43 (Trail)

If the edges in the walk are all distinct, it defines a **trail** in $G = (V, E)$

Definition 44 (Path)

If the vertices in the walk are all distinct, it defines a **path** in G

The sets of vertices and edges determined by a trail is a subgraph

Distance between two vertices

Definition 45 (Distance between two vertices)

The **distance** $d(u, v)$ in $G = (V, E)$ between two vertices u and v is the length of the shortest path linking u and v in G

If no such path exists, we assume $d(u, v) = \infty$

Circuit and cycle

Definition 46 (Circuit)

A trail linking u to v , containing at least 3 edges and in which $u = v$, is a **circuit**

Definition 47 (Cycle)

A circuit which does not repeat any vertices (except the first and the last) is a **cycle** (or **simple circuit**)

Definition 48 (Length of a cycle)

The **length of a cycle** is its number of edges

Definition 49 (Eulerian trail)

A walk in an undirected multigraph M that uses each edge **exactly once** is a **Eulerian trail** of M

Definition 50 (Traversable graph)

If a graph G has a Eulerian trail, then G is a **traversable graph**

Definition 51 (Eulerian circuit)

A circuit containing all the vertices and edges of a multigraph M is a **Eulerian circuit** of M

Definition 52 (Eulerian graph)

A graph (resp. multigraph) containing an Eulerian circuit is a **Eulerian graph** (resp. **Eulerian multigraph**)

Remember Euler's bridges of Königsberg?

Cross the 7 bridges in a single walk without recrossing any of them?

`FIGS_slides/bridge_color_with_graph`

Theorem 53

A multigraph M is traversable $\iff M$ is connected and has exactly two odd vertices

Furthermore, any Eulerian trail of M begins at one of the odd vertices and ends at the other odd vertex

Theorem 54

A multigraph M is Eulerian $\iff M$ is connected and every vertex of M is even

Fleury's algorithm to find a Eulerian *trail*

For a connected graph with exactly 2 odd vertices

- ▶ Start at one of the odd vertices
- ▶ Marking your path as you move from vertex to vertex, travel along any edges you wish, but DO NOT travel along an edge that is a bridge for the graph formed by the EDGES THAT HAVE YET TO BE TRAVELED – unless you have to
- ▶ Continue until every edge has been traveled

RESULT: a Eulerian trail

Fleury's algorithm to find a Eulerian *circuit*

For a connected graph with no odd vertices

- ▶ Pick any vertex as a starting point
- ▶ Marking your path as you move from vertex to vertex, travel along any edges you wish, but DO NOT travel along an edge that is a bridge for the graph formed by the EDGES THAT HAVE YET TO BE TRAVELED – unless you have to
- ▶ Continue until you return to your starting point

RESULT: a Eulerian circuit

Definition 55 (Hamiltonian path)

A path containing all vertices of a graph G is a **Hamiltonian path** of G

Definition 56 (Traceable graph)

If a graph G has an Hamiltonian path, then G is a **traceable graph**

Definition 57 (Hamiltonian cycle)

A cycle containing all vertices of a graph G is a **Hamiltonian cycle** of G

Definition 58 (Hamiltonian graph)

A graph containing a Hamiltonian cycle is a **Hamiltonian graph**

Theorem 59 (Dirac's theorem)

If G is a graph of order $p \geq 3$ such that $\deg(v) \geq p/2$ for every vertex v of G , then G is Hamiltonian

Theorem 60 (Ore's theorem)

If G is a graph of order $p \geq 3$ such that for all distinct nonadjacent vertices u and v of G ,

$$\deg(u) + \deg(v) \geq p$$

then G is Hamiltonian

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

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Graph colouring

Definition 61 (Complete graph)

A graph is complete if every two of its vertices are adjacent

Definition 62 (n -clique)

A simple, complete graph on n vertices is called an n -**clique** and is often denoted K_n

Note that a complete graph of order p is $(p - 1)$ -regular

Bipartite graph

Definition 63 (Bipartite graph)

A graph is **bipartite** if its vertices can be partitioned into two sets V_1 and V_2 , such that no two vertices in the same set are adjacent. This graph may be written $G = (V_1, V_2, E)$

Definition 64 (Complete bipartite graph)

A bipartite graph in which every two vertices from the 2 different partitions are adjacent is called a **complete bipartite graph**

We often denote $K_{p,q}$ a simple, complete bipartite graph with $|V_1| = p$ and $|V_2| = q$

Some specific graphs

Definition 65 (Tree)

Any connected graph that has no cycles is a **tree**

Definition 66 (Cycle C_n)

For $n \geq 3$, the **cycle** C_n is a connected graph of order n that is a cycle on n vertices

Definition 67 (Path P_n)

The **path** P_n is a connected graph that consists of $n \geq 2$ vertices and $n - 1$ edges. Two vertices of P_n have degree 1 and the rest are of degree 2

Definition 68 (Star S_n)

The **star** of order n is the complete bipartite graph $K_{1,n-1}$ (1 vertex of degree $n - 1$ and $n - 1$ vertices of degree 1)

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

Complete, bipartite and other notable graphs

Planar graphs

Graph colouring

Planar graph

Definition 69 (Planar graph)

A graph is **planar** if it *can be* drawn in the plane with no crossing edges (except at the vertices). Otherwise, it is **nonplanar**

Definition 70 (Plane graph)

A **plane graph** is a graph *that is drawn* in the plane with no crossing edges. (This is only possible if the graph is planar)

(To see the difference, have you ever played this game?)

Let G be a plane graph

- ▶ the connected parts of the plane are called **regions**
- ▶ vertices and edges that are incident with a region R make up a **boundary** of R

Theorem 71 (Euler's formula)

Let G be a connected plane graph with p vertices, q edges, and r regions, then

$$p - q + r = 2$$

Corollary 72

Let G be a plane graph with p vertices, q edges, r regions, and k connected components, then

$$p - q + r = k + 1$$

Theorem 73

Let G be a connected planar graph with p vertices and q edges, where $p \geq 3$, then

$$q \leq 3p - 6.$$

(a maximal connected planar graph with p vertices has $q = 3p - 6$ edges)

Corollary 74

*If G is a planar graph, then $\delta(G) \leq 5$, where $\delta(G)$ is the minimal degree of G .
(every planar graph contains a vertex of degree less than 6)*

Two well-known non-planar graphs

$K_{3,3}$ and K_5 are nonplanar

Theorem 75 (Kuratowski Theorem)

A graph G is planar $\iff$ it contains no subgraph isomorphic to K_5 or $K_{3,3}$ or any subdivision of K_5 or $K_{3,3}$

Note: If a graph G is nonplanar and G is a subgraph of G' , then G' is also nonplanar

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

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Complete, bipartite and other notable graphs

Planar graphs

Graph colouring

Definition 76 (Colouring of a graph G)

A **colouring** of a graph G is an assignment of colours to the vertices of G such that adjacent vertices have different colours

Definition 77 (n -colouring of G)

A **n -colouring** is a colouring of G using n colours

Definition 78 (n -colourable)

G is **n -colourable** if there exists a colouring of G that uses n colours

Definition 79 (Chromatic number)

The **chromatic number** $\chi(G)$ of a graph G is the minimal value n for which an n -colouring of G exists

Property 80

- ▶ $\chi(G) = 1 \iff G$ have no edges
- ▶ If $G = K_{n,m}$, then $\chi(G) = 2$
- ▶ If $G = K_n$, then $\chi(G) = n$
- ▶ For any graph G ,

$$\chi(G) \leq 1 + \Delta(G)$$

where $\Delta(G)$ is the maximum degree of G

- ▶ If G is a planar graph, then $\chi(G) \leq 4$

“Real life” problem

What is the minimal number of colours to colour all states in the map so that two adjacent states have different colours?

`FIGS_slides/USA-4-colours`

“Real life” problem

What is the minimal number of colours to colour all states in the map so that two adjacent states have different colours?

Mathematical representation:

- ▶ vertices correspond to the states
- ▶ vertices are adjacent $\iff$ the two states are adjacent (sharing an isolated point such as the “Four Corners” does not count)

Mathematical problem

What is the chromatic number of the graph associated to the map?

Welch-Powell algorithm for colouring a graph G

1. Order the vertices of G by decreasing degree. (Such an ordering may not be unique since some vertices may have the same degree)
2. Use one colour to paint the first vertex and to paint, in sequential order, each vertex on the list that is not adjacent to a vertex previously painted with this colour
3. Start again at the top of the list and repeat the process, painting previously unpainted vertices using a second colour
4. Repeat with additional colours until all vertices have been painted

Why use graphs/networks?

Binary relations

Undirected graphs

Directed graphs

Matrices associated to a graph/digraph

Trees

Why use graphs/networks?

Binary relations

Definitions

Definition 81 (Digraph)

A directed graph (or **digraph**) is a pair $G = (V, A)$ of sets such that

- ▶ V is a set of points: $V = \{v_1, v_2, v_3, \dots, v_p\}$
- ▶ A is a set of ordered pairs of V : $A = \{(v_i, v_j), (v_i, v_k), \dots, (v_n, v_p)\}$ or $A = \{v_i v_j, v_i v_k, \dots, v_n v_p\}$

Definition 82 (Vertex)

The elements of V are the vertices of the digraph G . V or $V(G)$ is the vertex set of the digraph G

Definition 83 (Arc)

The elements of A are the **arcs** (directed edges) of the digraph G . A or $A(G)$ is the arc set of the digraph G

Digraph and binary relation

A (simple) digraph D can be defined in term of a vertex set V and an irreflexive relation R over V

The defining relation R of the digraph G need not be symmetric

Directed network

Definition 84 (Directed network)

A directed network is a digraph together with a function f ,

$$f : A \rightarrow \mathbb{R},$$

which maps the arc set A into the set of real number. The value of the arc $uv \in A$ is $f(uv)$

Loops & Multiple arcs

Definition 85 (Loop)

A **loop** is an arc with both the same ends; *e.g.* (u, u) is a loop

Definition 86 (Multiple arcs)

Multiple arcs (or multi-arcs) are two or more arcs connecting the same two vertices

Multidigraph/Digraph

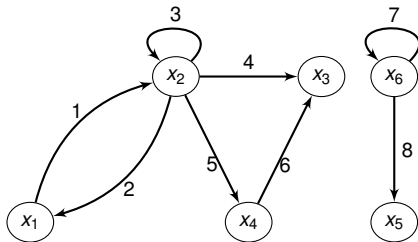
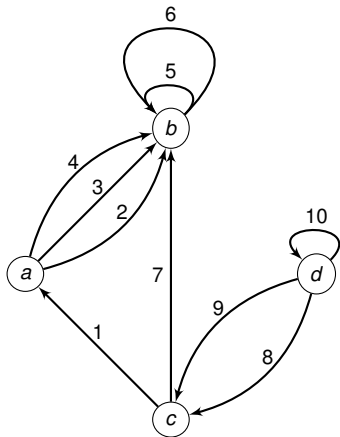
Definition 87 (Multidigraph)

A **multidigraph** is a digraph which allows repetition of arcs or loops

Definition 88 (Digraph)

In a digraph, no more than one arc can join any pair of vertices

Examples



Let $G = (V, A)$ be a digraph

Definition 89 (Arc endpoints)

For an arc $u = (x, y)$, vertex x is the **initial endpoint**, and vertex y is the **terminal endpoint**

Definition 90 (Predecessor - Successor)

If $(u, v) \in A(G)$ is an arc of G , then

- ▶ u is a **predecessor** of v
- ▶ v is a **successor** of u

Definition 91 (Neighbours of a vertex)

Let $x \in V$ be a vertex. The **neighbours** of x is the set $\Gamma(x) = \Gamma_G^+(x) \cup \Gamma_G^-(x)$, where $\Gamma_G^+(x)$ and $\Gamma_G^-(x)$ are, respectively, the set of successors and predecessors of x

Sources and sinks

Definition 92 (Directed away - Directed towards)

If $a = (u, v) \in A(G)$ is an arc of G , then

- ▶ the arc a is said to be **directed away** from u
- ▶ the arc a is said to be **directed towards** v

Definition 93 (Source - Sink)

- ▶ Any vertex which has no arcs directed towards it is a **source**
- ▶ Any vertex which has no arcs directed away from it is a **sink**

Adjacent arcs

Definition 94 (Adjacent arcs)

Two arcs are **adjacent** if they have at least one endpoint in common

Arcs incident to a subset of arcs

Definition 95 (Arc incident out of $X \subset A(G)$)

If the initial endpoint of an arc u belongs to $X \subset A(G)$ and if the terminal endpoint of arc u does not belong to X , then u is said to be **incident out of** X ; we write $u \in \omega^+(X)$

Similarly, we define an **arc incident into** X and the set $\omega^-(X)$

Finally, the set of arcs **incident to** X is denoted

$$\omega(X) = \omega^+(X) \cup \omega^-(X)$$

Definition 96 (Subgraph of G generated by $A \subset V$)

The **subgraph** of G generated by A is the graph with A as its vertex set and with all the arcs in G that have both their endpoints in A . If $G = (V, \Gamma)$ is a 1-graph, then the subgraph generated by A is the 1-graph $G_A = (A, \Gamma_A)$ where

$$\Gamma_A(x) = \Gamma(x) \cap A \quad (x \in A)$$

Definition 97 (Partial graph of G generated by $V \subset U$)

The graph (X, V) whose vertex set is X and whose arc set is V . In other words, it is graph G without the arcs $U - V$

Definition 98 (Partial subgraph of G)

A partial subgraph of G is the subgraph of a partial graph of G

Directed graphs

Degrees in digraphs

Walks, paths, etc.

Connectivity in digraphs

Orientable graphs

FIGS/random_ug_11_4.pdf

Degree

Let v be a vertex of a digraph $G = (V, A)$

Definition 99 (Outdegree of a vertex)

The number of arcs directed away from a vertex v , in a digraph is called the **outdegree** of v and is written $d_G^+(v)$

Definition 100 (Indegree of a vertex)

The number of arcs directed towards a vertex v , in a digraph is called the **indegree** of v and is written $d_G^-(v)$

Definition 101 (Degree)

For any vertex v in a digraph, the **degree** of v is defined as

$$d_G(v) = d_G^+(v) + d_G^-(v)$$

Theorem 102

For any (di)graph, the sum of the degrees of the vertices equals twice the number of edges (arcs)

Corollary 103

In any (di)graph, the sum of the degrees of the vertices is a nonnegative even integer

Theorem 104

If G is a digraph with vertex set $V(G) = \{v_1, \dots, v_p\}$ and q arcs, then

$$\sum_{i=1}^p d_G^+(v_i) = \sum_{i=1}^p d_G^-(v_i) = q$$

Definition 105 (Regular digraph)

A digraph G is r -regular if $d_G^+(v) = d_G^-(v) = r$ for all $v \in V(G)$

Symmetric/antisymmetric digraphs

Definition 106 (Symmetric digraph)

Let $G = (V, A)$ be a digraph with associated binary relation R . If R is *symmetric*, the digraph is symmetric

Definition 107 (Anti-symmetric digraph)

Let $G = (V, A)$ be a digraph with associated binary relation R . The digraph G is **anti-symmetric** if

$$xRy \implies y \not R x$$

Definition 108 (Symmetric multidigraph)

Let $G = (V, A)$ be a multidigraph. G is symmetric if $\forall x, y \in V(G)$, the number of arcs from x to y equals the number of arcs from y to x

Directed graphs

Degrees in digraphs

Walks, paths, etc.

Connectivity in digraphs

Orientable graphs

FIGS/random_g_10_4.pdf

Walks

Let $G = (V, A)$ be a digraph.

Definition 109 (Directed walk)

A **directed walk** in a digraph G is a non-empty alternating sequence $v_0 a_0 v_1 a_1 v_2 \dots a_{k-1} v_k$ of vertices and arcs in G such that $a_i = (v_i, v_{i+1})$ for all $i < k$. This walk begins with v_0 and ends with v_k

Definition 110 (Length of a directed walk)

The length of a directed walk is equal to the number of arcs in the directed walk

Definition 111 (Closed walk)

If $v_0 = v_k$, the walk is closed

Trails

Let $G = (V, A)$ be a digraph.

Definition 112 (Directed trail)

A directed walk in G in which all arcs are distinct is a **directed trail** in G

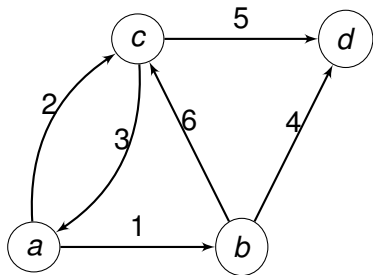
Definition 113 (Directed path)

A directed walk in G in which all vertices are distinct is a **directed path** in G

Definition 114 (Directed cycle)

A closed walk is a **directed cycle** if it contains at least three vertices and all its vertices are distinct except for $v_0 = v_k$

Examples of directed cycles



Cycles:

- ▶ $\mu^1 = (1, 6, 2) = [abca]$
- ▶ $\mu^2 = (1, 6, 3) = [abca]$
- ▶ $\mu^3 = (2, 3) = [aca]$
- ▶ $\mu^4 = (1, 4, 5, 2) = [abdca]$
- ▶ $\mu^5 = (6, 5, 4) = [acdb]$
- ▶ $\mu^6 = (1, 4, 5, 3) = [abdca]$

Directed graphs

Degrees in digraphs

Walks, paths, etc.

Connectivity in digraphs

Orientable graphs

FIGS/random_ug_10_1.pdf

Definitions

Definition 115 (Underlying graph)

Given a digraph, the undirected graph with each arc replaced by an edge is called the **underlying graph**

Definition 116 (Weakly connected digraph)

If the underlying graph is a connected graph, then the digraph is **weakly connected**

Definition 117 (Strongly connected digraph)

A digraph G is **strongly connected** if for every two distinct vertices u and v of G , there exists a directed path from u to v

Definition 118 (Disconnected digraph)

A digraph is said to be **disconnected** if it is not weakly connected

Strong connectedness is an equivalence relation

Denote $x \equiv y$ the relation “ $x = y$, or $x \neq y$ and there exists a directed path in G from x to y ”. $\equiv$ is an equivalence relation since

1. $x \equiv x$ [reflexivity]
2. $x \equiv y \implies y \equiv x$ [symmetry]
3. $x \equiv y, y \equiv z \implies x \equiv z$ [transitivity]

Definition 119 (Connected component of a graph)

Sets of the form

$$A(x_0) = \{x : x \in V, x \equiv x_0\}$$

are equivalence classes. They partition V into strongly connected sub-digraphs of G called **strongly connected components** (or **strong components**) of G

A strong component in G is a maximal strongly connected subdigraph of G

Theorem 120 (Properties)

Let $G = (V, A)$ be a digraph

- ▶ *If G is strongly connected, it has only one strongly connected component*
- ▶ *The strongly connected components partition the vertices $V(G)$, with every vertex in exactly one strongly connected component*

Algorithm for determining strongly connected components in

$$G = (V, A)$$

- ▶ Determine the strongly connected component $C(v)$ containing the vertex v ; if $V - C(v)$ is non-empty, re-do the same operation on the sub-digraph $G' = (V - C(v), A')$
- ▶ To determine $C(v)$, the strongly connected component containing v : let v be a vertex of a digraph, which is not already in any strongly connected component
 1. Mark the vertex v with $\pm$
 2. Mark with $+$ all successors (not already marked with $+$) of a vertex marked with $+$
 3. Mark with $-$ all predecessors (not already marked with $-$) of a vertex marked with $-$
 4. Repeat until no more possible marking with $+$ or $-$

All vertices marked with $\pm$ belong to the same strongly connected component $C(v)$ containing the vertex v

Condensation of a digraph

Definition 121 (Condensation of a digraph)

The condensation G^* of a digraph G is a digraph having as vertices the strongly connected components (SCC) of G and such that there exists an arc in G^* from a SCC C_i to another SCC C_j if there is an arc in G from some vertex of S_i to a vertex of S_j

Definition 122 (Articulation set)

For a connected graph, a set X of vertices is called an **articulation set** (or a **cutset**) if the subgraph of G generated by $V - X$ is not connected

Definition 123 (Stable set)

A set S of vertices is called a **stable set** if no arc joins two distinct vertices in S

Directed graphs

Degrees in digraphs

Walks, paths, etc.

Connectivity in digraphs

Orientable graphs

FIGS/random_graphs.py

Orientation

Definition 124 (Orienting a graph)

Given a connected *graph*, we describe the act of assigning a direction to each edge (edge $\rightarrow$ arc) as **orienting the graph**

Definition 125 (Strong orientation)

If the digraph resulting from orienting a graph is strongly connected, the orientation is a **strong orientation**

Orienable graph

Definition 126 (Orienable graph)

A connected graph G is **orientable** if it admits a strong orientation

Theorem 127

A connected graph $G = (V, E)$ is orientable $\iff G$ contains no bridges

(in other words, iff every edge is contained in a cycle)

Matrices associated to a graph/digraph

There are multiple matrices associated to a graph/digraph

The branch of graph theory that studies the properties of matrices derived from graphs and uses of these matrices in determining graph properties is *spectral graph theory*

Graphs greatly simplify some problems in linear algebra and vice versa

Matrices associated to a graph/digraph

Adjacency matrices

Other matrices associated to a graph/digraph

Linking graphs and linear algebra

FIGS/random_bg_18.pdf

Adjacency matrix (undirected case)

Let $G = (V, E)$ be a graph of order p and size q , with vertices $v_1, \dots, v_p$ and edges $e_1, \dots, e_q$

Definition 128 (Adjacency matrix)

The **adjacency matrix** is

$$M_A = M_A(G) = [m_{ij}]$$

is a $p \times p$ matrix in which

$$m_{ij} = \begin{cases} 1 & \text{if } v_i \text{ and } v_j \text{ are adjacent} \\ 0 & \text{otherwise} \end{cases}$$

Theorem 129 (Adjacency matrix and degree)

The sum of the entries in row i of the adjacency matrix is the degree of v_i in the graph

We often write $A(G)$ and, reciprocally, if A is an adjacency matrix, $G(A)$ the corresponding graph

G undirected $\implies A(G)$ symmetric

$A(G)$ has nonzero diagonal entries if G is not simple

Adjacency matrix (directed case)

Let $G = (V, A)$ be a digraph of order p with vertices $v_1, \dots, v_p$

Definition 130 (Adjacency matrix)

The **adjacency matrix** $M = M(G) = [m_{ij}]$ is a $p \times p$ matrix in which

$$m_{ij} = \begin{cases} 1 & \text{if arc } v_i v_j \in A \\ 0 & \text{otherwise} \end{cases}$$

Theorem 131 (Properties)

- ▶ *M is not necessarily symmetric*
- ▶ *The sum of any column of M is equal to the number of arcs directed towards v_j*
- ▶ *The sum of the entries in row i is equal to the number of arcs directed away from vertex v_i*
- ▶ *The (i, j) –entry of M^n is equal to the number of walks of length n from vertex v_i to v_j*

Definition 132 (Multiplicity of a pair)

The **multiplicity** of a pair x, y is the number $m_G^+(x, y)$ of arcs with initial endpoint x and terminal endpoint y . Let

$$m_G^-(x, y) = m_G^+(y, x)$$

$$m_G(x, y) = m_G^+(x, y) + m_G^-(x, y)$$

If $x \neq y$, then $m_G(x, y)$ is number of arcs with both x and y as endpoints. If $x = y$, then $m_G(x, y)$ equals twice the number of loops attached to vertex x . If $A, B \subset V$, $A \neq B$, let

$$m_G^+(A, B) = \{u : u \in U, u = (x, y), x \in A, y \in B\}$$

$$m_G(A, B) = m_G^+(A, B) + m_G^-(A, B)$$

Adjacency matrix of a multigraph

Definition 133 (Matrix associated with G)

If G has vertices $x_1, x_2, \dots, x_n$, then the **matrix associated** with G is

$$a_{ij} = m_G^+(x_i, x_j)$$

Definition 134 (Adjacency matrix)

The matrix $a_{ij} + a_{ji}$ is the **adjacency matrix** associated with G

Adjacency matrix (multigraph case)

Definition 135 (Adjacency matrix of a multigraph)

G an ℓ -graph, then the adjacency matrix $M_A = [m_{ij}]$ is defined as follows

$$m_{ij} = \begin{cases} k & \text{if arc there are } k \text{ arcs } (i, j) \in U \\ 0 & \text{otherwise} \end{cases}$$

with $k \leq \ell$

G undirected $\implies M_A(G)$ symmetric

$M_A(G)$ has nonzero diagonal entries if G is not simple.

Weighted adjacency matrices

Sometimes, adjacency matrices (typically for 1-graphs) have real entries, usually positive

This means that the arcs/edges have been given a weight

Theorem 136 (Number of walks of length n)

Let A be the adjacency matrix of a graph $G = (V(G), E(G))$, where $V(G) = \{v_1, v_2, \dots, v_p\}$. Then the (i, j) -entry of A^n , $n \geq 1$, is the number of different walks linking v_i to v_j of length n in G .

(two walks of the same length are equal if their edges occur in exactly the same order)

Example: let A be the adjacency matrix of a graph $G = (V(G), E(G))$.

- ▶ the (i, i) -entry of A^2 is equal to the degree of v_i .
- ▶ the (i, i) -entry of A^3 is equal to twice the number of C_3 containing v_i .

Matrices associated to a graph/digraph

Adjacency matrices

Other matrices associated to a graph/digraph

Linking graphs and linear algebra

FIGS/random_bg_19.pdf

Incidence matrix (undirected case)

Let $G = (V, E)$ be a graph of order p , and size q , with vertices $v_1, \dots, v_p$, and edges $e_1, \dots, e_q$

Definition 137 (Incidence matrix)

The incidence matrix is

$$B = B(G) = [b_{ij}]$$

is that $p \times q$ matrix in which

$$b_{ij} = \begin{cases} 1 & \text{if } v_i \text{ is incident with } e_j \\ 0 & \text{otherwise} \end{cases}$$

Theorem 138 (Incidence matrix and degrees)

The sum of the entries in row i of the incidence matrix is the degree of v_i in the graph

Incidence matrix (directed case)

Let $G = (V, A)$ be a digraph of order p and size q , with vertices $v_1, \dots, v_p$ and arcs $a_1, \dots, a_q$

Definition 139 (Incidence matrix)

The **incidence matrix** $B = B(G) = [b_{ij}]$ is a $p \times q$ matrix in which

$$b_{ij} = \begin{cases} 1 & \text{if arc } a_j \text{ is directed away from a vertex } v_i \\ -1 & \text{if arc } a_j \text{ is directed towards a vertex } v_i \\ 0 & \text{otherwise} \end{cases}$$

Spectrum of a graph

We will come back to this later, but for now..

Definition 140 (Spectrum of a graph)

The **spectrum** of a graph G is the spectrum (set of eigenvalues) of its associated adjacency matrix $M(G)$

This is regardless of the type of adjacency matrix or graph

Degree matrix

Definition 141 (Degree matrix)

The **degree** matrix $D = [d_{ij}]$ for G is a $n \times n$ diagonal matrix defined as

$$d_{ij} = \begin{cases} d_G(v_i) & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

In an undirected graph, this means that each loop increases the degree of a vertex by two

In a directed graph, the term “degree” may refer either to indegree (the number of incoming edges at each vertex) or outdegree (the number of outgoing edges at each vertex)

Laplacian matrix

Definition 142 (Laplacian matrix)

$G = (V, A)$ a simple graph with n vertices. The **Laplacian** matrix is

$$L = D(G) - M(G)$$

where $D(G)$ is the degree matrix and $M(G)$ is the adjacency matrix

Laplacian matrix (continued)

G simple graph $\implies M(G)$ only contains 1 or 0 and its diagonal elements are all 0

For directed graphs, either the indegree or outdegree is used, depending on the application

Elements of L are given by

$$\ell_{ij} = \begin{cases} d_G(v_i) & \text{if } i = j \\ -1 & \text{if } i \neq j \text{ and } v_i \text{ is adjacent to } v_j \\ 0 & \text{otherwise} \end{cases}$$

Distance matrix

Let G be a graph of order p with vertices $v_1, \dots, v_p$

Definition 143 (Distance matrix)

The distance matrix $\Delta(G) = [d_{ij}]$ is a $p \times p$ matrix in which

$$\delta_{ij} = d_G(v_i, v_j)$$

Note $\delta_{ii} = 0$ for $i = 1, \dots, p$

Property 144

- ▶ *M is not necessarily symmetric*
- ▶ *The sum of any column of M is equal to the number of arcs directed towards v_j*
- ▶ *The sum of the entries in row i is equal to the number of arcs directed away from vertex v_i*
- ▶ *The (i, j) –entry of M^n is equal to the number of walks of length n from vertex v_i to v_j*

Matrices associated to a graph/digraph

Adjacency matrices

Other matrices associated to a graph/digraph

Linking graphs and linear algebra

FIGS/random_bg_20.pdf

Counting paths

Theorem 145

G a digraph and $M_A(G)$ its adjacency matrix. Denote $P = [p_{ij}]$ the matrix $P = M_A^k$. Then p_{ij} is the number of distinct paths of length k from i to j in G

Definition 146 (Irreducible matrix)

A matrix $A \in \mathcal{M}_n$ is **reducible** if $\exists P \in \mathcal{M}_n$, permutation matrix, s.t. $P^T A P$ can be written in block triangular form. If no such P exists, A is **irreducible**

Theorem 147

A irreducible $\iff G(A)$ strongly connected

Theorem 148

Let A be the adjacency matrix of a graph G on p vertices. A graph G on p vertices is connected $\iff$

$$I + A + A^2 + \dots + A^{p-1} = C$$

has no zero entries

Theorem 149

Let M be the adjacency matrix of a digraph D on p vertices. A digraph D on p vertices is strongly connected $\iff$

$$I + M + M^2 + \dots + M^{p-1} = C$$

has no zero entries

Nonnegative matrix

$A = [a_{ij}] \in \mathcal{M}_n(\mathbb{R})$ **nonnegative** if $a_{ij} \geq 0 \forall i, j = 1, \dots, n$; $\mathbf{v} \in \mathbb{R}^n$ nonnegative if $v_i \geq 0 \forall i = 1, \dots, n$. **Spectral radius** of A

$$\rho(A) = \max_{\lambda \in \text{Sp}(A)} \{|\lambda|\}$$

$\text{Sp}(A)$ the **spectrum** of A

Perron-Frobenius (PF) theorem

Theorem 150 (PF – Nonnegative case)

$0 \leq A \in \mathcal{M}_n(\mathbb{R})$. Then $\exists \mathbf{v} \geq \mathbf{0}$ s.t.

$$A\mathbf{v} = \rho(A)\mathbf{v}$$

Theorem 151 (PF – Irreducible case)

Let $0 \leq A \in \mathcal{M}_n(\mathbb{R})$ irreducible. Then $\exists \mathbf{v} > \mathbf{0}$ s.t.

$$A\mathbf{v} = \rho(A)\mathbf{v}$$

$\rho(A) > 0$ and with algebraic multiplicity 1. No nonnegative eigenvector is associated to any other eigenvalue of A

Primitive matrices

Definition 152

$0 \leq A \in \mathcal{M}_n(\mathbb{R})$ **primitive** (with **primitivity index** $k \in \mathbb{N}_+^*$) if $\exists k \in \mathbb{N}_+^*$ s.t.

$$A^k > 0,$$

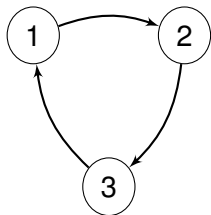
with k the smallest integer for which this is true. A **imprimitive** if it is not primitive

A primitive $\implies A$ irreducible; the converse is false

Theorem 153

$A \in \mathcal{M}_n(\mathbb{R})$ *irreducible and* $\exists i = 1, \dots, n$ s.t. $a_{ij} > 0 \implies A$ *primitive*

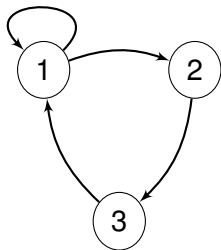
Here d is the index of imprimitivity (i.e., the number of eigenvalues that have the same modulus as $\lambda_p = \rho(A)$). If $d = 1$, then A is primitive. We have that $d = \gcd$ of all the lengths of closed walks in $G(A)$



Adjacency matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

Closed walks in $G(A)$ (lengths): $1 \rightarrow 1$ (3), $2 \rightarrow 2$ (3), $3 \rightarrow 3$ (3) $\implies \gcd = 3 \implies d = 3$ (here, all eigenvalues have modulus 1)



$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

Closed walk $1 \rightarrow 1$ has length 1 $\implies \gcd$ of lengths of closed walks is 1 $\implies A$ primitive

Let $\mathbf{0} \leq A \in \mathcal{M}_n$

Theorem 154

A primitive $\implies \exists 0 < k \leq (n-1)n^n$ such that $A^k > \mathbf{0}$

Theorem 155

If A is primitive and the shortest simple directed cycle in $G(A)$ has length s , then the primitivity index is $\leq n + s(n-1)$

Theorem 156

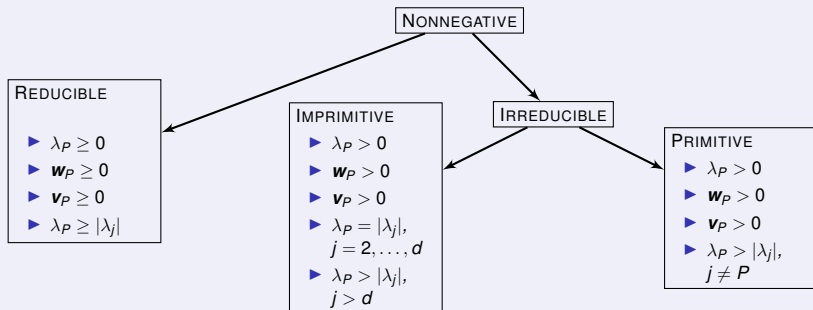
A primitive $\iff A^{n^2-2n+2} > \mathbf{0}$

Theorem 157

If A is irreducible and has d positive entries on the diagonal, then the primitivity index $\leq 2n - d - 1$

Theorem 158

$\mathbf{0} \leq A \in \mathcal{M}_n$, $\lambda_P = \rho(A)$ the Perron root of A , $\mathbf{v}_P$ and $\mathbf{w}_P$ the corresponding right and left Perron vectors of A , respectively, d the index of imprimitivity of A (with $d = 1$ when A is primitive) and $\lambda_j \in \sigma(A)$ the spectrum of A , with $j = 2, \dots, n$ unless otherwise specified (assuming $\lambda_1 = \lambda_P$)



Why use graphs/networks?

Binary relations

Undirected graphs

Directed graphs

`FIGS/random_bg_21.pdf`

Matrices associated to a graph/digraph

Trees

Why use graphs/networks?

Binary relations

Trees

Definition 159 (Forest, trees and branches)

- ▶ A connected graph with no cycle is a **tree**
- ▶ A tree is a connected acyclic graph, its edges are called **branches**
- ▶ A graph (connected or not) without any cycle is a **forest**. Each component is a tree

(A forest is a graph whose connected components are trees)

Is the “Karate graph” a tree?

```
is_acyclic(G)
```

```
## [1] FALSE
```

```
is_tree(G)
```

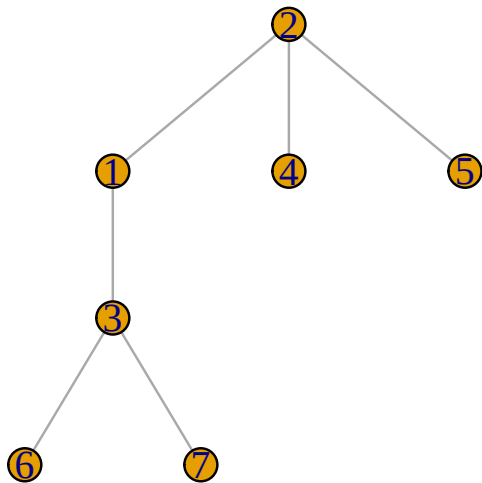
```
## [1] FALSE
```

So we need friend to play with!

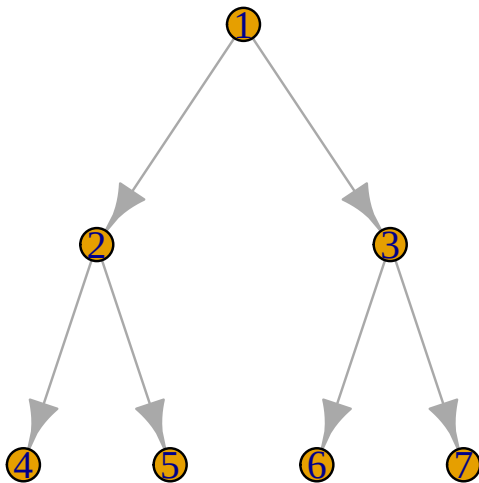
```
G_tu <- make_tree(7, 2, mode = "undirected")
```

```
G_td <- make_tree(7, 2)
```

An undirected tree



A (out) directed tree



Property 160

- ▶ *Every edge of a tree is a bridge*
- ▶ *Given two vertices u and v of a tree, there is an unique path linking u to v*
- ▶ *A tree with p vertices and q edges satisfies $q = p - 1$. Thus, a tree is minimally connected*

(First property: the deletion of any edge of a tree disconnects it)

Every edge of a tree is a bridge

```
E(G_tu)

## + 6/6 edges from c04e617:
## [1] 1--2 1--3 2--4 2--5 3--6 3--7

bridges(G_tu)

## + 6/6 edges from c04e617:
## [1] 2--4 2--5 1--2 3--6 3--7 1--3

all(sort(E(G_tu)) == sort(bridges(G_tu)))

## [1] TRUE
```


Spanning tree

Definition 161 (Spanning tree)

A **spanning tree** of a connected graph G is a subgraph of G that contains all the vertices of G and is a tree.

A graph may have many spanning trees

Minimal spanning tree

Definition 162 (Value of a spanning tree)

The **value of a spanning tree** T of order p is

$$\sum_{i=1}^{p-1} f(e_i)$$

where f is the function that maps the edge set into $\mathbb{R}$

Definition 163 (Minimal spanning tree)

Let G be an undirected network, and let T be a **minimal spanning tree** of G . Then T is a spanning tree whose the value is minimum

Algorithm to find a minimal spanning tree

Let $G = (V(G), E(G))$ be an undirected network and T be a minimal spanning tree

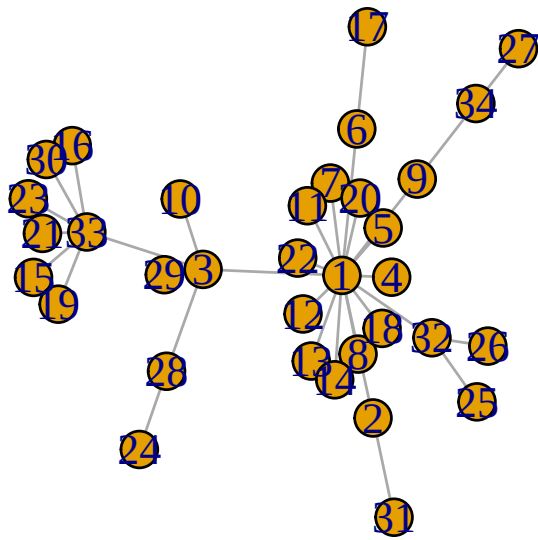
1. Sort the edges of G in increasing order by value
2. $T = (V(G), \emptyset)$
3. For each edge e in sorted order if the endpoints of e are disconnected in T add e to T

Finding a minimal spanning tree of the Karate graph

The function `mst` finds minimal spanning trees, using distances if no edge weights are provided

```
G_mst = mst(G)
```

A minimal spanning tree of the Karate graph



Minimal connector problem

- ▶ Model: a graph G such that edges represent all possible connections, and each edge has a positive value which represents its cost; an undirected network G
- ▶ Solution: a minimal spanning tree T of G
 - ▶ a spanning tree of G is a subgraph of G that contains all the vertices of G and is a tree.
 - ▶ the cost of the spanning tree is the sum of values of the edges of T
 - ▶ a spanning tree such that no other spanning tree has a smaller cost is a minimal spanning tree.

Theorem 164 (Characterisation of trees)

$H = (V, U)$ a graph of order $|V| = n > 2$. The following are equivalent and all characterise a tree :

- 1. H connected and has no cycles*
- 2. H has $n - 1$ arcs and no cycles*
- 3. H connected and has exactly $n - 1$ arcs*
- 4. H has no cycles, and if an arc is added to H , exactly one cycle is created*
- 5. H connected, and if any arc is removed, the remaining graph is not connected*
- 6. Every pair of vertices of H is connected by one and only one chain*

Definition 165 (Pendant vertex)

A vertex is **pendant** if it is adjacent to exactly one other vertex

Theorem 166

A tree of order $n \geq 2$ has at least two pendant vertices

Theorem 167

A graph $G = (V, U)$ has a partial graph that is a tree $\iff G$ connected

Recall that a partial graph is a graph generated by a subset of the arcs
(Definition 473 slide 384)

Spanning tree

The procedure in the proof of Theorem 573 gives a **spanning tree**

Can also build a spanning tree as follows:

- ▶ Consider any arc u_0
- ▶ Find arc u_1 that does not form a cycle with u_0
- ▶ Find arc u_2 that does not form a cycle with $\{u_0, u_1\}$
- ▶ Continue
- ▶ When you cannot continue anymore, you have a spanning tree

Theorem 168

G connected graph with ≥ 1 arc. TFAE

1. *G strongly connected*
2. *Every arc lies on a circuit*
3. *G contains no cocircuits*

Theorem 169

G graph with ≥ 1 arc. TFAE

- 1. G is a graph without circuits*
- 2. Each arc is contained in a cocircuit*

Theorem 170

If G is a strongly connected graph of order n, then G has a cycle basis of $\nu(G)$ circuits

Definition 171 (Node, anti-node, branch)

$G = (V, U)$ strongly connected without loops and > 1 vertex. For each $x \in V$, there is a path from it and a path to it so x has at least 2 incident arcs. Specifically,

- ▶ $x \in V$ with > 2 incident arcs is a **node**
- ▶ $x \in V$ with 2 incident arcs is an **anti-node**

A path whose only nodes are its endpoints is a **branch**

Definition 172 (Minimally connected graph)

G is **minimally connected** if it is strongly connected and removal of any arc destroys strong-connectedness

A minimally connected graph is 1-graph without loops

Definition 173 (Contraction)

$G = (V, U)$. The **contraction** of the set $A \subset V$ of vertices consists in replacing A by a single vertex a and replacing each arc into (resp. out of) A by an arc with same index into (resp. out of) a

Theorem 174

*G minimally connected, $A \subset V$ generating a strongly connected subgraph of G .
Then the contraction of A gives a minimally connected graph*

Theorem 175

G a minimally connected graph, G' be the minimally connected graph obtained by the contraction of an elementary circuit of G . Then

$$\nu(G) = \nu(G') + 1$$

Theorem 176

G minimally connected of order $n \geq 2 \implies G$ has ≥ 2 anti-nodes

Theorem 177

$G = (V, U)$. Then the graph C' obtained by contracting each strongly connected component of G contains no circuits

Arborescences

Definition 178 (Root)

Vertex $a \in V$ in $G = (V, U)$ is a **root** if all vertices of G can be reached by paths *starting* from a

Not all graphs have roots

Definition 179 (Quasi-strong connectedness)

G is **quasi-strongly connected** if $\forall x, y \in V$, exists $z \in V$ (denoted $z(x, y)$ to emphasize dependence on x, y) from which there is a path to x and a path to y

Strongly connected $\implies$ quasi-strongly connected (take $z(x, y) = x$); converse not true

Quasi-strongly connected $\implies$ connected

Arborescence

Definition 180 (Arborescence)

An **arborescence** is a tree that has a root

Lemma 181

$G = (V, U)$ has a root $\iff G$ quasi-strongly connected

Theorem 182

H graph of order $n > 1$. TFAE (and all characterise an arborescence)

- 1. H quasi-strongly connected without cycles*
- 2. H quasi-strongly connected with $n - 1$ arcs*
- 3. H tree having a root a*
- 4. $\exists a \in V$ s.t. all other vertices are connected with a by 1 and only 1 path from a*
- 5. H quasi-strongly connected and loses quasi-strong connectedness if any arc is removed*
- 6. H quasi-strongly connected and $\exists a \in V$ s.t.*

$$d_H^-(a) = 0$$

$$d_H^-(x) = 1 \quad \forall x \neq a$$

- 7. H has no cycles and $\exists a \in V$ s.t.*

$$d_H^-(a) = 0$$

$$d_H^-(x) = 1 \quad \forall x \neq a$$

Theorem 183

G has a partial graph that is an arborescence $\iff G$ quasi-strongly connected

Theorem 184

$G = (V, E)$ simple connected graph and $x_1 \in V$. It is possible to direct all edges of E so that the resulting graph $G_0 = (V, U)$ has a spanning tree H s.t.

- 1. H is an arborescence with root x_1*
- 2. The cycles associated with H are circuits*
- 3. The only elementary circuits of G_0 are the cycles associated with H*

Counting trees

Proposition 185

X a set with n distinct objects, $n_1, \dots, n_p$ nonnegative integers s.t. $n_1 + \dots + n_p = n$. The number of ways to place the n objects into p boxes $X_1, \dots, X_p$ containing $n_1, \dots, n_p$ objects respectively is

$$\binom{n}{n_1, \dots, n_p} = \frac{n!}{n_1! \cdots n_p!}$$

Proposition 186 (Multinomial formula)

Let $a_1, \dots, a_p \in \mathbb{R}$ be p real numbers, then

$$(a_1 + \dots + a_p)^n = \sum_{n_1, \dots, n_p \geq 0} \binom{n}{n_1, \dots, n_p} (a_1)^{n_1} \cdots (a_p)^{n_p}$$

Theorem 187

Denote $T(n; d_1, \dots, d_n)$ the number of distinct trees H with vertices $x_1, \dots, x_n$ and with degrees $d_H(x_1) = d_1, \dots, d_H(x_n) = d_n$. Then

$$T(n; d_1, \dots, d_n) = \binom{n-2}{d_1-1, \dots, d_n-1}$$

Theorem 188

The number of different trees with vertices $x_1, \dots, x_n$ is n^{n-2}

There is a whole industry of similar results (as well as for arborescences), but we will stop here. The main point is that we are talking about a large number of possibilities..

Why use graphs/networks?

Binary relations

Undirected graphs

Directed graphs

`FIGS/random_bg_22.pdf`

Matrices associated to a graph/digraph

Trees

Why use graphs/networks?

Binary relations

Graphs versus networks

Mostly a terminology difference:

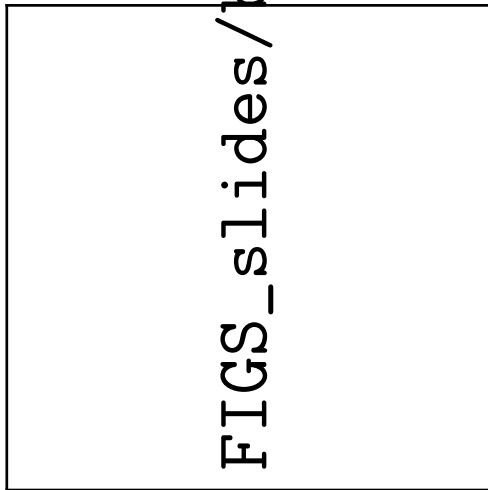
- ▶ graphs in the mathematical world
- ▶ networks elsewhere

I will mostly say *graphs* (this is a math course) but might oscillate

Beware: language is not consistent, so make sure you read the definitions at the start of whatever source you are using

The genesis of graphs – Euler's bridges of Königsberg

Cross the 7 bridges in a single walk without recrossing any of them?



The genesis of graphs – Euler's bridges of Königsberg

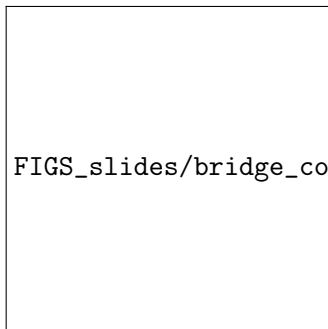
Cross the 7 bridges in a single walk without recrossing any of them?



FIGS_slides/bridge_color_with_graph

The genesis of graphs – Euler's bridges of Königsberg

Cross the 7 bridges in a single walk without recrossing any of them?



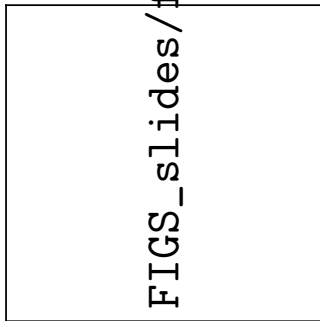
FIGS_slides/bridge_color_with_graph

Mathematical problem

Is it possible to find a *trail* containing all *edges* of the graph?

Finding a cycle with all vertices

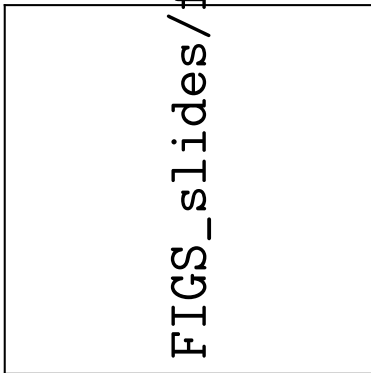
A salesperson must visit a couple of cities for their job. Is it possible for them to plan a round trip using highways enabling them to visit each specified city exactly once?



- ▶ vertices correspond to cities
- ▶ two vertices are connected iff a highway connects the corresponding cities and does not pass through any other city.

How far is it to drive through n cities?

What is the minimal length of driving needed to drive through n cities?



- ▶ vertices correspond to the cities
- ▶ all cities are connected; each edge has a value assigned to it

Mathematical problem

What is the minimal spanning tree associated to the graph?

Graphs/networks encode relations

Graphs are used in a variety of contexts because they encode *relations* between objects

Many objects in the world have relations... so graphs are quite easy to find

We will see many examples later, for now we cover the mathematical background

Graphs vs digraphs vs multigraphs vs multidigraphs vs ...

Name-wise and notation-wise, this domain is a bit of a mess

- ▶ The vertex set V is essentially the only constant
- ▶ *Undirected graph* $G = (V, E)$, where E are the *edges*
- ▶ *Undirected multigraph* $G_M = (V, E)$
- ▶ *Directed graph* (or *digraph*) $G = (V, A)$, where A are the *arcs*
- ▶ *Directed multigraph* (or *multidigraph*) $G_M = (V, A)$
- ▶ Any of the above is called a *graph* and is denoted $G = (V, X)$, when we seek generality

And just to confuse the whole thing more: we often say *graph* for *unoriented graph*

Why use graphs/networks?

Binary relations

Undirected graphs

Directed graphs

`FIGS/random_bg_23.pdf`

Matrices associated to a graph/digraph

Trees

Why use graphs/networks?

Binary relations

Binary relation

Definition 189 (Binary relation)

- ▶ A **binary relation** is an arbitrary association of elements of one set with elements of another (maybe the same) set
- ▶ A binary relation over the sets X and Y is defined as a subset of the Cartesian product $X \times Y = \{(x, y) | x \in X, y \in Y\}$
- ▶ $(x, y) \in R$ is read “ x is R -related to y ” and is denoted xRy
- ▶ If $(x, y) \notin R$, we write “not xRy ” or $x \not R y$

Definition 190 (Properties of binary relations)

A binary relation R over a set X is

- ▶ **Reflexive** if $\forall x \in X, xRx$
- ▶ **Irreflexive** if there does not exist $x \in X$ such that xRx
- ▶ **Symmetric** if $xRy \Rightarrow yRx$
- ▶ **Asymmetric** if $xRy \Rightarrow y \not R x$
- ▶ **Antisymmetric** if xRy and $yRx \Rightarrow x = y$
- ▶ **Transitive** if xRy and $yRz \Rightarrow xRz$
- ▶ **Total** (or **complete**) if $\forall x, y \in X, xRy$ or yRx

Definition 191 (Equivalence relation)

A relation that is reflexive ($\forall x \in X, xRx$), symmetric ($xRy \Rightarrow yRx$) and transitive (xRy and $yRz \Rightarrow xRz$) is an **equivalence relation**

Definition 192 (Partial order)

A relation that is reflexive ($\forall x \in X, xRx$), antisymmetric (xRy and $yRx \Rightarrow x = y$) and transitive (xRy and $yRz \Rightarrow xRz$) is a **partial order**

Definition 193 (Total order)

A partial order that is total ($\forall x, y \in X, xRy$ or yRx) is a **total order**

Why use graphs/networks?

Binary relations

Undirected graphs

Directed graphs

`FIGS/random_bg_24.pdf`

Matrices associated to a graph/digraph

Trees

Why use graphs/networks?

Binary relations

The `igraph` library

Throughout these slides, we use the package `igraph`

I illustrate the functions that can be used to study some of the mathematical notions I introduce

I use mostly examples from the `igraph` documentation

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

Complete, bipartite and other notable graphs

Planar graphs

Graph colouring

Graph

Intuitively: a graph is a set of points, and a set of relations between the points

The points are called the *vertices* of the graph and the relations are the *edges* of the graph

We can also think of the relations as being one directional, in which case the relations are the *arcs* of the digraph (a contraction of “directed graph”)

Graph, vertex and edge

Definition 194 (Graph)

An **undirected graph** is a pair $G = (V, E)$ of sets such that

- ▶ V is a set of points: $V = \{v_1, \dots, v_p\}$
- ▶ E is a set of 2-element subsets of V : $E = \{\{v_i, v_j\}, \{v_i, v_k\}, \dots, \{v_n, v_p\}\}$ or $E = \{v_i v_j, v_i v_k, \dots, v_n v_p\}$

Definition 195 (Vertex)

The elements of V are the **vertices** (or nodes, or points) of the graph G . V (or $V(G)$) is the vertex set of the graph G

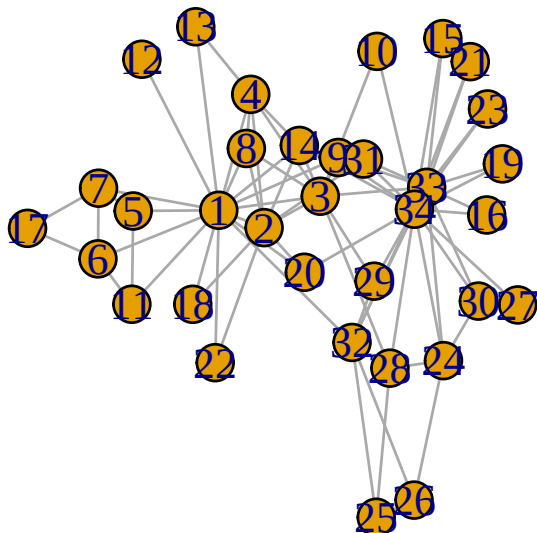
Definition 196 (Edge)

The elements of E are the **edges** (or lines) of the graph G . E (or $E(G)$) is the edge set of the graph G

Setting up graphs in igraph

```
G <- make_empty_graph()
G <- make_graph(edges = c(1, 2, 1, 5), n = 10,
                 directed = FALSE)
G <- make_graph("Zachary")
plot(G, main = "Data on a Karate club")
```

Data on a Karate club



Order and Size

Definition 197 (Order of a graph)

The number of vertices in G is the **order** of G . Using the notation $|V(G)|$ for the *cardinality* of $V(G)$,

$$|V(G)| = \text{order of } G$$

Definition 198 (Size of a graph)

The number of edges in G is the **size** of G ,

$$|E(G)| = \text{size of } G$$

- ▶ A graph having order p and size q is called a (p, q) –graph
- ▶ A graph is finite if $|V(G)| < \infty$

Some simple measures

```
V(G)
```

```
## + 34/34 vertices, from 87688e5:
```

```
## [1] 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
```

```
## [26] 26 27 28 29 30 31 32 33 34
```

```
head(E(G))
```

```
## + 6/78 edges from 87688e5:
```

```
## [1] 1--2 1--3 1--4 1--5 1--6 1--7
```

```
gorder(G)
```

```
## [1] 34
```

```
gsize(G)
```

```
## [1] 78
```

Incident – Adjacent

Definition 199 (Incident)

- ▶ A vertex v is **incident** with an edge e if $v \in e$; then e is an edge at v
- ▶ If $e = uv \in E(G)$, then u and v are each incident with e
- ▶ The two vertices incident with an edge are its ends
- ▶ An edge $e = uv$ is incident with both vertices u and v

Definition 200 (Adjacent)

- ▶ Two vertices u and v are **adjacent** in a graph G if $uv \in E(G)$
- ▶ If uv and uw are distinct edges (i.e. $v \neq w$) of a graph G , then uv and uw are adjacent edges

Incident vertices

```
incident(G, 1)
```

```
## + 16/78 edges from 87688e5:
```

```
## [1] 1-- 2 1-- 3 1-- 4 1-- 5 1-- 6 1-- 7 1-- 8 1-- 9 1--11 1--12 1--13 1--
```

```
## [13] 1--18 1--20 1--22 1--32
```

```
incident_edges(G, c(1, 2))
```

```
## [[1]]
```

```
## + 16/78 edges from 87688e5:
```

```
## [1] 1-- 2 1-- 3 1-- 4 1-- 5 1-- 6 1-- 7 1-- 8 1-- 9 1--11 1--12 1--13 1--
```

```
## [13] 1--18 1--20 1--22 1--32
```

```
##
```

```
## [[2]]
```

```
## + 9/78 edges from 87688e5:
```

```
## [1] 1-- 2 2-- 3 2-- 4 2-- 8 2--14 2--18 2--20 2--22 2--31
```

Adjacent vertices

```
adjacent_vertices(G, v = 1)

## [[1]]
## + 16/34 vertices, from 87688e5:
## [1]  2  3  4  5  6  7  8  9 11 12 13 14 18 20 22 32

adjacent_vertices(G, v = c(1, 2))

## [[1]]
## + 16/34 vertices, from 87688e5:
## [1]  2  3  4  5  6  7  8  9 11 12 13 14 18 20 22 32
##
## [[2]]
## + 9/34 vertices, from 87688e5:
## [1]  1  3  4  8 14 18 20 22 31
```

Definition 201 (Multiple edge)

Multiple edges are two or more edges connecting the same two vertices within a multigraph

Definition 202 (Loop)

A **loop** is an edge with both the same ends; *e.g.* $\{u, u\}$ is a loop

Definition 203 (Simple graph)

A **simple graph** is a graph which contains no loops or multiple edges

Definition 204 (Multigraph)

A **multigraph** is a graph which can contain multiple edges or loops

Testing for these properties

```
any_multiple(G)
```

```
## [1] FALSE
```

```
any_loop(G)
```

```
## [1] FALSE
```

```
is_simple(G)
```

```
## [1] TRUE
```

Graph and binary relations

A simple graph G can be defined in term of a vertex set V and a binary relation over V that is

- ▶ irreflexive ($\forall u \in V, u \not R u$)
- ▶ symmetric ($\forall u, v \in V, u R v \implies v R u$)

The set of edges $E(G)$ is the set of symmetric pairs in R

If R is not irreflexive, the graph is not simple

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

Complete, bipartite and other notable graphs

Planar graphs

Graph colouring

Definition 205 (Degree of a vertex)

Let v be a vertex of $G = (V, E)$.

- ▶ The number of edges of G incident with v is the **degree** of v in G
- ▶ The number of edges of G at v is the **degree** of v in G
- ▶ The degree of v in G is noted $d_G(v)$ or $\deg_G(v)$

Theorem 206

Let G be a (p, q) –graph with vertices $v_1, \dots, v_p$, then

$$\sum_{i=1}^p d_G(v_i) = 2q$$

Degree

```
degree(G)
```

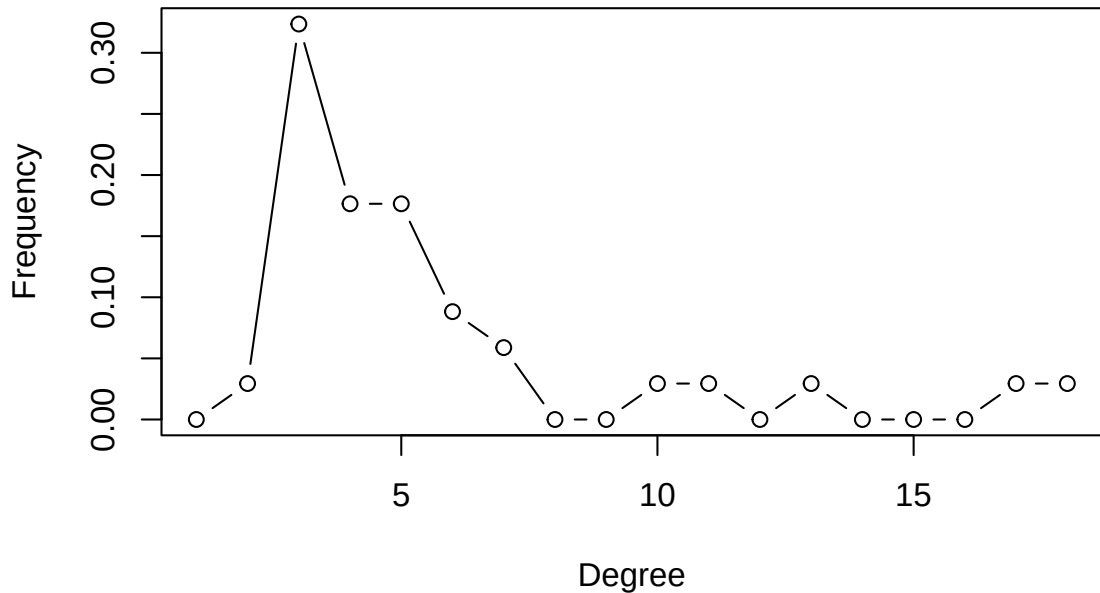
```
## [1] 16  9 10  6  3  4  4  4  5  2  3  1  2  5  2  2  2  2  2  3  2  2  2
## [26]  3  2  4  3  4  4  6 12 17
```

```
degree_distribution(G)
```

```
## [1] 0.00000000 0.02941176 0.32352941 0.17647059 0.17647059 0.08823529
## [7] 0.05882353 0.00000000 0.00000000 0.02941176 0.02941176 0.00000000
## [13] 0.02941176 0.00000000 0.00000000 0.00000000 0.02941176 0.02941176
```

```
plot(degree_distribution(G),
     type = "b",
     xlab = "Degree", ylab = "Frequency",
     main = "Degree distribution of the Karate graph")
```

Degree distribution of the Karate graph



Definition 207 (Odd vertex)

A vertex is an **odd vertex** if its degree is odd

Definition 208 (Even vertex)

A vertex is called **even vertex** if its degree is even

Theorem 209

Every graph contains an even number of odd vertices

Illustration of recent results

Theorem 394 states that a (p, q) -graph has $\sum_{i=1}^p d_G(v_i) = 2q$

```
sum(degree(G)) == 2*length(E(G))  
## [1] TRUE
```

Theorem 397 states that every graph contains an even number of odd vertices

```
sum(pracma::mod(degree(G), 2))  
## [1] 12
```

(`mode(x, 2)` returns 1 if x is odd so `sum` counts how many are odd)

Regular graph

Definition 210 (Regular graph)

If all the vertices of G have the same degree k , then the graph G is k -regular

```
length(unique(degree(G))) == 1
```

```
## [1] FALSE
```

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

Complete, bipartite and other notable graphs

Planar graphs

Graph colouring

Isomorphic graphs

Definition 211 (Isomorphic graphs)

Let $G_1 = (V(G_1), E(G_1))$ and $G_2 = (V(G_2), E(G_2))$ be two graphs. G_1 and G_2 are **isomorphic** if there exists an isomorphism ϕ from G_1 to G_2 , that is defined as an injective mapping $\phi : V(G_1) \rightarrow V(G_2)$ such that two vertices u_1 and v_1 are adjacent in $G_1 \iff$ the vertices $\phi(u_1)$ and $\phi(v_1)$ are adjacent in G_2

If ϕ is an isomorphism from G_1 to G_2 , then the inverse mapping ϕ^{-1} from $V(G_2)$ to $V(G_1)$ also satisfies the definition of an isomorphism. As a consequence, if G_1 and G_2 are isomorphic graphs, then

- ▶ G_1 is isomorphic to G_2
- ▶ G_2 is isomorphic to G_1

Theorem 212

The relation “is isomorphic to” is an equivalence relation on the set of all graphs

Theorem 213

If G_1 and G_2 are isomorphic graphs, then the degrees of vertices of G_1 are exactly the degrees of vertices of G_2

Testing isomorphism

Create two isomorphic graphs by permuting the vertices of the first, then test if they are isomorphic

```
g1 <- sample_pa(30, m = 2, directed = FALSE)
g2 <- permute(g1, sample(vcount(g1)))
# should be TRUE
isomorphic(g1, g2)

## [1] TRUE
```

Undirected graphs

Undirected graph

Degree of a vertex

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Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

Complete, bipartite and other notable graphs

Planar graphs

Graph colouring

Subgraph

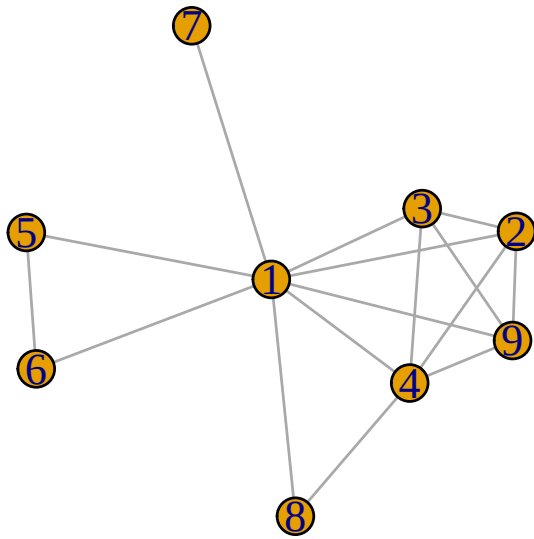
Definition 214 (Subgraph)

Let $G = (V, E)$ be a graph. A graph $H = (V(H), E(H))$ is a **subgraph** of G if $V(H) \subseteq V$ and $E(H) \subseteq E$

Extracting a subgraph

```
G_sub = induced_subgraph(G, c(1:5, 11:14))  
plot(G_sub, main = "A subgraph of the Karate graph")
```


A subgraph of the Karate graph



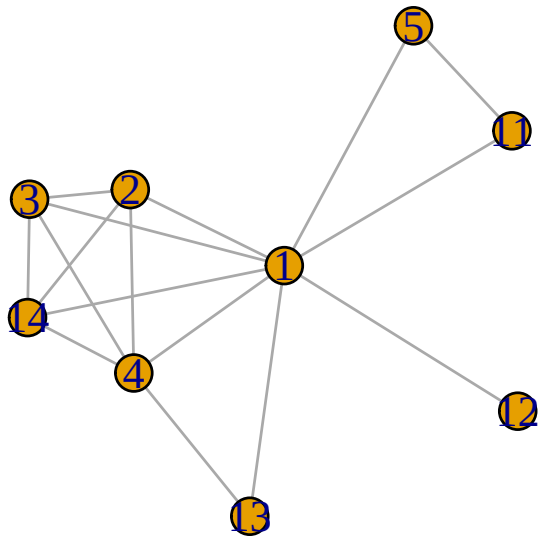
Naming vertices

Note that vertices (and as a consequence, edges) will be relabelled, so G_{sub} has vertices labelled 1 to 9

You can name vertices if you want to avoid this

```
V(G)$name = 1:length(V(G))  
G_sub = induced_subgraph(G, c(1:5, 11:14))  
plot(G_sub, labels = V(G)$name,  
      main = "Subgraph of the Karate graph preserving names")
```

Subgraph of the Karate graph preserving names



Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two graphs

Definition 215 (Union of G_1 and G_2)

$$G_1 \cup G_2 = (V_1 \cup V_2, E_1 \cup E_2)$$

Definition 216 (Intersection of G_1 and G_2)

$$G_1 \cap G_2 = (V_1 \cap V_2, E_1 \cap E_2)$$

Definition 217 (Disjoint graphs)

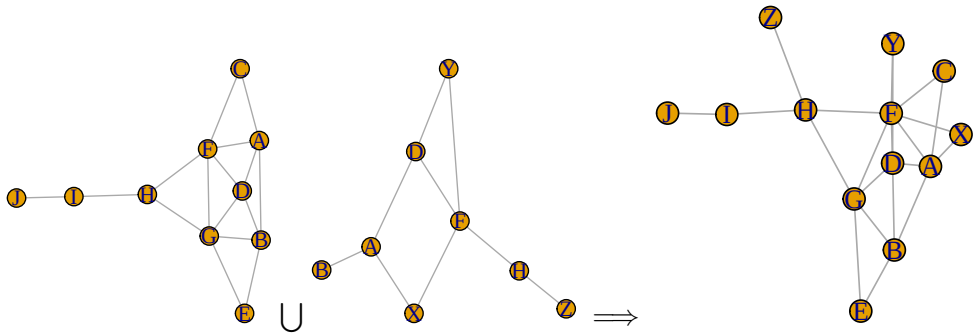
If $G_1 \cap G_2 = (\emptyset, \emptyset) = \emptyset$ (empty graph) then G_1 and G_2 are **disjoint**

Definition 218 (Complement of G_1)

The **complement** $\bar{G}_1$ of G_1 is the graph on V_1 , with the edge set
 $E(\bar{G}_1) = [V_1]^2 \setminus E_1$ ($e \in E(\bar{G}_1) \iff e \notin E_1$)

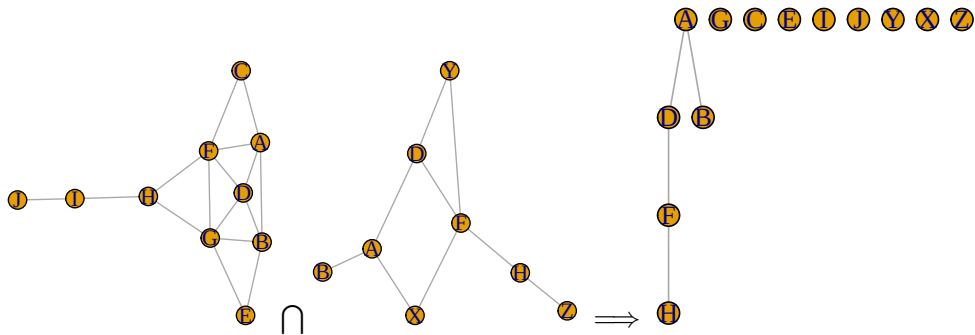
Union

```
net1 <- graph_from_literal(  
  D - A:B:F:G, A - C - F - A, B - E - G - B, A - B, F - G,  
  H - F:G, H - I - J  
)  
net2 <- graph_from_literal(D - A:F:Y, B - A - X - F - H - Z, F - Y)  
print_all(net1 %u% net2)  
  
## IGRAPH 302700a UN-- 13 21 --  
## + attr: name (v/c)  
## + edges from 302700a (vertex names):  
## [1] I--J H--Z H--I G--H G--E F--X F--Y F--H F--C F--G B--E B--G A--X A--  
## [16] A--B D--Y D--G D--F D--B D--A
```



Intersection of two graphs

```
G_inter = net1 %s% net2  
plot(G_inter)
```



Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

Complete, bipartite and other notable graphs

Planar graphs

Graph colouring

Connected vertices and graph, components

Definition 219 (Connected vertices)

Two vertices u and v in a graph G are **connected** if $u = v$, or if $u \neq v$ and there exists a path in G that links u and v

(For *path*, see Definition 420 later)

Definition 220 (Connected graph)

A graph is **connected** if every two vertices of G are connected; otherwise, G is **disconnected**

A necessary condition for connectedness

Theorem 221

A connected graph on p vertices has at least $p - 1$ edges

In other words, a connected graph G of order p has $\text{size}(G) \geq p - 1$

Connectedness is an equivalence relation

Denote $x \equiv y$ the relation “ $x = y$, or $x \neq y$ and there exists a path in G connecting x and y ”. $\equiv$ is an equivalence relation since

1. $x \equiv x$ [reflexivity]
2. $x \equiv y \implies y \equiv x$ [symmetry]
3. $x \equiv y, y \equiv z \implies x \equiv z$ [transitivity]

Definition 222 (Connected component of a graph)

The classes of the equivalence relation $\equiv$ partition V into connected sub-graphs of G called **connected components** (or **components** for short) of G

A connected subgraph H of a graph G is a component of G if H is not contained in any connected subgraph of G having more vertices or edges than H

Vertex deletion & cut vertices

Definition 223 (Vertex deletion)

If $v \in V(G)$ is a vertex of G , the graph $G - v$ is the graph formed from G by removing v and all edges incident with v

Definition 224 (Cut-vertices)

Let G be a connected graph. Then v is a **cut-vertex** G if $G - v$ is disconnected

Edge deletion & bridges

Definition 225 (Edge deletion)

If e is an edge of G , the graph $G - e$ is the graph formed from G by removing e from G

Definition 226 (Bridge)

An edge e in a connected graph G is a **bridge** if $G - e$ is disconnected

Theorem 227

Let G be a connected graph. An edge e of G is a bridge of $G \iff e$ does not lie on any cycle of G

(For *cycle*, see Definition 423 later)

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

Complete, bipartite and other notable graphs

Planar graphs

Graph colouring

Walk

Definition 228 (Walk)

A **walk** in a graph $G = (V, E)$ is a non-empty alternating sequence $v_0 e_0 v_1 e_1 v_2 \dots e_{k-1} v_k$ of vertices and edges in G such that $e_i = \{v_i, v_{i+1}\}$ for all $i < k$. This walk begins with v_0 and ends with v_k

Definition 229 (Length of a walk)

The **length** of a walk is equal to the number of edges in the walk

Definition 230 (Closed walk)

If $v_0 = v_k$, the walk is **closed**

Trail and path

Definition 231 (Trail)

If the edges in the walk are all distinct, it defines a **trail** in $G = (V, E)$

Definition 232 (Path)

If the vertices in the walk are all distinct, it defines a **path** in G

The sets of vertices and edges determined by a trail is a subgraph

Distance between two vertices

Definition 233 (Distance between two vertices)

The **distance** $d(u, v)$ in $G = (V, E)$ between two vertices u and v is the length of the shortest path linking u and v in G

If no such path exists, we assume $d(u, v) = \infty$

Circuit and cycle

Definition 234 (Circuit)

A trail linking u to v , containing at least 3 edges and in which $u = v$, is a **circuit**

Definition 235 (Cycle)

A circuit which does not repeat any vertices (except the first and the last) is a **cycle** (or **simple circuit**)

Definition 236 (Length of a cycle)

The **length of a cycle** is its number of edges

Definition 237 (Eulerian trail)

A walk in an undirected multigraph M that uses each edge **exactly once** is a **Eulerian trail** of M

Definition 238 (Traversable graph)

If a graph G has a Eulerian trail, then G is a **traversable graph**

Definition 239 (Eulerian circuit)

A circuit containing all the vertices and edges of a multigraph M is a **Eulerian circuit** of M

Definition 240 (Eulerian graph)

A graph (resp. multigraph) containing an Eulerian circuit is a **Eulerian graph** (resp. **Eulerian multigraph**)

Remember Euler's bridges of Königsberg?

Cross the 7 bridges in a single walk without recrossing any of them?

`FIGS_slides/bridge_color_with_graph`

Theorem 241

A multigraph M is traversable $\iff M$ is connected and has exactly two odd vertices

Furthermore, any Eulerian trail of M begins at one of the odd vertices and ends at the other odd vertex

Theorem 242

A multigraph M is Eulerian $\iff M$ is connected and every vertex of M is even

Fleury's algorithm to find a Eulerian *trail*

For a connected graph with exactly 2 odd vertices

- ▶ Start at one of the odd vertices
- ▶ Marking your path as you move from vertex to vertex, travel along any edges you wish, but DO NOT travel along an edge that is a bridge for the graph formed by the EDGES THAT HAVE YET TO BE TRAVELED – unless you have to
- ▶ Continue until every edge has been traveled

RESULT: a Eulerian trail

Fleury's algorithm to find a Eulerian *circuit*

For a connected graph with no odd vertices

- ▶ Pick any vertex as a starting point
- ▶ Marking your path as you move from vertex to vertex, travel along any edges you wish, but DO NOT travel along an edge that is a bridge for the graph formed by the EDGES THAT HAVE YET TO BE TRAVELED – unless you have to
- ▶ Continue until you return to your starting point

RESULT: a Eulerian circuit

Definition 243 (Hamiltonian path)

A path containing all vertices of a graph G is a **Hamiltonian path** of G

Definition 244 (Traceable graph)

If a graph G has an Hamiltonian path, then G is a **traceable graph**

Definition 245 (Hamiltonian cycle)

A cycle containing all vertices of a graph G is a **Hamiltonian cycle** of G

Definition 246 (Hamiltonian graph)

A graph containing a Hamiltonian cycle is a **Hamiltonian graph**

Theorem 247 (Dirac's theorem)

If G is a graph of order $p \geq 3$ such that $\deg(v) \geq p/2$ for every vertex v of G , then G is Hamiltonian

Theorem 248 (Ore's theorem)

If G is a graph of order $p \geq 3$ such that for all distinct nonadjacent vertices u and v of G ,

$$\deg(u) + \deg(v) \geq p$$

then G is Hamiltonian

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

Complete, bipartite and other notable graphs

Planar graphs

Graph colouring

Definition 249 (Complete graph)

A graph is complete if every two of its vertices are adjacent

Definition 250 (n -clique)

A simple, complete graph on n vertices is called an n -**clique** and is often denoted K_n

Note that a complete graph of order p is $(p - 1)$ -regular

Bipartite graph

Definition 251 (Bipartite graph)

A graph is **bipartite** if its vertices can be partitioned into two sets V_1 and V_2 , such that no two vertices in the same set are adjacent. This graph may be written $G = (V_1, V_2, E)$

Definition 252 (Complete bipartite graph)

A bipartite graph in which every two vertices from the 2 different partitions are adjacent is called a **complete bipartite graph**

We often denote $K_{p,q}$ a simple, complete bipartite graph with $|V_1| = p$ and $|V_2| = q$

Some specific graphs

Definition 253 (Tree)

Any connected graph that has no cycles is a **tree**

Definition 254 (Cycle C_n)

For $n \geq 3$, the **cycle** C_n is a connected graph of order n that is a cycle on n vertices

Definition 255 (Path P_n)

The **path** P_n is a connected graph that consists of $n \geq 2$ vertices and $n - 1$ edges. Two vertices of P_n have degree 1 and the rest are of degree 2

Definition 256 (Star S_n)

The **star** of order n is the complete bipartite graph $K_{1,n-1}$ (1 vertex of degree $n - 1$ and $n - 1$ vertices of degree 1)

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

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Planar graphs

Graph colouring

Planar graph

Definition 257 (Planar graph)

A graph is **planar** if it *can be* drawn in the plane with no crossing edges (except at the vertices). Otherwise, it is **nonplanar**

Definition 258 (Plane graph)

A **plane graph** is a graph *that is drawn* in the plane with no crossing edges. (This is only possible if the graph is planar)

(To see the difference, have you ever played this game?)

Let G be a plane graph

- ▶ the connected parts of the plane are called **regions**
- ▶ vertices and edges that are incident with a region R make up a **boundary** of R

Theorem 259 (Euler's formula)

Let G be a connected plane graph with p vertices, q edges, and r regions, then

$$p - q + r = 2$$

Corollary 260

Let G be a plane graph with p vertices, q edges, r regions, and k connected components, then

$$p - q + r = k + 1$$

Theorem 261

Let G be a connected planar graph with p vertices and q edges, where $p \geq 3$, then

$$q \leq 3p - 6.$$

(a maximal connected planar graph with p vertices has $q = 3p - 6$ edges)

Corollary 262

*If G is a planar graph, then $\delta(G) \leq 5$, where $\delta(G)$ is the minimal degree of G .
(every planar graph contains a vertex of degree less than 6)*

Two well-known non-planar graphs

$K_{3,3}$ and K_5 are nonplanar

Theorem 263 (Kuratowski Theorem)

A graph G is planar $\iff$ it contains no subgraph isomorphic to K_5 or $K_{3,3}$ or any subdivision of K_5 or $K_{3,3}$

Note: If a graph G is nonplanar and G is a subgraph of G' , then G' is also nonplanar

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

Complete, bipartite and other notable graphs

Planar graphs

Graph colouring

Definition 264 (Colouring of a graph G)

A **colouring** of a graph G is an assignment of colours to the vertices of G such that adjacent vertices have different colours

Definition 265 (n -colouring of G)

A **n -colouring** is a colouring of G using n colours

Definition 266 (n -colourable)

G is **n -colourable** if there exists a colouring of G that uses n colours

Definition 267 (Chromatic number)

The **chromatic number** $\chi(G)$ of a graph G is the minimal value n for which an n -colouring of G exists

Property 268

- ▶ $\chi(G) = 1 \iff G$ have no edges
- ▶ If $G = K_{n,m}$, then $\chi(G) = 2$
- ▶ If $G = K_n$, then $\chi(G) = n$
- ▶ For any graph G ,

$$\chi(G) \leq 1 + \Delta(G)$$

where $\Delta(G)$ is the maximum degree of G

- ▶ If G is a planar graph, then $\chi(G) \leq 4$

“Real life” problem

What is the minimal number of colours to colour all states in the map so that two adjacent states have different colours?

`FIGS_slides/USA-4-colours`

“Real life” problem

What is the minimal number of colours to colour all states in the map so that two adjacent states have different colours?

Mathematical representation:

- ▶ vertices correspond to the states
- ▶ vertices are adjacent $\iff$ the two states are adjacent (sharing an isolated point such as the “Four Corners” does not count)

Mathematical problem

What is the chromatic number of the graph associated to the map?

Welch-Powell algorithm for colouring a graph G

1. Order the vertices of G by decreasing degree. (Such an ordering may not be unique since some vertices may have the same degree)
2. Use one colour to paint the first vertex and to paint, in sequential order, each vertex on the list that is not adjacent to a vertex previously painted with this colour
3. Start again at the top of the list and repeat the process, painting previously unpainted vertices using a second colour
4. Repeat with additional colours until all vertices have been painted

Why use graphs/networks?

Binary relations

Undirected graphs

Directed graphs

`FIGS/random_bg_34.pdf`

Matrices associated to a graph/digraph

Trees

Why use graphs/networks?

Binary relations

Definitions

Definition 269 (Digraph)

A directed graph (or **digraph**) is a pair $G = (V, A)$ of sets such that

- ▶ V is a set of points: $V = \{v_1, v_2, v_3, \dots, v_p\}$
- ▶ A is a set of ordered pairs of V : $A = \{(v_i, v_j), (v_i, v_k), \dots, (v_n, v_p)\}$ or $A = \{v_i v_j, v_i v_k, \dots, v_n v_p\}$

Definition 270 (Vertex)

The elements of V are the vertices of the digraph G . V or $V(G)$ is the vertex set of the digraph G

Definition 271 (Arc)

The elements of A are the **arcs** (directed edges) of the digraph G . A or $A(G)$ is the arc set of the digraph G

Digraph and binary relation

A (simple) digraph D can be defined in term of a vertex set V and an irreflexive relation R over V

The defining relation R of the digraph G need not be symmetric

Directed network

Definition 272 (Directed network)

A directed network is a digraph together with a function f ,

$$f : A \rightarrow \mathbb{R},$$

which maps the arc set A into the set of real number. The value of the arc $uv \in A$ is $f(uv)$

Loops & Multiple arcs

Definition 273 (Loop)

A **loop** is an arc with both the same ends; *e.g.* (u, u) is a loop

Definition 274 (Multiple arcs)

Multiple arcs (or multi-arcs) are two or more arcs connecting the same two vertices

Multidigraph/Digraph

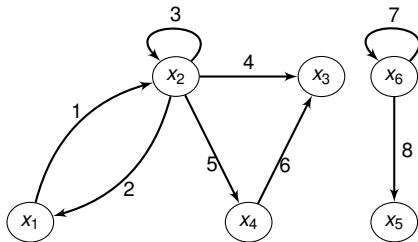
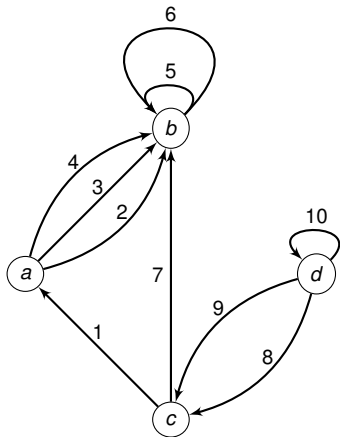
Definition 275 (Multidigraph)

A **multidigraph** is a digraph which allows repetition of arcs or loops

Definition 276 (Digraph)

In a digraph, no more than one arc can join any pair of vertices

Examples



Let $G = (V, A)$ be a digraph

Definition 277 (Arc endpoints)

For an arc $u = (x, y)$, vertex x is the **initial endpoint**, and vertex y is the **terminal endpoint**

Definition 278 (Predecessor - Successor)

If $(u, v) \in A(G)$ is an arc of G , then

- ▶ u is a **predecessor** of v
- ▶ v is a **successor** of u

Definition 279 (Neighbours of a vertex)

Let $x \in V$ be a vertex. The **neighbours** of x is the set $\Gamma(x) = \Gamma_G^+(x) \cup \Gamma_G^-(x)$, where $\Gamma_G^+(x)$ and $\Gamma_G^-(x)$ are, respectively, the set of successors and predecessors of x

Sources and sinks

Definition 280 (Directed away - Directed towards)

If $a = (u, v) \in A(G)$ is an arc of G , then

- ▶ the arc a is said to be **directed away** from u
- ▶ the arc a is said to be **directed towards** v

Definition 281 (Source - Sink)

- ▶ Any vertex which has no arcs directed towards it is a **source**
- ▶ Any vertex which has no arcs directed away from it is a **sink**

Adjacent arcs

Definition 282 (Adjacent arcs)

Two arcs are **adjacent** if they have at least one endpoint in common

Arcs incident to a subset of arcs

Definition 283 (Arc incident out of $X \subset A(G)$)

If the initial endpoint of an arc u belongs to $X \subset A(G)$ and if the terminal endpoint of arc u does not belong to X , then u is said to be **incident out of** X ; we write $u \in \omega^+(X)$

Similarly, we define an **arc incident into** X and the set $\omega^-(X)$

Finally, the set of arcs **incident to** X is denoted

$$\omega(X) = \omega^+(X) \cup \omega^-(X)$$

Definition 284 (Subgraph of G generated by $A \subset V$)

The **subgraph** of G generated by A is the graph with A as its vertex set and with all the arcs in G that have both their endpoints in A . If $G = (V, \Gamma)$ is a 1-graph, then the subgraph generated by A is the 1-graph $G_A = (A, \Gamma_A)$ where

$$\Gamma_A(x) = \Gamma(x) \cap A \quad (x \in A)$$

Definition 285 (Partial graph of G generated by $V \subset U$)

The graph (X, V) whose vertex set is X and whose arc set is V . In other words, it is graph G without the arcs $U - V$

Definition 286 (Partial subgraph of G)

A partial subgraph of G is the subgraph of a partial graph of G

Directed graphs

Degrees in digraphs

Walks, paths, etc.

Connectivity in digraphs

Orientable graphs

FIGS/random_ug_00.pdf

Degree

Let v be a vertex of a digraph $G = (V, A)$

Definition 287 (Outdegree of a vertex)

The number of arcs directed away from a vertex v , in a digraph is called the **outdegree** of v and is written $d_G^+(v)$

Definition 288 (Indegree of a vertex)

The number of arcs directed towards a vertex v , in a digraph is called the **indegree** of v and is written $d_G^-(v)$

Definition 289 (Degree)

For any vertex v in a digraph, the **degree** of v is defined as

$$d_G(v) = d_G^+(v) + d_G^-(v)$$

Theorem 290

For any (di)graph, the sum of the degrees of the vertices equals twice the number of edges (arcs)

Corollary 291

In any (di)graph, the sum of the degrees of the vertices is a nonnegative even integer

Theorem 292

If G is a digraph with vertex set $V(G) = \{v_1, \dots, v_p\}$ and q arcs, then

$$\sum_{i=1}^p d_G^+(v_i) = \sum_{i=1}^p d_G^-(v_i) = q$$

Definition 293 (Regular digraph)

A digraph G is r -regular if $d_G^+(v) = d_G^-(v) = r$ for all $v \in V(G)$

Symmetric/antisymmetric digraphs

Definition 294 (Symmetric digraph)

Let $G = (V, A)$ be a digraph with associated binary relation R . If R is *symmetric*, the digraph is symmetric

Definition 295 (Anti-symmetric digraph)

Let $G = (V, A)$ be a digraph with associated binary relation R . The digraph G is **anti-symmetric** if

$$xRy \implies y \not R x$$

Definition 296 (Symmetric multidigraph)

Let $G = (V, A)$ be a multidigraph. G is symmetric if $\forall x, y \in V(G)$, the number of arcs from x to y equals the number of arcs from y to x

Directed graphs

Degrees in digraphs

Walks, paths, etc.

Connectivity in digraphs

Orientable graphs

FIGS/random_ug_00.pdf

Walks

Let $G = (V, A)$ be a digraph.

Definition 297 (Directed walk)

A **directed walk** in a digraph G is a non-empty alternating sequence $v_0 a_0 v_1 a_1 v_2 \dots a_{k-1} v_k$ of vertices and arcs in G such that $a_i = (v_i, v_{i+1})$ for all $i < k$. This walk begins with v_0 and ends with v_k

Definition 298 (Length of a directed walk)

The length of a directed walk is equal to the number of arcs in the directed walk

Definition 299 (Closed walk)

If $v_0 = v_k$, the walk is closed

Trails

Let $G = (V, A)$ be a digraph.

Definition 300 (Directed trail)

A directed walk in G in which all arcs are distinct is a **directed trail** in G

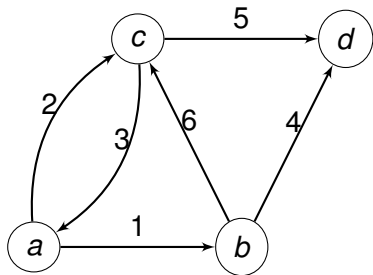
Definition 301 (Directed path)

A directed walk in G in which all vertices are distinct is a **directed path** in G

Definition 302 (Directed cycle)

A closed walk is a **directed cycle** if it contains at least three vertices and all its vertices are distinct except for $v_0 = v_k$

Examples of directed cycles



Cycles:

- ▶ $\mu^1 = (1, 6, 2) = [abca]$
- ▶ $\mu^2 = (1, 6, 3) = [abca]$
- ▶ $\mu^3 = (2, 3) = [aca]$
- ▶ $\mu^4 = (1, 4, 5, 2) = [abdca]$
- ▶ $\mu^5 = (6, 5, 4) = [acdb]$
- ▶ $\mu^6 = (1, 4, 5, 3) = [abdca]$

Directed graphs

Degrees in digraphs

Walks, paths, etc.

Connectivity in digraphs

Orientable graphs

FIGS/random_ug_07.pdf

Definitions

Definition 303 (Underlying graph)

Given a digraph, the undirected graph with each arc replaced by an edge is called the **underlying graph**

Definition 304 (Weakly connected digraph)

If the underlying graph is a connected graph, then the digraph is **weakly connected**

Definition 305 (Strongly connected digraph)

A digraph G is **strongly connected** if for every two distinct vertices u and v of G , there exists a directed path from u to v

Definition 306 (Disconnected digraph)

A digraph is said to be **disconnected** if it is not weakly connected

Strong connectedness is an equivalence relation

Denote $x \equiv y$ the relation “ $x = y$, or $x \neq y$ and there exists a directed path in G from x to y ”. $\equiv$ is an equivalence relation since

1. $x \equiv x$ [reflexivity]
2. $x \equiv y \implies y \equiv x$ [symmetry]
3. $x \equiv y, y \equiv z \implies x \equiv z$ [transitivity]

Definition 307 (Connected component of a graph)

Sets of the form

$$A(x_0) = \{x : x \in V, x \equiv x_0\}$$

are equivalence classes. They partition V into strongly connected sub-digraphs of G called **strongly connected components** (or **strong components**) of G

A strong component in G is a maximal strongly connected subdigraph of G

Theorem 308 (Properties)

Let $G = (V, A)$ be a digraph

- ▶ *If G is strongly connected, it has only one strongly connected component*
- ▶ *The strongly connected components partition the vertices $V(G)$, with every vertex in exactly one strongly connected component*

Algorithm for determining strongly connected components in

$$G = (V, A)$$

- ▶ Determine the strongly connected component $C(v)$ containing the vertex v ; if $V - C(v)$ is non-empty, re-do the same operation on the sub-digraph $G' = (V - C(v), A')$
- ▶ To determine $C(v)$, the strongly connected component containing v : let v be a vertex of a digraph, which is not already in any strongly connected component
 1. Mark the vertex v with $\pm$
 2. Mark with $+$ all successors (not already marked with $+$) of a vertex marked with $+$
 3. Mark with $-$ all predecessors (not already marked with $-$) of a vertex marked with $-$
 4. Repeat until no more possible marking with $+$ or $-$

All vertices marked with $\pm$ belong to the same strongly connected component $C(v)$ containing the vertex v

Condensation of a digraph

Definition 309 (Condensation of a digraph)

The condensation G^* of a digraph G is a digraph having as vertices the strongly connected components (SCC) of G and such that there exists an arc in G^* from a SCC C_i to another SCC C_j if there is an arc in G from some vertex of S_i to a vertex of S_j

Definition 310 (Articulation set)

For a connected graph, a set X of vertices is called an **articulation set** (or a **cutset**) if the subgraph of G generated by $V - X$ is not connected

Definition 311 (Stable set)

A set S of vertices is called a **stable set** if no arc joins two distinct vertices in S

Directed graphs

Degrees in digraphs

Walks, paths, etc.

Connectivity in digraphs

Orientable graphs

FIGS/random-graphs.pdf

Orientation

Definition 312 (Orienting a graph)

Given a connected *graph*, we describe the act of assigning a direction to each edge (edge $\rightarrow$ arc) as **orienting the graph**

Definition 313 (Strong orientation)

If the digraph resulting from orienting a graph is strongly connected, the orientation is a **strong orientation**

Orienable graph

Definition 314 (Orienable graph)

A connected graph G is **orientable** if it admits a strong orientation

Theorem 315

A connected graph $G = (V, E)$ is orientable $\iff G$ contains no bridges

(in other words, iff every edge is contained in a cycle)

Matrices associated to a graph/digraph

There are multiple matrices associated to a graph/digraph

The branch of graph theory that studies the properties of matrices derived from graphs and uses of these matrices in determining graph properties is *spectral graph theory*

Graphs greatly simplify some problems in linear algebra and vice versa

Matrices associated to a graph/digraph

Adjacency matrices

Other matrices associated to a graph/digraph

Linking graphs and linear algebra

FIGS/random_bg_39.pdf

Adjacency matrix (undirected case)

Let $G = (V, E)$ be a graph of order p and size q , with vertices $v_1, \dots, v_p$ and edges $e_1, \dots, e_q$

Definition 316 (Adjacency matrix)

The **adjacency matrix** is

$$M_A = M_A(G) = [m_{ij}]$$

is a $p \times p$ matrix in which

$$m_{ij} = \begin{cases} 1 & \text{if } v_i \text{ and } v_j \text{ are adjacent} \\ 0 & \text{otherwise} \end{cases}$$

Theorem 317 (Adjacency matrix and degree)

The sum of the entries in row i of the adjacency matrix is the degree of v_i in the graph

We often write $A(G)$ and, reciprocally, if A is an adjacency matrix, $G(A)$ the corresponding graph

G undirected $\implies A(G)$ symmetric

$A(G)$ has nonzero diagonal entries if G is not simple

Adjacency matrix (directed case)

Let $G = (V, A)$ be a digraph of order p with vertices $v_1, \dots, v_p$

Definition 318 (Adjacency matrix)

The **adjacency matrix** $M = M(G) = [m_{ij}]$ is a $p \times p$ matrix in which

$$m_{ij} = \begin{cases} 1 & \text{if arc } v_i v_j \in A \\ 0 & \text{otherwise} \end{cases}$$

Theorem 319 (Properties)

- ▶ *M is not necessarily symmetric*
- ▶ *The sum of any column of M is equal to the number of arcs directed towards v_j*
- ▶ *The sum of the entries in row i is equal to the number of arcs directed away from vertex v_i*
- ▶ *The (i, j) –entry of M^n is equal to the number of walks of length n from vertex v_i to v_j*

Definition 320 (Multiplicity of a pair)

The **multiplicity** of a pair x, y is the number $m_G^+(x, y)$ of arcs with initial endpoint x and terminal endpoint y . Let

$$m_G^-(x, y) = m_G^+(y, x)$$

$$m_G(x, y) = m_G^+(x, y) + m_G^-(x, y)$$

If $x \neq y$, then $m_G(x, y)$ is number of arcs with both x and y as endpoints. If $x = y$, then $m_G(x, y)$ equals twice the number of loops attached to vertex x . If $A, B \subset V$, $A \neq B$, let

$$m_G^+(A, B) = \{u : u \in U, u = (x, y), x \in A, y \in B\}$$

$$m_G(A, B) = m_G^+(A, B) + m_G^-(A, B)$$

Adjacency matrix of a multigraph

Definition 321 (Matrix associated with G)

If G has vertices $x_1, x_2, \dots, x_n$, then the **matrix associated** with G is

$$a_{ij} = m_G^+(x_i, x_j)$$

Definition 322 (Adjacency matrix)

The matrix $a_{ij} + a_{ji}$ is the **adjacency matrix** associated with G

Adjacency matrix (multigraph case)

Definition 323 (Adjacency matrix of a multigraph)

G an ℓ -graph, then the adjacency matrix $M_A = [m_{ij}]$ is defined as follows

$$m_{ij} = \begin{cases} k & \text{if arc there are } k \text{ arcs } (i, j) \in U \\ 0 & \text{otherwise} \end{cases}$$

with $k \leq \ell$

G undirected $\implies M_A(G)$ symmetric

$M_A(G)$ has nonzero diagonal entries if G is not simple.

Weighted adjacency matrices

Sometimes, adjacency matrices (typically for 1-graphs) have real entries, usually positive

This means that the arcs/edges have been given a weight

Theorem 324 (Number of walks of length n)

Let A be the adjacency matrix of a graph $G = (V(G), E(G))$, where $V(G) = \{v_1, v_2, \dots, v_p\}$. Then the (i, j) -entry of A^n , $n \geq 1$, is the number of different walks linking v_i to v_j of length n in G .

(two walks of the same length are equal if their edges occur in exactly the same order)

Example: let A be the adjacency matrix of a graph $G = (V(G), E(G))$.

- ▶ the (i, i) -entry of A^2 is equal to the degree of v_i .
- ▶ the (i, i) -entry of A^3 is equal to twice the number of C_3 containing v_i .

Matrices associated to a graph/digraph

Adjacency matrices

Other matrices associated to a graph/digraph

Linking graphs and linear algebra

FIGS/random_bg_40.pdf

Incidence matrix (undirected case)

Let $G = (V, E)$ be a graph of order p , and size q , with vertices $v_1, \dots, v_p$, and edges $e_1, \dots, e_q$

Definition 325 (Incidence matrix)

The incidence matrix is

$$B = B(G) = [b_{ij}]$$

is that $p \times q$ matrix in which

$$b_{ij} = \begin{cases} 1 & \text{if } v_i \text{ is incident with } e_j \\ 0 & \text{otherwise} \end{cases}$$

Theorem 326 (Incidence matrix and degrees)

The sum of the entries in row i of the incidence matrix is the degree of v_i in the graph

Incidence matrix (directed case)

Let $G = (V, A)$ be a digraph of order p and size q , with vertices $v_1, \dots, v_p$ and arcs $a_1, \dots, a_q$

Definition 327 (Incidence matrix)

The **incidence matrix** $B = B(G) = [b_{ij}]$ is a $p \times q$ matrix in which

$$b_{ij} = \begin{cases} 1 & \text{if arc } a_j \text{ is directed away from a vertex } v_i \\ -1 & \text{if arc } a_j \text{ is directed towards a vertex } v_i \\ 0 & \text{otherwise} \end{cases}$$

Spectrum of a graph

We will come back to this later, but for now..

Definition 328 (Spectrum of a graph)

The **spectrum** of a graph G is the spectrum (set of eigenvalues) of its associated adjacency matrix $M(G)$

This is regardless of the type of adjacency matrix or graph

Degree matrix

Definition 329 (Degree matrix)

The **degree** matrix $D = [d_{ij}]$ for G is a $n \times n$ diagonal matrix defined as

$$d_{ij} = \begin{cases} d_G(v_i) & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

In an undirected graph, this means that each loop increases the degree of a vertex by two

In a directed graph, the term “degree” may refer either to indegree (the number of incoming edges at each vertex) or outdegree (the number of outgoing edges at each vertex)

Laplacian matrix

Definition 330 (Laplacian matrix)

$G = (V, A)$ a simple graph with n vertices. The **Laplacian** matrix is

$$L = D(G) - M(G)$$

where $D(G)$ is the degree matrix and $M(G)$ is the adjacency matrix

Laplacian matrix (continued)

G simple graph $\implies M(G)$ only contains 1 or 0 and its diagonal elements are all 0

For directed graphs, either the indegree or outdegree is used, depending on the application

Elements of L are given by

$$\ell_{ij} = \begin{cases} d_G(v_i) & \text{if } i = j \\ -1 & \text{if } i \neq j \text{ and } v_i \text{ is adjacent to } v_j \\ 0 & \text{otherwise} \end{cases}$$

Distance matrix

Let G be a graph of order p with vertices $v_1, \dots, v_p$

Definition 331 (Distance matrix)

The distance matrix $\Delta(G) = [d_{ij}]$ is a $p \times p$ matrix in which

$$\delta_{ij} = d_G(v_i, v_j)$$

Note $\delta_{ii} = 0$ for $i = 1, \dots, p$

Property 332

- ▶ *M is not necessarily symmetric*
- ▶ *The sum of any column of M is equal to the number of arcs directed towards v_j*
- ▶ *The sum of the entries in row i is equal to the number of arcs directed away from vertex v_i*
- ▶ *The (i, j) –entry of M^n is equal to the number of walks of length n from vertex v_i to v_j*

Matrices associated to a graph/digraph

Adjacency matrices

Other matrices associated to a graph/digraph

Linking graphs and linear algebra

FIGS/random_bg_41.pdf

Counting paths

Theorem 333

G a digraph and $M_A(G)$ its adjacency matrix. Denote $P = [p_{ij}]$ the matrix $P = M_A^k$. Then p_{ij} is the number of distinct paths of length k from i to j in G

Definition 334 (Irreducible matrix)

A matrix $A \in \mathcal{M}_n$ is **reducible** if $\exists P \in \mathcal{M}_n$, permutation matrix, s.t. $P^T A P$ can be written in block triangular form. If no such P exists, A is **irreducible**

Theorem 335

A irreducible $\iff G(A)$ strongly connected

Theorem 336

Let A be the adjacency matrix of a graph G on p vertices. A graph G on p vertices is connected $\iff$

$$I + A + A^2 + \cdots + A^{p-1} = C$$

has no zero entries

Theorem 337

Let M be the adjacency matrix of a digraph D on p vertices. A digraph D on p vertices is strongly connected $\iff$

$$I + M + M^2 + \cdots + M^{p-1} = C$$

has no zero entries

Nonnegative matrix

$A = [a_{ij}] \in \mathcal{M}_n(\mathbb{R})$ **nonnegative** if $a_{ij} \geq 0 \ \forall i, j = 1, \dots, n$; $\mathbf{v} \in \mathbb{R}^n$ nonnegative if $v_i \geq 0 \ \forall i = 1, \dots, n$. **Spectral radius** of A

$$\rho(A) = \max_{\lambda \in \text{Sp}(A)} \{|\lambda|\}$$

$\text{Sp}(A)$ the **spectrum** of A

Perron-Frobenius (PF) theorem

Theorem 338 (PF – Nonnegative case)

$0 \leq A \in \mathcal{M}_n(\mathbb{R})$. Then $\exists \mathbf{v} \geq \mathbf{0}$ s.t.

$$A\mathbf{v} = \rho(A)\mathbf{v}$$

Theorem 339 (PF – Irreducible case)

Let $0 \leq A \in \mathcal{M}_n(\mathbb{R})$ irreducible. Then $\exists \mathbf{v} > \mathbf{0}$ s.t.

$$A\mathbf{v} = \rho(A)\mathbf{v}$$

$\rho(A) > 0$ and with algebraic multiplicity 1. No nonnegative eigenvector is associated to any other eigenvalue of A

Primitive matrices

Definition 340

$0 \leq A \in \mathcal{M}_n(\mathbb{R})$ **primitive** (with **primitivity index** $k \in \mathbb{N}_+^*$) if $\exists k \in \mathbb{N}_+^*$ s.t.

$$A^k > 0,$$

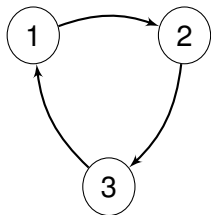
with k the smallest integer for which this is true. A **imprimitive** if it is not primitive

A primitive $\implies A$ irreducible; the converse is false

Theorem 341

$A \in \mathcal{M}_n(\mathbb{R})$ *irreducible and* $\exists i = 1, \dots, n$ s.t. $a_{ij} > 0 \implies A$ *primitive*

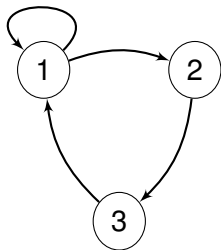
Here d is the index of imprimitivity (i.e., the number of eigenvalues that have the same modulus as $\lambda_p = \rho(A)$). If $d = 1$, then A is primitive. We have that $d = \gcd$ of all the lengths of closed walks in $G(A)$



Adjacency matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

Closed walks in $G(A)$ (lengths): $1 \rightarrow 1$ (3), $2 \rightarrow 2$ (3), $3 \rightarrow 3$ (3) $\implies \gcd = 3 \implies d = 3$ (here, all eigenvalues have modulus 1)



$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

Closed walk $1 \rightarrow 1$ has length 1 $\implies \gcd$ of lengths of closed walks is 1 $\implies A$ primitive

Let $\mathbf{0} \leq A \in \mathcal{M}_n$

Theorem 342

A primitive $\implies \exists 0 < k \leq (n-1)n^n$ such that $A^k > \mathbf{0}$

Theorem 343

If A is primitive and the shortest simple directed cycle in $G(A)$ has length s , then the primitivity index is $\leq n + s(n-1)$

Theorem 344

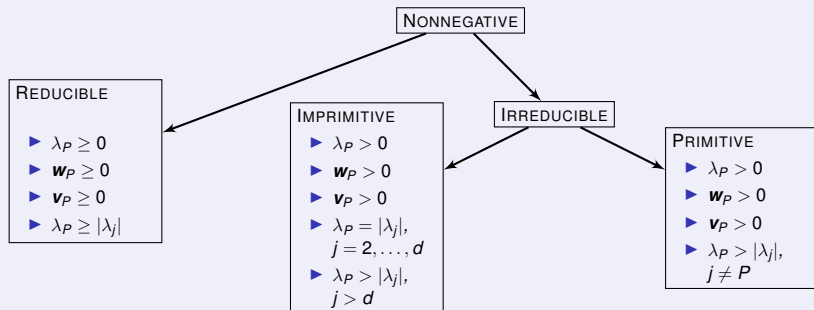
A primitive $\iff A^{n^2-2n+2} > \mathbf{0}$

Theorem 345

If A is irreducible and has d positive entries on the diagonal, then the primitivity index $\leq 2n - d - 1$

Theorem 346

$\mathbf{0} \leq A \in \mathcal{M}_n$, $\lambda_P = \rho(A)$ the Perron root of A , $\mathbf{v}_P$ and $\mathbf{w}_P$ the corresponding right and left Perron vectors of A , respectively, d the index of imprimitivity of A (with $d = 1$ when A is primitive) and $\lambda_j \in \sigma(A)$ the spectrum of A , with $j = 2, \dots, n$ unless otherwise specified (assuming $\lambda_1 = \lambda_P$)



Why use graphs/networks?

Binary relations

Undirected graphs

Directed graphs

`FIGS/random_bg_42.pdf`

Matrices associated to a graph/digraph

Trees

Why use graphs/networks?

Binary relations

Trees

Definition 347 (Forest, trees and branches)

- ▶ A connected graph with no cycle is a **tree**
- ▶ A tree is a connected acyclic graph, its edges are called **branches**
- ▶ A graph (connected or not) without any cycle is a **forest**. Each component is a tree

(A forest is a graph whose connected components are trees)

Is the “Karate graph” a tree?

```
is_acyclic(G)
```

```
## [1] FALSE
```

```
is_tree(G)
```

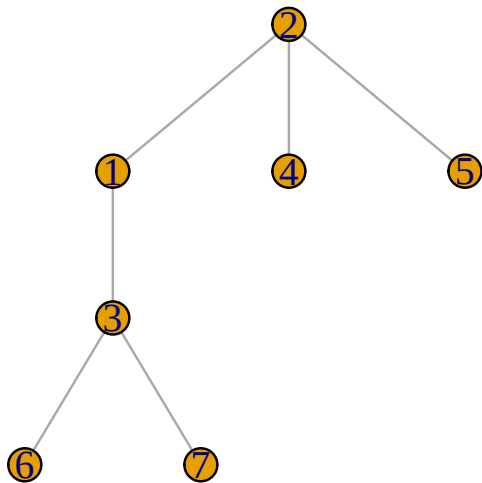
```
## [1] FALSE
```

So we need friend to play with!

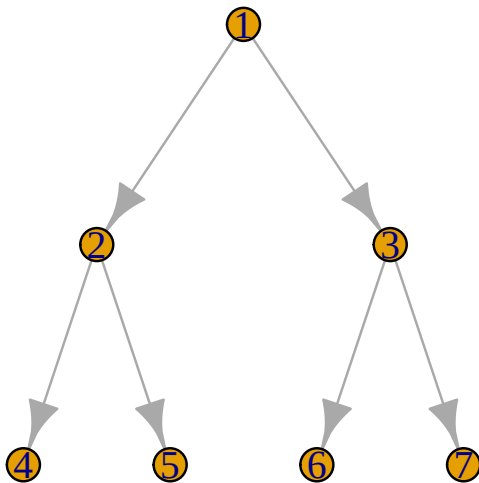
```
G_tu <- make_tree(7, 2, mode = "undirected")
```

```
G_td <- make_tree(7, 2)
```

An undirected tree



A (out) directed tree



Property 348

- ▶ *Every edge of a tree is a bridge*
- ▶ *Given two vertices u and v of a tree, there is an unique path linking u to v*
- ▶ *A tree with p vertices and q edges satisfies $q = p - 1$. Thus, a tree is minimally connected*

(First property: the deletion of any edge of a tree disconnects it)

Every edge of a tree is a bridge

```
E(G_tu)

## + 6/6 edges from e484df2:
## [1] 1--2 1--3 2--4 2--5 3--6 3--7

bridges(G_tu)

## + 6/6 edges from e484df2:
## [1] 2--4 2--5 1--2 3--6 3--7 1--3

all(sort(E(G_tu)) == sort(bridges(G_tu)))

## [1] TRUE
```

Spanning tree

Definition 349 (Spanning tree)

A **spanning tree** of a connected graph G is a subgraph of G that contains all the vertices of G and is a tree.

A graph may have many spanning trees

Minimal spanning tree

Definition 350 (Value of a spanning tree)

The **value of a spanning tree** T of order p is

$$\sum_{i=1}^{p-1} f(e_i)$$

where f is the function that maps the edge set into $\mathbb{R}$

Definition 351 (Minimal spanning tree)

Let G be an undirected network, and let T be a **minimal spanning tree** of G . Then T is a spanning tree whose the value is minimum

Algorithm to find a minimal spanning tree

Let $G = (V(G), E(G))$ be an undirected network and T be a minimal spanning tree

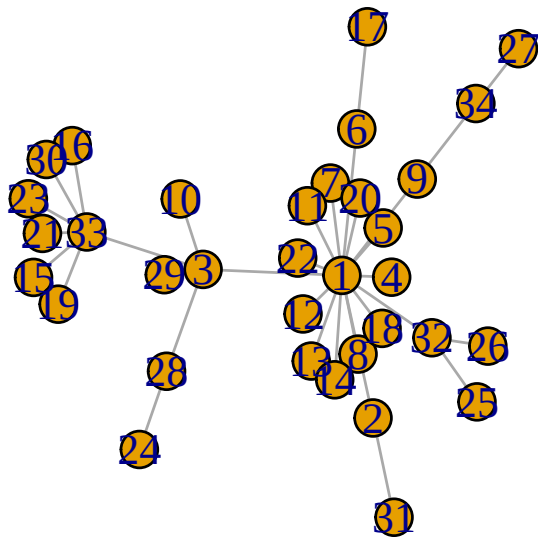
1. Sort the edges of G in increasing order by value
2. $T = (V(G), \emptyset)$
3. For each edge e in sorted order if the endpoints of e are disconnected in T add e to T

Finding a minimal spanning tree of the Karate graph

The function `mst` finds minimal spanning trees, using distances if no edge weights are provided

```
G_mst = mst(G)
```


A minimal spanning tree of the Karate graph



Minimal connector problem

- ▶ Model: a graph G such that edges represent all possible connections, and each edge has a positive value which represents its cost; an undirected network G
- ▶ Solution: a minimal spanning tree T of G
 - ▶ a spanning tree of G is a subgraph of G that contains all the vertices of G and is a tree.
 - ▶ the cost of the spanning tree is the sum of values of the edges of T
 - ▶ a spanning tree such that no other spanning tree has a smaller cost is a minimal spanning tree.

Theorem 352 (Characterisation of trees)

$H = (V, U)$ a graph of order $|V| = n > 2$. The following are equivalent and all characterise a tree :

- 1. H connected and has no cycles*
- 2. H has $n - 1$ arcs and no cycles*
- 3. H connected and has exactly $n - 1$ arcs*
- 4. H has no cycles, and if an arc is added to H , exactly one cycle is created*
- 5. H connected, and if any arc is removed, the remaining graph is not connected*
- 6. Every pair of vertices of H is connected by one and only one chain*

Definition 353 (Pendant vertex)

A vertex is **pendant** if it is adjacent to exactly one other vertex

Theorem 354

A tree of order $n \geq 2$ has at least two pendant vertices

Theorem 355

A graph $G = (V, U)$ has a partial graph that is a tree $\iff G$ connected

Recall that a partial graph is a graph generated by a subset of the arcs
(Definition 473 slide 384)

Spanning tree

The procedure in the proof of Theorem 573 gives a **spanning tree**

Can also build a spanning tree as follows:

- ▶ Consider any arc u_0
- ▶ Find arc u_1 that does not form a cycle with u_0
- ▶ Find arc u_2 that does not form a cycle with $\{u_0, u_1\}$
- ▶ Continue
- ▶ When you cannot continue anymore, you have a spanning tree

Theorem 356

G connected graph with ≥ 1 arc. TFAE

1. *G strongly connected*
2. *Every arc lies on a circuit*
3. *G contains no cocircuits*

Theorem 357

G graph with ≥ 1 arc. TFAE

1. *G is a graph without circuits*
2. *Each arc is contained in a cocircuit*

Theorem 358

If G is a strongly connected graph of order n , then G has a cycle basis of $\nu(G)$ circuits

Definition 359 (Node, anti-node, branch)

$G = (V, U)$ strongly connected without loops and > 1 vertex. For each $x \in V$, there is a path from it and a path to it so x has at least 2 incident arcs. Specifically,

- ▶ $x \in V$ with > 2 incident arcs is a **node**
- ▶ $x \in V$ with 2 incident arcs is an **anti-node**

A path whose only nodes are its endpoints is a **branch**

Definition 360 (Minimally connected graph)

G is **minimally connected** if it is strongly connected and removal of any arc destroys strong-connectedness

A minimally connected graph is 1-graph without loops

Definition 361 (Contraction)

$G = (V, U)$. The **contraction** of the set $A \subset V$ of vertices consists in replacing A by a single vertex a and replacing each arc into (resp. out of) A by an arc with same index into (resp. out of) a

Theorem 362

*G minimally connected, $A \subset V$ generating a strongly connected subgraph of G .
Then the contraction of A gives a minimally connected graph*

Theorem 363

G a minimally connected graph, G' be the minimally connected graph obtained by the contraction of an elementary circuit of G . Then

$$\nu(G) = \nu(G') + 1$$

Theorem 364

G minimally connected of order $n \geq 2 \implies G$ has ≥ 2 anti-nodes

Theorem 365

$G = (V, U)$. Then the graph C' obtained by contracting each strongly connected component of G contains no circuits

Arborescences

Definition 366 (Root)

Vertex $a \in V$ in $G = (V, U)$ is a **root** if all vertices of G can be reached by paths *starting* from a

Not all graphs have roots

Definition 367 (Quasi-strong connectedness)

G is **quasi-strongly connected** if $\forall x, y \in V$, exists $z \in V$ (denoted $z(x, y)$ to emphasize dependence on x, y) from which there is a path to x and a path to y

Strongly connected $\implies$ quasi-strongly connected (take $z(x, y) = x$); converse not true

Quasi-strongly connected $\implies$ connected

Arborescence

Definition 368 (Arborescence)

An **arborescence** is a tree that has a root

Lemma 369

$G = (V, U)$ has a root $\iff G$ quasi-strongly connected

Theorem 370

H graph of order $n > 1$. TFAE (and all characterise an arborescence)

1. *H quasi-strongly connected without cycles*
2. *H quasi-strongly connected with $n - 1$ arcs*
3. *H tree having a root a*
4. $\exists a \in V$ s.t. *all other vertices are connected with a by 1 and only 1 path from a*
5. *H quasi-strongly connected and loses quasi-strong connectedness if any arc is removed*
6. *H quasi-strongly connected and $\exists a \in V$ s.t.*

$$d_H^-(a) = 0$$

$$d_H^-(x) = 1 \quad \forall x \neq a$$

7. *H has no cycles and $\exists a \in V$ s.t.*

$$d_H^-(a) = 0$$

$$d_H^-(x) = 1 \quad \forall x \neq a$$

Theorem 371

G has a partial graph that is an arborescence $\iff G$ quasi-strongly connected

Theorem 372

$G = (V, E)$ simple connected graph and $x_1 \in V$. It is possible to direct all edges of E so that the resulting graph $G_0 = (V, U)$ has a spanning tree H s.t.

- 1. H is an arborescence with root x_1*
- 2. The cycles associated with H are circuits*
- 3. The only elementary circuits of G_0 are the cycles associated with H*

Counting trees

Proposition 373

*X a set with n distinct objects, $n_1, \dots, n_p$ nonnegative integers s.t.
 $n_1 + \dots + n_p = n$. The number of ways to place the n objects into p boxes
 $X_1, \dots, X_p$ containing $n_1, \dots, n_p$ objects respectively is*

$$\binom{n}{n_1, \dots, n_p} = \frac{n!}{n_1! \cdots n_p!}$$

Proposition 374 (Multinomial formula)

Let $a_1, \dots, a_p \in \mathbb{R}$ be p real numbers, then

$$(a_1 + \dots + a_p)^n = \sum_{n_1, \dots, n_p \geq 0} \binom{n}{n_1, \dots, n_p} (a_1)^{n_1} \cdots (a_p)^{n_p}$$

Theorem 375

Denote $T(n; d_1, \dots, d_n)$ the number of distinct trees H with vertices $x_1, \dots, x_n$ and with degrees $d_H(x_1) = d_1, \dots, d_H(x_n) = d_n$. Then

$$T(n; d_1, \dots, d_n) = \binom{n-2}{d_1-1, \dots, d_n-1}$$

Theorem 376

The number of different trees with vertices $x_1, \dots, x_n$ is n^{n-2}

There is a whole industry of similar results (as well as for arborescences), but we will stop here. The main point is that we are talking about a large number of possibilities..

Why use graphs/networks?

Binary relations

Undirected graphs

Directed graphs

`FIGS/random_bg_43.pdf`

Matrices associated to a graph/digraph

Trees

Why use graphs/networks?

Binary relations

Graphs versus networks

Mostly a terminology difference:

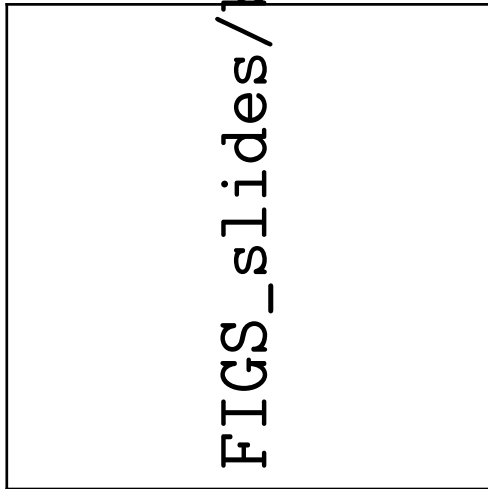
- ▶ graphs in the mathematical world
- ▶ networks elsewhere

I will mostly say *graphs* (this is a math course) but might oscillate

Beware: language is not consistent, so make sure you read the definitions at the start of whatever source you are using

The genesis of graphs – Euler's bridges of Königsberg

Cross the 7 bridges in a single walk without recrossing any of them?



The genesis of graphs – Euler's bridges of Königsberg

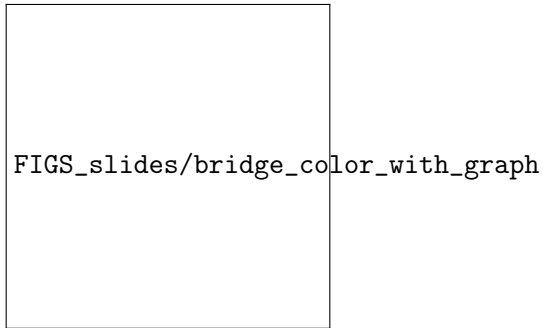
Cross the 7 bridges in a single walk without recrossing any of them?



FIGS_slides/bridge_color_with_graph

The genesis of graphs – Euler's bridges of Königsberg

Cross the 7 bridges in a single walk without recrossing any of them?

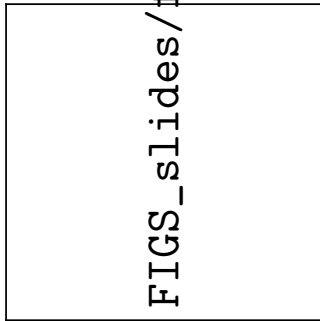


Mathematical problem

Is it possible to find a *trail* containing all *edges* of the graph?

Finding a cycle with all vertices

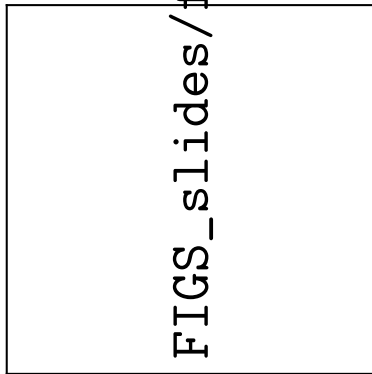
A salesperson must visit a couple of cities for their job. Is it possible for them to plan a round trip using highways enabling them to visit each specified city exactly once?



- ▶ vertices correspond to cities
- ▶ two vertices are connected iff a highway connects the corresponding cities and does not pass through any other city.

How far is it to drive through n cities?

What is the minimal length of driving needed to drive through n cities?



- ▶ vertices correspond to the cities
- ▶ all cities are connected; each edge has a value assigned to it

Mathematical problem

What is the minimal spanning tree associated to the graph?

Graphs/networks encode relations

Graphs are used in a variety of contexts because they encode *relations* between objects

Many objects in the world have relations... so graphs are quite easy to find

We will see many examples later, for now we cover the mathematical background

Graphs vs digraphs vs multigraphs vs multidigraphs vs ...

Name-wise and notation-wise, this domain is a bit of a mess

- ▶ The vertex set V is essentially the only constant
- ▶ *Undirected graph* $G = (V, E)$, where E are the *edges*
- ▶ *Undirected multigraph* $G_M = (V, E)$
- ▶ *Directed graph* (or *digraph*) $G = (V, A)$, where A are the *arcs*
- ▶ *Directed multigraph* (or *multidigraph*) $G_M = (V, A)$
- ▶ Any of the above is called a *graph* and is denoted $G = (V, X)$, when we seek generality

And just to confuse the whole thing more: we often say *graph* for *unoriented graph*

Why use graphs/networks?

Binary relations

Undirected graphs

Directed graphs

`FIGS/random_bg_44.pdf`

Matrices associated to a graph/digraph

Trees

Why use graphs/networks?

Binary relations

Binary relation

Definition 377 (Binary relation)

- ▶ A **binary relation** is an arbitrary association of elements of one set with elements of another (maybe the same) set
- ▶ A binary relation over the sets X and Y is defined as a subset of the Cartesian product $X \times Y = \{(x, y) | x \in X, y \in Y\}$
- ▶ $(x, y) \in R$ is read “ x is R -related to y ” and is denoted xRy
- ▶ If $(x, y) \notin R$, we write “not xRy ” or $x \not R y$

Definition 378 (Properties of binary relations)

A binary relation R over a set X is

- ▶ **Reflexive** if $\forall x \in X, xRx$
- ▶ **Irreflexive** if there does not exist $x \in X$ such that xRx
- ▶ **Symmetric** if $xRy \Rightarrow yRx$
- ▶ **Asymmetric** if $xRy \Rightarrow y \not R x$
- ▶ **Antisymmetric** if xRy and $yRx \Rightarrow x = y$
- ▶ **Transitive** if xRy and $yRz \Rightarrow xRz$
- ▶ **Total** (or **complete**) if $\forall x, y \in X, xRy$ or yRx

Definition 379 (Equivalence relation)

A relation that is reflexive ($\forall x \in X, xRx$), symmetric ($xRy \Rightarrow yRx$) and transitive (xRy and $yRz \Rightarrow xRz$) is an **equivalence relation**

Definition 380 (Partial order)

A relation that is reflexive ($\forall x \in X, xRx$), antisymmetric (xRy and $yRx \Rightarrow x = y$) and transitive (xRy and $yRz \Rightarrow xRz$) is a **partial order**

Definition 381 (Total order)

A partial order that is total ($\forall x, y \in X, xRy$ or yRx) is a **total order**

Why use graphs/networks?

Binary relations

Undirected graphs

Directed graphs

`FIGS/random_bg_45.pdf`

Matrices associated to a graph/digraph

Trees

Why use graphs/networks?

Binary relations

The `igraph` library

Throughout these slides, we use the package `igraph`

I illustrate the functions that can be used to study some of the mathematical notions I introduce

I use mostly examples from the `igraph` documentation

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

Complete, bipartite and other notable graphs

Planar graphs

Graph colouring

Graph

Intuitively: a graph is a set of points, and a set of relations between the points

The points are called the *vertices* of the graph and the relations are the *edges* of the graph

We can also think of the relations as being one directional, in which case the relations are the *arcs* of the digraph (a contraction of “directed graph”)

Graph, vertex and edge

Definition 382 (Graph)

An **undirected graph** is a pair $G = (V, E)$ of sets such that

- ▶ V is a set of points: $V = \{v_1, \dots, v_p\}$
- ▶ E is a set of 2-element subsets of V : $E = \{\{v_i, v_j\}, \{v_i, v_k\}, \dots, \{v_n, v_p\}\}$ or $E = \{v_i v_j, v_i v_k, \dots, v_n v_p\}$

Definition 383 (Vertex)

The elements of V are the **vertices** (or nodes, or points) of the graph G . V (or $V(G)$) is the vertex set of the graph G

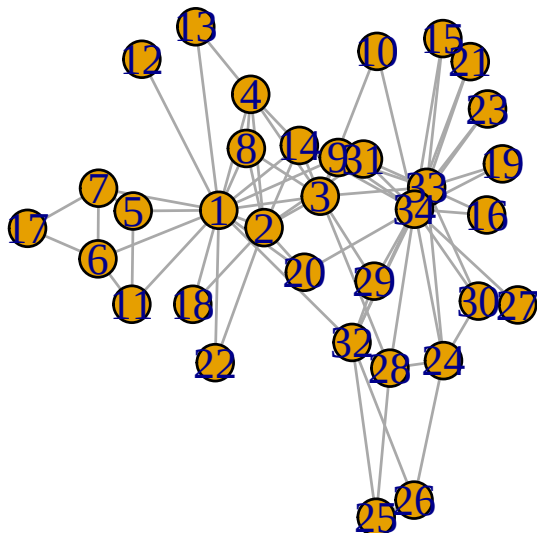
Definition 384 (Edge)

The elements of E are the **edges** (or lines) of the graph G . E (or $E(G)$) is the edge set of the graph G

Setting up graphs in igraph

```
G <- make_empty_graph()
G <- make_graph(edges = c(1, 2, 1, 5), n = 10,
                 directed = FALSE)
G <- make_graph("Zachary")
plot(G, main = "Data on a Karate club")
```

Data on a Karate club



Order and Size

Definition 385 (Order of a graph)

The number of vertices in G is the **order** of G . Using the notation $|V(G)|$ for the *cardinality* of $V(G)$,

$$|V(G)| = \text{order of } G$$

Definition 386 (Size of a graph)

The number of edges in G is the **size** of G ,

$$|E(G)| = \text{size of } G$$

- ▶ A graph having order p and size q is called a (p, q) –graph
- ▶ A graph is finite if $|V(G)| < \infty$

Some simple measures

```
V(G)
```

```
## + 34/34 vertices, from 1a947e8:
```

```
## [1] 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
```

```
## [26] 26 27 28 29 30 31 32 33 34
```

```
head(E(G))
```

```
## + 6/78 edges from 1a947e8:
```

```
## [1] 1--2 1--3 1--4 1--5 1--6 1--7
```

```
gorder(G)
```

```
## [1] 34
```

```
gsize(G)
```

```
## [1] 78
```


Incident – Adjacent

Definition 387 (Incident)

- ▶ A vertex v is **incident** with an edge e if $v \in e$; then e is an edge at v
- ▶ If $e = uv \in E(G)$, then u and v are each incident with e
- ▶ The two vertices incident with an edge are its ends
- ▶ An edge $e = uv$ is incident with both vertices u and v

Definition 388 (Adjacent)

- ▶ Two vertices u and v are **adjacent** in a graph G if $uv \in E(G)$
- ▶ If uv and uw are distinct edges (i.e. $v \neq w$) of a graph G , then uv and uw are adjacent edges

Incident vertices

```
incident(G, 1)
```

```
## + 16/78 edges from 1a947e8:
```

```
## [1] 1-- 2 1-- 3 1-- 4 1-- 5 1-- 6 1-- 7 1-- 8 1-- 9 1--11 1--12 1--13 1--
```

```
## [13] 1--18 1--20 1--22 1--32
```

```
incident_edges(G, c(1, 2))
```

```
## [[1]]
```

```
## + 16/78 edges from 1a947e8:
```

```
## [1] 1-- 2 1-- 3 1-- 4 1-- 5 1-- 6 1-- 7 1-- 8 1-- 9 1--11 1--12 1--13 1--
```

```
## [13] 1--18 1--20 1--22 1--32
```

```
##
```

```
## [[2]]
```

```
## + 9/78 edges from 1a947e8:
```

```
## [1] 1-- 2 2-- 3 2-- 4 2-- 8 2--14 2--18 2--20 2--22 2--31
```

Adjacent vertices

```
adjacent_vertices(G, v = 1)

## [[1]]
## + 16/34 vertices, from 1a947e8:
## [1]  2  3  4  5  6  7  8  9 11 12 13 14 18 20 22 32

adjacent_vertices(G, v = c(1, 2))

## [[1]]
## + 16/34 vertices, from 1a947e8:
## [1]  2  3  4  5  6  7  8  9 11 12 13 14 18 20 22 32
##
## [[2]]
## + 9/34 vertices, from 1a947e8:
## [1]  1  3  4  8 14 18 20 22 31
```

Definition 389 (Multiple edge)

Multiple edges are two or more edges connecting the same two vertices within a multigraph

Definition 390 (Loop)

A **loop** is an edge with both the same ends; *e.g.* $\{u, u\}$ is a loop

Definition 391 (Simple graph)

A **simple graph** is a graph which contains no loops or multiple edges

Definition 392 (Multigraph)

A **multigraph** is a graph which can contain multiple edges or loops

Testing for these properties

```
any_multiple(G)
```

```
## [1] FALSE
```

```
any_loop(G)
```

```
## [1] FALSE
```

```
is_simple(G)
```

```
## [1] TRUE
```

Graph and binary relations

A simple graph G can be defined in term of a vertex set V and a binary relation over V that is

- ▶ irreflexive ($\forall u \in V, u \not R u$)
- ▶ symmetric ($\forall u, v \in V, u R v \implies v R u$)

The set of edges $E(G)$ is the set of symmetric pairs in R

If R is not irreflexive, the graph is not simple

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

Complete, bipartite and other notable graphs

Planar graphs

Graph colouring

Definition 393 (Degree of a vertex)

Let v be a vertex of $G = (V, E)$.

- ▶ The number of edges of G incident with v is the **degree** of v in G
- ▶ The number of edges of G at v is the **degree** of v in G
- ▶ The degree of v in G is noted $d_G(v)$ or $\deg_G(v)$

Theorem 394

Let G be a (p, q) –graph with vertices $v_1, \dots, v_p$, then

$$\sum_{i=1}^p d_G(v_i) = 2q$$

Degree

```
degree(G)
```

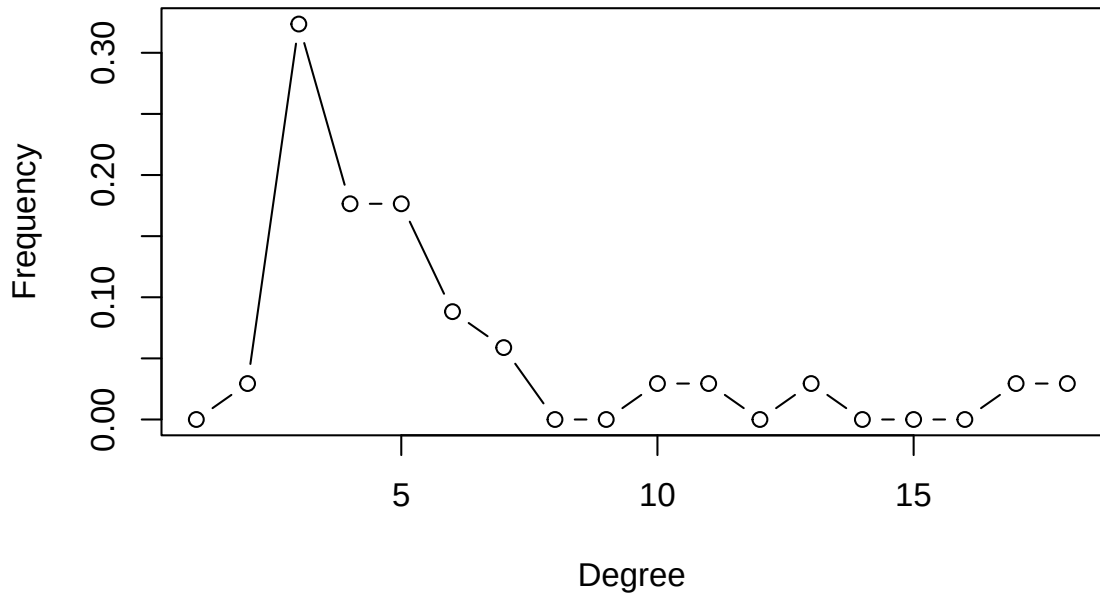
```
## [1] 16 9 10 6 3 4 4 4 5 2 3 1 2 5 2 2 2 2 2 3 2 2 2
## [26] 3 2 4 3 4 4 6 12 17
```

```
degree_distribution(G)
```

```
## [1] 0.00000000 0.02941176 0.32352941 0.17647059 0.17647059 0.08823529
## [7] 0.05882353 0.00000000 0.00000000 0.02941176 0.02941176 0.00000000
## [13] 0.02941176 0.00000000 0.00000000 0.00000000 0.02941176 0.02941176
```

```
plot(degree_distribution(G),
     type = "b",
     xlab = "Degree", ylab = "Frequency",
     main = "Degree distribution of the Karate graph")
```

Degree distribution of the Karate graph



Definition 395 (Odd vertex)

A vertex is an **odd vertex** if its degree is odd

Definition 396 (Even vertex)

A vertex is called **even vertex** if its degree is even

Theorem 397

Every graph contains an even number of odd vertices

Illustration of recent results

Theorem 394 states that a (p, q) -graph has $\sum_{i=1}^p d_G(v_i) = 2q$

```
sum(degree(G)) == 2*length(E(G))  
## [1] TRUE
```

Theorem 397 states that every graph contains an even number of odd vertices

```
sum(pracma::mod(degree(G),2))  
## [1] 12
```

(mode(x,2) returns 1 if x is odd so sum counts how many are odd)

Regular graph

Definition 398 (Regular graph)

If all the vertices of G have the same degree k , then the graph G is k -regular

```
length(unique(degree(G))) == 1
```

```
## [1] FALSE
```

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

Complete, bipartite and other notable graphs

Planar graphs

Graph colouring

Isomorphic graphs

Definition 399 (Isomorphic graphs)

Let $G_1 = (V(G_1), E(G_1))$ and $G_2 = (V(G_2), E(G_2))$ be two graphs. G_1 and G_2 are **isomorphic** if there exists an isomorphism ϕ from G_1 to G_2 , that is defined as an injective mapping $\phi : V(G_1) \rightarrow V(G_2)$ such that two vertices u_1 and v_1 are adjacent in $G_1 \iff$ the vertices $\phi(u_1)$ and $\phi(v_1)$ are adjacent in G_2

If ϕ is an isomorphism from G_1 to G_2 , then the inverse mapping ϕ^{-1} from $V(G_2)$ to $V(G_1)$ also satisfies the definition of an isomorphism. As a consequence, if G_1 and G_2 are isomorphic graphs, then

- ▶ G_1 is isomorphic to G_2
- ▶ G_2 is isomorphic to G_1

Theorem 400

The relation “is isomorphic to” is an equivalence relation on the set of all graphs

Theorem 401

If G_1 and G_2 are isomorphic graphs, then the degrees of vertices of G_1 are exactly the degrees of vertices of G_2

Testing isomorphism

Create two isomorphic graphs by permuting the vertices of the first, then test if they are isomorphic

```
g1 <- sample_pa(30, m = 2, directed = FALSE)
g2 <- permute(g1, sample(vcount(g1)))
# should be TRUE
isomorphic(g1, g2)

## [1] TRUE
```

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

Complete, bipartite and other notable graphs

Planar graphs

Graph colouring

Subgraph

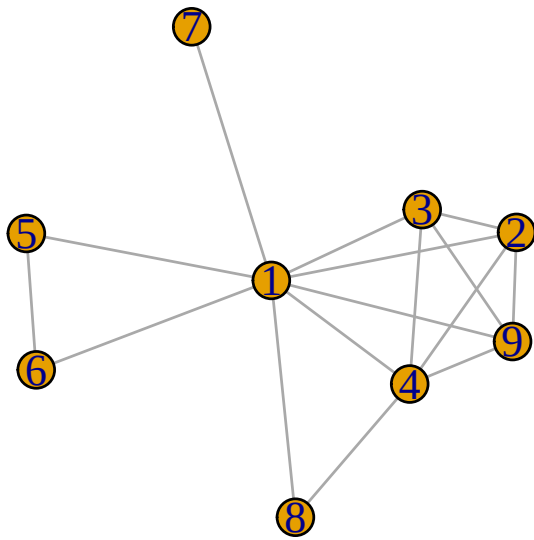
Definition 402 (Subgraph)

Let $G = (V, E)$ be a graph. A graph $H = (V(H), E(H))$ is a **subgraph** of G if $V(H) \subseteq V$ and $E(H) \subseteq E$

Extracting a subgraph

```
G_sub = induced_subgraph(G, c(1:5, 11:14))  
plot(G_sub, main = "A subgraph of the Karate graph")
```

A subgraph of the Karate graph



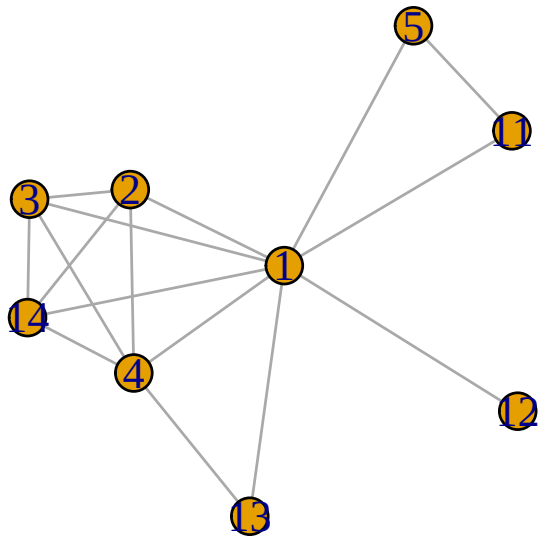
Naming vertices

Note that vertices (and as a consequence, edges) will be relabelled, so G_{sub} has vertices labelled 1 to 9

You can name vertices if you want to avoid this

```
V(G)$name = 1:length(V(G))  
G_sub = induced_subgraph(G, c(1:5, 11:14))  
plot(G_sub, labels = V(G)$name,  
      main = "Subgraph of the Karate graph preserving names")
```

Subgraph of the Karate graph preserving names



Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two graphs

Definition 403 (Union of G_1 and G_2)

$$G_1 \cup G_2 = (V_1 \cup V_2, E_1 \cup E_2)$$

Definition 404 (Intersection of G_1 and G_2)

$$G_1 \cap G_2 = (V_1 \cap V_2, E_1 \cap E_2)$$

Definition 405 (Disjoint graphs)

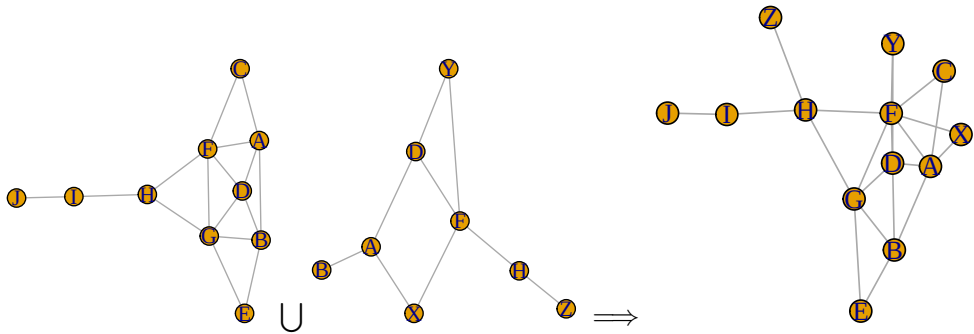
If $G_1 \cap G_2 = (\emptyset, \emptyset) = \emptyset$ (empty graph) then G_1 and G_2 are **disjoint**

Definition 406 (Complement of G_1)

The **complement** $\bar{G}_1$ of G_1 is the graph on V_1 , with the edge set
 $E(\bar{G}_1) = [V_1]^2 \setminus E_1$ ($e \in E(\bar{G}_1) \iff e \notin E_1$)

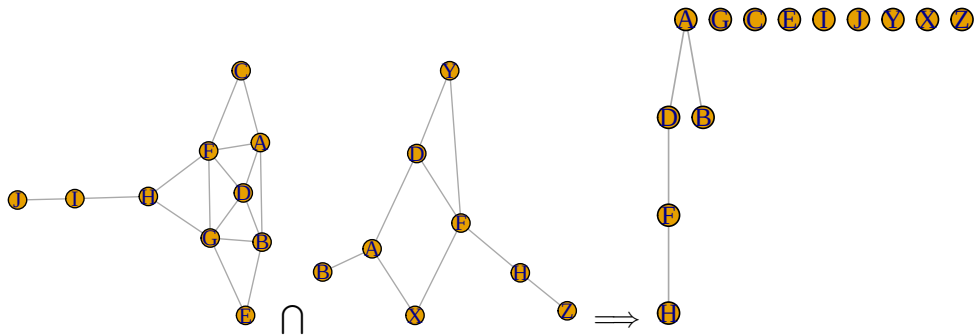
Union

```
net1 <- graph_from_literal(  
  D - A:B:F:G, A - C - F - A, B - E - G - B, A - B, F - G,  
  H - F:G, H - I - J  
)  
net2 <- graph_from_literal(D - A:F:Y, B - A - X - F - H - Z, F - Y)  
print_all(net1 %u% net2)  
  
## IGRAPH 3d1d163 UN-- 13 21 --  
## + attr: name (v/c)  
## + edges from 3d1d163 (vertex names):  
## [1] I--J H--Z H--I G--H G--E F--X F--Y F--H F--C F--G B--E B--G A--X A--  
## [16] A--B D--Y D--G D--F D--B D--A
```



Intersection of two graphs

```
G_inter = net1 %s% net2  
plot(G_inter)
```



Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

Complete, bipartite and other notable graphs

Planar graphs

Graph colouring

Connected vertices and graph, components

Definition 407 (Connected vertices)

Two vertices u and v in a graph G are **connected** if $u = v$, or if $u \neq v$ and there exists a path in G that links u and v

(For *path*, see Definition 420 later)

Definition 408 (Connected graph)

A graph is **connected** if every two vertices of G are connected; otherwise, G is **disconnected**

A necessary condition for connectedness

Theorem 409

A connected graph on p vertices has at least $p - 1$ edges

In other words, a connected graph G of order p has $\text{size}(G) \geq p - 1$

Connectedness is an equivalence relation

Denote $x \equiv y$ the relation “ $x = y$, or $x \neq y$ and there exists a path in G connecting x and y ”. $\equiv$ is an equivalence relation since

1. $x \equiv x$ [reflexivity]
2. $x \equiv y \implies y \equiv x$ [symmetry]
3. $x \equiv y, y \equiv z \implies x \equiv z$ [transitivity]

Definition 410 (Connected component of a graph)

The classes of the equivalence relation $\equiv$ partition V into connected sub-graphs of G called **connected components** (or **components** for short) of G

A connected subgraph H of a graph G is a component of G if H is not contained in any connected subgraph of G having more vertices or edges than H

Vertex deletion & cut vertices

Definition 411 (Vertex deletion)

If $v \in V(G)$ is a vertex of G , the graph $G - v$ is the graph formed from G by removing v and all edges incident with v

Definition 412 (Cut-vertices)

Let G be a connected graph. Then v is a **cut-vertex** G if $G - v$ is disconnected

Edge deletion & bridges

Definition 413 (Edge deletion)

If e is an edge of G , the graph $G - e$ is the graph formed from G by removing e from G

Definition 414 (Bridge)

An edge e in a connected graph G is a **bridge** if $G - e$ is disconnected

Theorem 415

Let G be a connected graph. An edge e of G is a bridge of $G \iff e$ does not lie on any cycle of G

(For *cycle*, see Definition 423 later)

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

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Planar graphs

Graph colouring

Walk

Definition 416 (Walk)

A **walk** in a graph $G = (V, E)$ is a non-empty alternating sequence $v_0 e_0 v_1 e_1 v_2 \dots e_{k-1} v_k$ of vertices and edges in G such that $e_i = \{v_i, v_{i+1}\}$ for all $i < k$. This walk begins with v_0 and ends with v_k

Definition 417 (Length of a walk)

The **length** of a walk is equal to the number of edges in the walk

Definition 418 (Closed walk)

If $v_0 = v_k$, the walk is **closed**

Trail and path

Definition 419 (Trail)

If the edges in the walk are all distinct, it defines a **trail** in $G = (V, E)$

Definition 420 (Path)

If the vertices in the walk are all distinct, it defines a **path** in G

The sets of vertices and edges determined by a trail is a subgraph

Distance between two vertices

Definition 421 (Distance between two vertices)

The **distance** $d(u, v)$ in $G = (V, E)$ between two vertices u and v is the length of the shortest path linking u and v in G

If no such path exists, we assume $d(u, v) = \infty$

Circuit and cycle

Definition 422 (Circuit)

A trail linking u to v , containing at least 3 edges and in which $u = v$, is a **circuit**

Definition 423 (Cycle)

A circuit which does not repeat any vertices (except the first and the last) is a **cycle** (or **simple circuit**)

Definition 424 (Length of a cycle)

The **length of a cycle** is its number of edges

Definition 425 (Eulerian trail)

A walk in an undirected multigraph M that uses each edge **exactly once** is a **Eulerian trail** of M

Definition 426 (Traversable graph)

If a graph G has a Eulerian trail, then G is a **traversable graph**

Definition 427 (Eulerian circuit)

A circuit containing all the vertices and edges of a multigraph M is a **Eulerian circuit** of M

Definition 428 (Eulerian graph)

A graph (resp. multigraph) containing an Eulerian circuit is a **Eulerian graph** (resp. **Eulerian multigraph**)

Remember Euler's bridges of Königsberg?

Cross the 7 bridges in a single walk without recrossing any of them?

`FIGS_slides/bridge_color_with_graph`

Theorem 429

A multigraph M is traversable $\iff M$ is connected and has exactly two odd vertices

Furthermore, any Eulerian trail of M begins at one of the odd vertices and ends at the other odd vertex

Theorem 430

A multigraph M is Eulerian $\iff M$ is connected and every vertex of M is even

Fleury's algorithm to find a Eulerian *trail*

For a connected graph with exactly 2 odd vertices

- ▶ Start at one of the odd vertices
- ▶ Marking your path as you move from vertex to vertex, travel along any edges you wish, but DO NOT travel along an edge that is a bridge for the graph formed by the EDGES THAT HAVE YET TO BE TRAVELED – unless you have to
- ▶ Continue until every edge has been traveled

RESULT: a Eulerian trail

Fleury's algorithm to find a Eulerian *circuit*

For a connected graph with no odd vertices

- ▶ Pick any vertex as a starting point
- ▶ Marking your path as you move from vertex to vertex, travel along any edges you wish, but DO NOT travel along an edge that is a bridge for the graph formed by the EDGES THAT HAVE YET TO BE TRAVELED – unless you have to
- ▶ Continue until you return to your starting point

RESULT: a Eulerian circuit

Definition 431 (Hamiltonian path)

A path containing all vertices of a graph G is a **Hamiltonian path** of G

Definition 432 (Traceable graph)

If a graph G has an Hamiltonian path, then G is a **traceable graph**

Definition 433 (Hamiltonian cycle)

A cycle containing all vertices of a graph G is a **Hamiltonian cycle** of G

Definition 434 (Hamiltonian graph)

A graph containing a Hamiltonian cycle is a **Hamiltonian graph**

Theorem 435 (Dirac's theorem)

If G is a graph of order $p \geq 3$ such that $\deg(v) \geq p/2$ for every vertex v of G , then G is Hamiltonian

Theorem 436 (Ore's theorem)

If G is a graph of order $p \geq 3$ such that for all distinct nonadjacent vertices u and v of G ,

$$\deg(u) + \deg(v) \geq p$$

then G is Hamiltonian

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

Complete, bipartite and other notable graphs

Planar graphs

Graph colouring

Definition 437 (Complete graph)

A graph is complete if every two of its vertices are adjacent

Definition 438 (n -clique)

A simple, complete graph on n vertices is called an n -clique and is often denoted K_n

Note that a complete graph of order p is $(p - 1)$ -regular

Bipartite graph

Definition 439 (Bipartite graph)

A graph is **bipartite** if its vertices can be partitioned into two sets V_1 and V_2 , such that no two vertices in the same set are adjacent. This graph may be written $G = (V_1, V_2, E)$

Definition 440 (Complete bipartite graph)

A bipartite graph in which every two vertices from the 2 different partitions are adjacent is called a **complete bipartite graph**

We often denote $K_{p,q}$ a simple, complete bipartite graph with $|V_1| = p$ and $|V_2| = q$

Some specific graphs

Definition 441 (Tree)

Any connected graph that has no cycles is a **tree**

Definition 442 (Cycle C_n)

For $n \geq 3$, the **cycle** C_n is a connected graph of order n that is a cycle on n vertices

Definition 443 (Path P_n)

The **path** P_n is a connected graph that consists of $n \geq 2$ vertices and $n - 1$ edges. Two vertices of P_n have degree 1 and the rest are of degree 2

Definition 444 (Star S_n)

The **star** of order n is the complete bipartite graph $K_{1,n-1}$ (1 vertex of degree $n - 1$ and $n - 1$ vertices of degree 1)

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

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Graph colouring

Planar graph

Definition 445 (Planar graph)

A graph is **planar** if it *can be* drawn in the plane with no crossing edges (except at the vertices). Otherwise, it is **nonplanar**

Definition 446 (Plane graph)

A **plane graph** is a graph *that is drawn* in the plane with no crossing edges. (This is only possible if the graph is planar)

(To see the difference, have you ever played this game?)

Let G be a plane graph

- ▶ the connected parts of the plane are called **regions**
- ▶ vertices and edges that are incident with a region R make up a **boundary** of R

Theorem 447 (Euler's formula)

Let G be a connected plane graph with p vertices, q edges, and r regions, then

$$p - q + r = 2$$

Corollary 448

Let G be a plane graph with p vertices, q edges, r regions, and k connected components, then

$$p - q + r = k + 1$$

Theorem 449

Let G be a connected planar graph with p vertices and q edges, where $p \geq 3$, then

$$q \leq 3p - 6.$$

(a maximal connected planar graph with p vertices has $q = 3p - 6$ edges)

Corollary 450

*If G is a planar graph, then $\delta(G) \leq 5$, where $\delta(G)$ is the minimal degree of G .
(every planar graph contains a vertex of degree less than 6)*

Two well-known non-planar graphs

$K_{3,3}$ and K_5 are nonplanar

Theorem 451 (Kuratowski Theorem)

A graph G is planar $\iff$ it contains no subgraph isomorphic to K_5 or $K_{3,3}$ or any subdivision of K_5 or $K_{3,3}$

Note: If a graph G is nonplanar and G is a subgraph of G' , then G' is also nonplanar

Undirected graphs

Undirected graph

Degree of a vertex

Isomorphic graphs

Subgraphs, unions of graphs

Connectedness

Walks, trails, paths

Complete, bipartite and other notable graphs

Planar graphs

Graph colouring

Definition 452 (Colouring of a graph G)

A **colouring** of a graph G is an assignment of colours to the vertices of G such that adjacent vertices have different colours

Definition 453 (n -colouring of G)

A **n -colouring** is a colouring of G using n colours

Definition 454 (n -colourable)

G is **n -colourable** if there exists a colouring of G that uses n colours

Definition 455 (Chromatic number)

The **chromatic number** $\chi(G)$ of a graph G is the minimal value n for which an n -colouring of G exists

Property 456

- ▶ $\chi(G) = 1 \iff G$ have no edges
- ▶ If $G = K_{n,m}$, then $\chi(G) = 2$
- ▶ If $G = K_n$, then $\chi(G) = n$
- ▶ For any graph G ,

$$\chi(G) \leq 1 + \Delta(G)$$

where $\Delta(G)$ is the maximum degree of G

- ▶ If G is a planar graph, then $\chi(G) \leq 4$

“Real life” problem

What is the minimal number of colours to colour all states in the map so that two adjacent states have different colours?

`FIGS_slides/USA-4-colours`

“Real life” problem

What is the minimal number of colours to colour all states in the map so that two adjacent states have different colours?

Mathematical representation:

- ▶ vertices correspond to the states
- ▶ vertices are adjacent $\iff$ the two states are adjacent (sharing an isolated point such as the “Four Corners” does not count)

Mathematical problem

What is the chromatic number of the graph associated to the map?

Welch-Powell algorithm for colouring a graph G

1. Order the vertices of G by decreasing degree. (Such an ordering may not be unique since some vertices may have the same degree)
2. Use one colour to paint the first vertex and to paint, in sequential order, each vertex on the list that is not adjacent to a vertex previously painted with this colour
3. Start again at the top of the list and repeat the process, painting previously unpainted vertices using a second colour
4. Repeat with additional colours until all vertices have been painted

Why use graphs/networks?

Binary relations

Undirected graphs

Directed graphs

`FIGS/random_bg_55.pdf`

Matrices associated to a graph/digraph

Trees

Why use graphs/networks?

Binary relations

Definitions

Definition 457 (Digraph)

A directed graph (or **digraph**) is a pair $G = (V, A)$ of sets such that

- ▶ V is a set of points: $V = \{v_1, v_2, v_3, \dots, v_p\}$
- ▶ A is a set of ordered pairs of V : $A = \{(v_i, v_j), (v_i, v_k), \dots, (v_n, v_p)\}$ or $A = \{v_i v_j, v_i v_k, \dots, v_n v_p\}$

Definition 458 (Vertex)

The elements of V are the vertices of the digraph G . V or $V(G)$ is the vertex set of the digraph G

Definition 459 (Arc)

The elements of A are the **arcs** (directed edges) of the digraph G . A or $A(G)$ is the arc set of the digraph G

Digraph and binary relation

A (simple) digraph D can be defined in term of a vertex set V and an irreflexive relation R over V

The defining relation R of the digraph G need not be symmetric

Directed network

Definition 460 (Directed network)

A directed network is a digraph together with a function f ,

$$f : A \rightarrow \mathbb{R},$$

which maps the arc set A into the set of real number. The value of the arc $uv \in A$ is $f(uv)$

Loops & Multiple arcs

Definition 461 (Loop)

A **loop** is an arc with both the same ends; *e.g.* (u, u) is a loop

Definition 462 (Multiple arcs)

Multiple arcs (or multi-arcs) are two or more arcs connecting the same two vertices

Multidigraph/Digraph

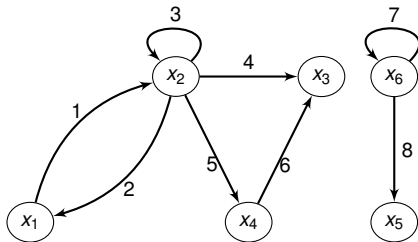
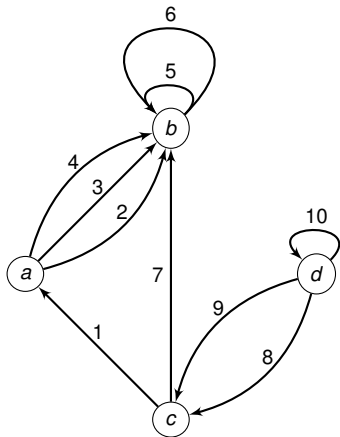
Definition 463 (Multidigraph)

A **multidigraph** is a digraph which allows repetition of arcs or loops

Definition 464 (Digraph)

In a digraph, no more than one arc can join any pair of vertices

Examples



Let $G = (V, A)$ be a digraph

Definition 465 (Arc endpoints)

For an arc $u = (x, y)$, vertex x is the **initial endpoint**, and vertex y is the **terminal endpoint**

Definition 466 (Predecessor - Successor)

If $(u, v) \in A(G)$ is an arc of G , then

- ▶ u is a **predecessor** of v
- ▶ v is a **successor** of u

Definition 467 (Neighbours of a vertex)

Let $x \in V$ be a vertex. The **neighbours** of x is the set $\Gamma(x) = \Gamma_G^+(x) \cup \Gamma_G^-(x)$, where $\Gamma_G^+(x)$ and $\Gamma_G^-(x)$ are, respectively, the set of successors and predecessors of x

Sources and sinks

Definition 468 (Directed away - Directed towards)

If $a = (u, v) \in A(G)$ is an arc of G , then

- ▶ the arc a is said to be **directed away** from u
- ▶ the arc a is said to be **directed towards** v

Definition 469 (Source - Sink)

- ▶ Any vertex which has no arcs directed towards it is a **source**
- ▶ Any vertex which has no arcs directed away from it is a **sink**

Adjacent arcs

Definition 470 (Adjacent arcs)

Two arcs are **adjacent** if they have at least one endpoint in common

Arcs incident to a subset of arcs

Definition 471 (Arc incident out of $X \subset A(G)$)

If the initial endpoint of an arc u belongs to $X \subset A(G)$ and if the terminal endpoint of arc u does not belong to X , then u is said to be **incident out of** X ; we write $u \in \omega^+(X)$

Similarly, we define an **arc incident into** X and the set $\omega^-(X)$

Finally, the set of arcs **incident to** X is denoted

$$\omega(X) = \omega^+(X) \cup \omega^-(X)$$

Definition 472 (Subgraph of G generated by $A \subset V$)

The **subgraph** of G generated by A is the graph with A as its vertex set and with all the arcs in G that have both their endpoints in A . If $G = (V, \Gamma)$ is a 1-graph, then the subgraph generated by A is the 1-graph $G_A = (A, \Gamma_A)$ where

$$\Gamma_A(x) = \Gamma(x) \cap A \quad (x \in A)$$

Definition 473 (Partial graph of G generated by $V \subset U$)

The graph (X, V) whose vertex set is X and whose arc set is V . In other words, it is graph G without the arcs $U - V$

Definition 474 (Partial subgraph of G)

A partial subgraph of G is the subgraph of a partial graph of G

Directed graphs

Degrees in digraphs

Walks, paths, etc.

Connectivity in digraphs

Orientable graphs

FIGS/random_ug_00_001

Degree

Let v be a vertex of a digraph $G = (V, A)$

Definition 475 (Outdegree of a vertex)

The number of arcs directed away from a vertex v , in a digraph is called the **outdegree** of v and is written $d_G^+(v)$

Definition 476 (Indegree of a vertex)

The number of arcs directed towards a vertex v , in a digraph is called the **indegree** of v and is written $d_G^-(v)$

Definition 477 (Degree)

For any vertex v in a digraph, the **degree** of v is defined as

$$d_G(v) = d_G^+(v) + d_G^-(v)$$

Theorem 478

For any (di)graph, the sum of the degrees of the vertices equals twice the number of edges (arcs)

Corollary 479

In any (di)graph, the sum of the degrees of the vertices is a nonnegative even integer

Theorem 480

If G is a digraph with vertex set $V(G) = \{v_1, \dots, v_p\}$ and q arcs, then

$$\sum_{i=1}^p d_G^+(v_i) = \sum_{i=1}^p d_G^-(v_i) = q$$

Definition 481 (Regular digraph)

A digraph G is r -regular if $d_G^+(v) = d_G^-(v) = r$ for all $v \in V(G)$

Symmetric/antisymmetric digraphs

Definition 482 (Symmetric digraph)

Let $G = (V, A)$ be a digraph with associated binary relation R . If R is *symmetric*, the digraph is symmetric

Definition 483 (Anti-symmetric digraph)

Let $G = (V, A)$ be a digraph with associated binary relation R . The digraph G is **anti-symmetric** if

$$xRy \implies y \not R x$$

Definition 484 (Symmetric multidigraph)

Let $G = (V, A)$ be a multidigraph. G is symmetric if $\forall x, y \in V(G)$, the number of arcs from x to y equals the number of arcs from y to x

Directed graphs

Degrees in digraphs

Walks, paths, etc.

Connectivity in digraphs

Orientable graphs

FIGS/random_ug_07.pdf

Walks

Let $G = (V, A)$ be a digraph.

Definition 485 (Directed walk)

A **directed walk** in a digraph G is a non-empty alternating sequence $v_0 a_0 v_1 a_1 v_2 \dots a_{k-1} v_k$ of vertices and arcs in G such that $a_i = (v_i, v_{i+1})$ for all $i < k$. This walk begins with v_0 and ends with v_k

Definition 486 (Length of a directed walk)

The length of a directed walk is equal to the number of arcs in the directed walk

Definition 487 (Closed walk)

If $v_0 = v_k$, the walk is closed

Trails

Let $G = (V, A)$ be a digraph.

Definition 488 (Directed trail)

A directed walk in G in which all arcs are distinct is a **directed trail** in G

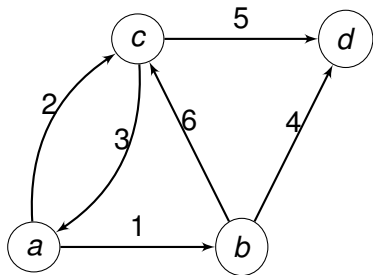
Definition 489 (Directed path)

A directed walk in G in which all vertices are distinct is a **directed path** in G

Definition 490 (Directed cycle)

A closed walk is a **directed cycle** if it contains at least three vertices and all its vertices are distinct except for $v_0 = v_k$

Examples of directed cycles



Cycles:

- ▶ $\mu^1 = (1, 6, 2) = [abca]$
- ▶ $\mu^2 = (1, 6, 3) = [abca]$
- ▶ $\mu^3 = (2, 3) = [aca]$
- ▶ $\mu^4 = (1, 4, 5, 2) = [abdca]$
- ▶ $\mu^5 = (6, 5, 4) = [acdb]$
- ▶ $\mu^6 = (1, 4, 5, 3) = [abdca]$

Directed graphs

Degrees in digraphs

Walks, paths, etc.

Connectivity in digraphs

Orientable graphs

FIGS/random_ug_00_001

Definitions

Definition 491 (Underlying graph)

Given a digraph, the undirected graph with each arc replaced by an edge is called the **underlying graph**

Definition 492 (Weakly connected digraph)

If the underlying graph is a connected graph, then the digraph is **weakly connected**

Definition 493 (Strongly connected digraph)

A digraph G is **strongly connected** if for every two distinct vertices u and v of G , there exists a directed path from u to v

Definition 494 (Disconnected digraph)

A digraph is said to be **disconnected** if it is not weakly connected

Strong connectedness is an equivalence relation

Denote $x \equiv y$ the relation “ $x = y$, or $x \neq y$ and there exists a directed path in G from x to y ”. $\equiv$ is an equivalence relation since

1. $x \equiv x$ [reflexivity]
2. $x \equiv y \implies y \equiv x$ [symmetry]
3. $x \equiv y, y \equiv z \implies x \equiv z$ [transitivity]

Definition 495 (Connected component of a graph)

Sets of the form

$$A(x_0) = \{x : x \in V, x \equiv x_0\}$$

are equivalence classes. They partition V into strongly connected sub-digraphs of G called **strongly connected components** (or **strong components**) of G

A strong component in G is a maximal strongly connected subdigraph of G

Theorem 496 (Properties)

Let $G = (V, A)$ be a digraph

- ▶ *If G is strongly connected, it has only one strongly connected component*
- ▶ *The strongly connected components partition the vertices $V(G)$, with every vertex in exactly one strongly connected component*

Algorithm for determining strongly connected components in

$$G = (V, A)$$

- ▶ Determine the strongly connected component $C(v)$ containing the vertex v ; if $V - C(v)$ is non-empty, re-do the same operation on the sub-digraph $G' = (V - C(v), A')$
- ▶ To determine $C(v)$, the strongly connected component containing v : let v be a vertex of a digraph, which is not already in any strongly connected component
 1. Mark the vertex v with $\pm$
 2. Mark with $+$ all successors (not already marked with $+$) of a vertex marked with $+$
 3. Mark with $-$ all predecessors (not already marked with $-$) of a vertex marked with $-$
 4. Repeat until no more possible marking with $+$ or $-$

All vertices marked with $\pm$ belong to the same strongly connected component $C(v)$ containing the vertex v

Condensation of a digraph

Definition 497 (Condensation of a digraph)

The condensation G^* of a digraph G is a digraph having as vertices the strongly connected components (SCC) of G and such that there exists an arc in G^* from a SCC C_i to another SCC C_j if there is an arc in G from some vertex of S_i to a vertex of S_j

Definition 498 (Articulation set)

For a connected graph, a set X of vertices is called an **articulation set** (or a **cutset**) if the subgraph of G generated by $V - X$ is not connected

Definition 499 (Stable set)

A set S of vertices is called a **stable set** if no arc joins two distinct vertices in S

Directed graphs

Degrees in digraphs

Walks, paths, etc.

Connectivity in digraphs

Orientable graphs

FIGS/random-graphs

Orientation

Definition 500 (Orienting a graph)

Given a connected *graph*, we describe the act of assigning a direction to each edge (edge $\rightarrow$ arc) as **orienting the graph**

Definition 501 (Strong orientation)

If the digraph resulting from orienting a graph is strongly connected, the orientation is a **strong orientation**

Orientable graph

Definition 502 (Orientable graph)

A connected graph G is **orientable** if it admits a strong orientation

Theorem 503

A connected graph $G = (V, E)$ is orientable $\iff G$ contains no bridges

(in other words, iff every edge is contained in a cycle)

Matrices associated to a graph/digraph

There are multiple matrices associated to a graph/digraph

The branch of graph theory that studies the properties of matrices derived from graphs and uses of these matrices in determining graph properties is *spectral graph theory*

Graphs greatly simplify some problems in linear algebra and vice versa

Matrices associated to a graph/digraph

Adjacency matrices

Other matrices associated to a graph/digraph

Linking graphs and linear algebra

FIGS/random_bg_60.pdf

Adjacency matrix (undirected case)

Let $G = (V, E)$ be a graph of order p and size q , with vertices $v_1, \dots, v_p$ and edges $e_1, \dots, e_q$

Definition 504 (Adjacency matrix)

The **adjacency matrix** is

$$M_A = M_A(G) = [m_{ij}]$$

is a $p \times p$ matrix in which

$$m_{ij} = \begin{cases} 1 & \text{if } v_i \text{ and } v_j \text{ are adjacent} \\ 0 & \text{otherwise} \end{cases}$$

Theorem 505 (Adjacency matrix and degree)

The sum of the entries in row i of the adjacency matrix is the degree of v_i in the graph

We often write $A(G)$ and, reciprocally, if A is an adjacency matrix, $G(A)$ the corresponding graph

G undirected $\implies A(G)$ symmetric

$A(G)$ has nonzero diagonal entries if G is not simple

Adjacency matrix (directed case)

Let $G = (V, A)$ be a digraph of order p with vertices $v_1, \dots, v_p$

Definition 506 (Adjacency matrix)

The **adjacency matrix** $M = M(G) = [m_{ij}]$ is a $p \times p$ matrix in which

$$m_{ij} = \begin{cases} 1 & \text{if arc } v_i v_j \in A \\ 0 & \text{otherwise} \end{cases}$$

Theorem 507 (Properties)

- ▶ *M is not necessarily symmetric*
- ▶ *The sum of any column of M is equal to the number of arcs directed towards v_j*
- ▶ *The sum of the entries in row i is equal to the number of arcs directed away from vertex v_i*
- ▶ *The (i, j) –entry of M^n is equal to the number of walks of length n from vertex v_i to v_j*

Definition 508 (Multiplicity of a pair)

The **multiplicity** of a pair x, y is the number $m_G^+(x, y)$ of arcs with initial endpoint x and terminal endpoint y . Let

$$m_G^-(x, y) = m_G^+(y, x)$$

$$m_G(x, y) = m_G^+(x, y) + m_G^-(x, y)$$

If $x \neq y$, then $m_G(x, y)$ is number of arcs with both x and y as endpoints. If $x = y$, then $m_G(x, y)$ equals twice the number of loops attached to vertex x . If $A, B \subset V$, $A \neq B$, let

$$m_G^+(A, B) = \{u : u \in U, u = (x, y), x \in A, y \in B\}$$

$$m_G(A, B) = m_G^+(A, B) + m_G^-(A, B)$$

Adjacency matrix of a multigraph

Definition 509 (Matrix associated with G)

If G has vertices $x_1, x_2, \dots, x_n$, then the **matrix associated** with G is

$$a_{ij} = m_G^+(x_i, x_j)$$

Definition 510 (Adjacency matrix)

The matrix $a_{ij} + a_{ji}$ is the **adjacency matrix** associated with G

Adjacency matrix (multigraph case)

Definition 511 (Adjacency matrix of a multigraph)

G an ℓ -graph, then the adjacency matrix $M_A = [m_{ij}]$ is defined as follows

$$m_{ij} = \begin{cases} k & \text{if arc there are } k \text{ arcs } (i, j) \in U \\ 0 & \text{otherwise} \end{cases}$$

with $k \leq \ell$

G undirected $\implies M_A(G)$ symmetric

$M_A(G)$ has nonzero diagonal entries if G is not simple.

Weighted adjacency matrices

Sometimes, adjacency matrices (typically for 1-graphs) have real entries, usually positive

This means that the arcs/edges have been given a weight

Theorem 512 (Number of walks of length n)

Let A be the adjacency matrix of a graph $G = (V(G), E(G))$, where $V(G) = \{v_1, v_2, \dots, v_p\}$. Then the (i, j) -entry of A^n , $n \geq 1$, is the number of different walks linking v_i to v_j of length n in G .

(two walks of the same length are equal if their edges occur in exactly the same order)

Example: let A be the adjacency matrix of a graph $G = (V(G), E(G))$.

- ▶ the (i, i) -entry of A^2 is equal to the degree of v_i .
- ▶ the (i, i) -entry of A^3 is equal to twice the number of C_3 containing v_i .

Matrices associated to a graph/digraph

Adjacency matrices

Other matrices associated to a graph/digraph

Linking graphs and linear algebra

FIGS/random_bg_61.pdf

Incidence matrix (undirected case)

Let $G = (V, E)$ be a graph of order p , and size q , with vertices $v_1, \dots, v_p$, and edges $e_1, \dots, e_q$

Definition 513 (Incidence matrix)

The incidence matrix is

$$B = B(G) = [b_{ij}]$$

is that $p \times q$ matrix in which

$$b_{ij} = \begin{cases} 1 & \text{if } v_i \text{ is incident with } e_j \\ 0 & \text{otherwise} \end{cases}$$

Theorem 514 (Incidence matrix and degrees)

The sum of the entries in row i of the incidence matrix is the degree of v_i in the graph

Incidence matrix (directed case)

Let $G = (V, A)$ be a digraph of order p and size q , with vertices $v_1, \dots, v_p$ and arcs $a_1, \dots, a_q$

Definition 515 (Incidence matrix)

The **incidence matrix** $B = B(G) = [b_{ij}]$ is a $p \times q$ matrix in which

$$b_{ij} = \begin{cases} 1 & \text{if arc } a_j \text{ is directed away from a vertex } v_i \\ -1 & \text{if arc } a_j \text{ is directed towards a vertex } v_i \\ 0 & \text{otherwise} \end{cases}$$

Spectrum of a graph

We will come back to this later, but for now..

Definition 516 (Spectrum of a graph)

The **spectrum** of a graph G is the spectrum (set of eigenvalues) of its associated adjacency matrix $M(G)$

This is regardless of the type of adjacency matrix or graph

Degree matrix

Definition 517 (Degree matrix)

The **degree** matrix $D = [d_{ij}]$ for G is a $n \times n$ diagonal matrix defined as

$$d_{ij} = \begin{cases} d_G(v_i) & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

In an undirected graph, this means that each loop increases the degree of a vertex by two

In a directed graph, the term “degree” may refer either to indegree (the number of incoming edges at each vertex) or outdegree (the number of outgoing edges at each vertex)

Laplacian matrix

Definition 518 (Laplacian matrix)

$G = (V, A)$ a simple graph with n vertices. The **Laplacian** matrix is

$$L = D(G) - M(G)$$

where $D(G)$ is the degree matrix and $M(G)$ is the adjacency matrix

Laplacian matrix (continued)

G simple graph $\implies M(G)$ only contains 1 or 0 and its diagonal elements are all 0

For directed graphs, either the indegree or outdegree is used, depending on the application

Elements of L are given by

$$\ell_{ij} = \begin{cases} d_G(v_i) & \text{if } i = j \\ -1 & \text{if } i \neq j \text{ and } v_i \text{ is adjacent to } v_j \\ 0 & \text{otherwise} \end{cases}$$

Distance matrix

Let G be a graph of order p with vertices $v_1, \dots, v_p$

Definition 519 (Distance matrix)

The distance matrix $\Delta(G) = [d_{ij}]$ is a $p \times p$ matrix in which

$$\delta_{ij} = d_G(v_i, v_j)$$

Note $\delta_{ii} = 0$ for $i = 1, \dots, p$

Property 520

- ▶ *M is not necessarily symmetric*
- ▶ *The sum of any column of M is equal to the number of arcs directed towards v_j*
- ▶ *The sum of the entries in row i is equal to the number of arcs directed away from vertex v_i*
- ▶ *The (i, j) –entry of M^n is equal to the number of walks of length n from vertex v_i to v_j*

Matrices associated to a graph/digraph

Adjacency matrices

Other matrices associated to a graph/digraph

Linking graphs and linear algebra

FIGS/random_bg_62.pdf

Counting paths

Theorem 521

G a digraph and $M_A(G)$ its adjacency matrix. Denote $P = [p_{ij}]$ the matrix $P = M_A^k$. Then p_{ij} is the number of distinct paths of length k from i to j in G

Definition 522 (Irreducible matrix)

A matrix $A \in \mathcal{M}_n$ is **reducible** if $\exists P \in \mathcal{M}_n$, permutation matrix, s.t. $P^T A P$ can be written in block triangular form. If no such P exists, A is **irreducible**

Theorem 523

A irreducible $\iff G(A)$ strongly connected

Theorem 524

Let A be the adjacency matrix of a graph G on p vertices. A graph G on p vertices is connected $\iff$

$$I + A + A^2 + \cdots + A^{p-1} = C$$

has no zero entries

Theorem 525

Let M be the adjacency matrix of a digraph D on p vertices. A digraph D on p vertices is strongly connected $\iff$

$$I + M + M^2 + \cdots + M^{p-1} = C$$

has no zero entries

Nonnegative matrix

$A = [a_{ij}] \in \mathcal{M}_n(\mathbb{R})$ **nonnegative** if $a_{ij} \geq 0 \forall i, j = 1, \dots, n$; $\mathbf{v} \in \mathbb{R}^n$ nonnegative if $v_i \geq 0 \forall i = 1, \dots, n$. **Spectral radius** of A

$$\rho(A) = \max_{\lambda \in \text{Sp}(A)} \{|\lambda|\}$$

$\text{Sp}(A)$ the **spectrum** of A

Perron-Frobenius (PF) theorem

Theorem 526 (PF – Nonnegative case)

$0 \leq A \in \mathcal{M}_n(\mathbb{R})$. Then $\exists \mathbf{v} \geq \mathbf{0}$ s.t.

$$A\mathbf{v} = \rho(A)\mathbf{v}$$

Theorem 527 (PF – Irreducible case)

Let $0 \leq A \in \mathcal{M}_n(\mathbb{R})$ irreducible. Then $\exists \mathbf{v} > \mathbf{0}$ s.t.

$$A\mathbf{v} = \rho(A)\mathbf{v}$$

$\rho(A) > 0$ and with algebraic multiplicity 1. No nonnegative eigenvector is associated to any other eigenvalue of A

Primitive matrices

Definition 528

$0 \leq A \in \mathcal{M}_n(\mathbb{R})$ **primitive** (with **primitivity index** $k \in \mathbb{N}_+^*$) if $\exists k \in \mathbb{N}_+^*$ s.t.

$$A^k > 0,$$

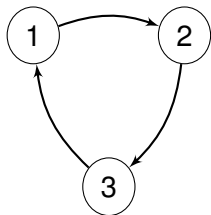
with k the smallest integer for which this is true. A **imprimitive** if it is not primitive

A primitive $\implies A$ irreducible; the converse is false

Theorem 529

$A \in \mathcal{M}_n(\mathbb{R})$ *irreducible and* $\exists i = 1, \dots, n$ s.t. $a_{ij} > 0 \implies A$ *primitive*

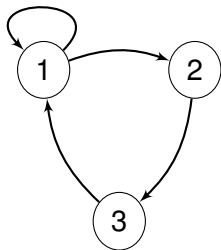
Here d is the index of imprimitivity (i.e., the number of eigenvalues that have the same modulus as $\lambda_p = \rho(A)$). If $d = 1$, then A is primitive. We have that $d = \gcd$ of all the lengths of closed walks in $G(A)$



Adjacency matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

Closed walks in $G(A)$ (lengths): $1 \rightarrow 1$ (3), $2 \rightarrow 2$ (3), $3 \rightarrow 3$ (3) $\implies \gcd = 3 \implies d = 3$ (here, all eigenvalues have modulus 1)



$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

Closed walk $1 \rightarrow 1$ has length 1 $\implies \gcd$ of lengths of closed walks is 1 $\implies A$ primitive

Let $\mathbf{0} \leq A \in \mathcal{M}_n$

Theorem 530

A primitive $\implies \exists 0 < k \leq (n-1)n^n$ such that $A^k > \mathbf{0}$

Theorem 531

If A is primitive and the shortest simple directed cycle in $G(A)$ has length s , then the primitivity index is $\leq n + s(n-1)$

Theorem 532

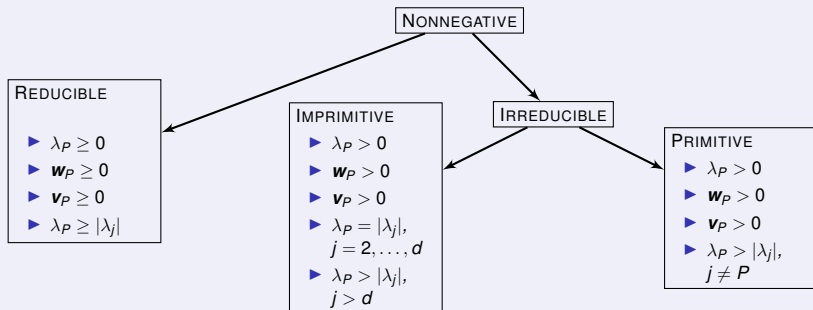
A primitive $\iff A^{n^2-2n+2} > \mathbf{0}$

Theorem 533

If A is irreducible and has d positive entries on the diagonal, then the primitivity index $\leq 2n - d - 1$

Theorem 534

$\mathbf{0} \leq A \in \mathcal{M}_n$, $\lambda_P = \rho(A)$ the Perron root of A , $\mathbf{v}_P$ and $\mathbf{w}_P$ the corresponding right and left Perron vectors of A , respectively, d the index of imprimitivity of A (with $d = 1$ when A is primitive) and $\lambda_j \in \sigma(A)$ the spectrum of A , with $j = 2, \dots, n$ unless otherwise specified (assuming $\lambda_1 = \lambda_P$)



Why use graphs/networks?

Binary relations

Undirected graphs

Directed graphs

`FIGS/random_bg_63.pdf`

Matrices associated to a graph/digraph

Trees

Why use graphs/networks?

Binary relations

Trees

Definition 535 (Forest, trees and branches)

- ▶ A connected graph with no cycle is a **tree**
- ▶ A tree is a connected acyclic graph, its edges are called **branches**
- ▶ A graph (connected or not) without any cycle is a **forest**. Each component is a tree

(A forest is a graph whose connected components are trees)

Is the “Karate graph” a tree?

```
is_acyclic(G)
```

```
## [1] FALSE
```

```
is_tree(G)
```

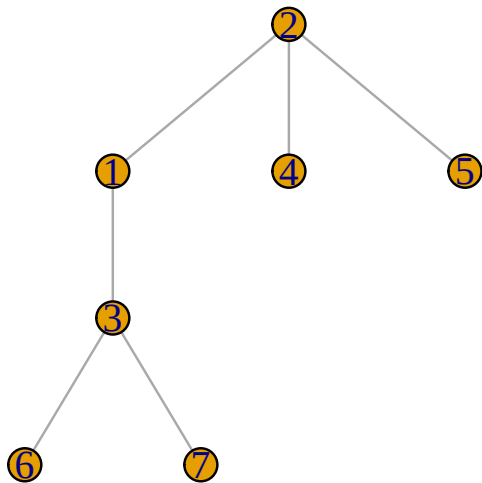
```
## [1] FALSE
```

So we need friend to play with!

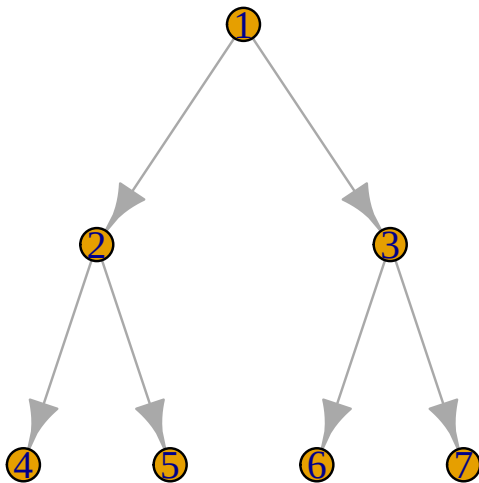
```
G_tu <- make_tree(7, 2, mode = "undirected")
```

```
G_td <- make_tree(7, 2)
```

An undirected tree



A (out) directed tree



Property 536

- ▶ *Every edge of a tree is a bridge*
- ▶ *Given two vertices u and v of a tree, there is an unique path linking u to v*
- ▶ *A tree with p vertices and q edges satisfies $q = p - 1$. Thus, a tree is minimally connected*

(First property: the deletion of any edge of a tree disconnects it)

Every edge of a tree is a bridge

```
E(G_tu)

## + 6/6 edges from 18cd90b:
## [1] 1--2 1--3 2--4 2--5 3--6 3--7

bridges(G_tu)

## + 6/6 edges from 18cd90b:
## [1] 2--4 2--5 1--2 3--6 3--7 1--3

all(sort(E(G_tu)) == sort(bridges(G_tu)))

## [1] TRUE
```


Spanning tree

Definition 537 (Spanning tree)

A **spanning tree** of a connected graph G is a subgraph of G that contains all the vertices of G and is a tree.

A graph may have many spanning trees

Minimal spanning tree

Definition 538 (Value of a spanning tree)

The **value of a spanning tree** T of order p is

$$\sum_{i=1}^{p-1} f(e_i)$$

where f is the function that maps the edge set into $\mathbb{R}$

Definition 539 (Minimal spanning tree)

Let G be an undirected network, and let T be a **minimal spanning tree** of G . Then T is a spanning tree whose the value is minimum

Algorithm to find a minimal spanning tree

Let $G = (V(G), E(G))$ be an undirected network and T be a minimal spanning tree

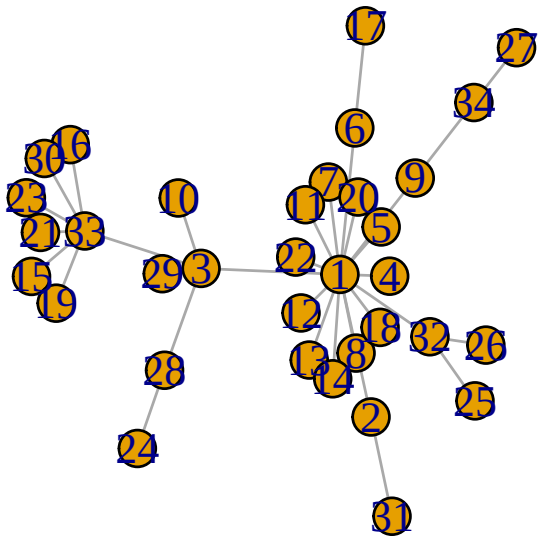
1. Sort the edges of G in increasing order by value
2. $T = (V(G), \emptyset)$
3. For each edge e in sorted order if the endpoints of e are disconnected in T add e to T

Finding a minimal spanning tree of the Karate graph

The function `mst` finds minimal spanning trees, using distances if no edge weights are provided

```
G_mst = mst(G)
```

A minimal spanning tree of the Karate graph



Minimal connector problem

- ▶ Model: a graph G such that edges represent all possible connections, and each edge has a positive value which represents its cost; an undirected network G
- ▶ Solution: a minimal spanning tree T of G
 - ▶ a spanning tree of G is a subgraph of G that contains all the vertices of G and is a tree.
 - ▶ the cost of the spanning tree is the sum of values of the edges of T
 - ▶ a spanning tree such that no other spanning tree has a smaller cost is a minimal spanning tree.

Theorem 540 (Characterisation of trees)

$H = (V, U)$ a graph of order $|V| = n > 2$. The following are equivalent and all characterise a tree :

- 1. H connected and has no cycles*
- 2. H has $n - 1$ arcs and no cycles*
- 3. H connected and has exactly $n - 1$ arcs*
- 4. H has no cycles, and if an arc is added to H , exactly one cycle is created*
- 5. H connected, and if any arc is removed, the remaining graph is not connected*
- 6. Every pair of vertices of H is connected by one and only one chain*

Definition 541 (Pendant vertex)

A vertex is **pendant** if it is adjacent to exactly one other vertex

Theorem 542

A tree of order $n \geq 2$ has at least two pendant vertices

Theorem 543

A graph $G = (V, U)$ has a partial graph that is a tree $\iff G$ connected

Recall that a partial graph is a graph generated by a subset of the arcs
(Definition 473 slide 384)

Spanning tree

The procedure in the proof of Theorem 573 gives a **spanning tree**

Can also build a spanning tree as follows:

- ▶ Consider any arc u_0
- ▶ Find arc u_1 that does not form a cycle with u_0
- ▶ Find arc u_2 that does not form a cycle with $\{u_0, u_1\}$
- ▶ Continue
- ▶ When you cannot continue anymore, you have a spanning tree

Theorem 544

G connected graph with ≥ 1 arc. TFAE

1. *G strongly connected*
2. *Every arc lies on a circuit*
3. *G contains no cocircuits*

Theorem 545

G graph with ≥ 1 arc. TFAE

1. *G is a graph without circuits*
2. *Each arc is contained in a cocircuit*

Theorem 546

If G is a strongly connected graph of order n, then G has a cycle basis of $\nu(G)$ circuits

Definition 547 (Node, anti-node, branch)

$G = (V, U)$ strongly connected without loops and > 1 vertex. For each $x \in V$, there is a path from it and a path to it so x has at least 2 incident arcs. Specifically,

- ▶ $x \in V$ with > 2 incident arcs is a **node**
- ▶ $x \in V$ with 2 incident arcs is an **anti-node**

A path whose only nodes are its endpoints is a **branch**

Definition 548 (Minimally connected graph)

G is **minimally connected** if it is strongly connected and removal of any arc destroys strong-connectedness

A minimally connected graph is 1-graph without loops

Definition 549 (Contraction)

$G = (V, U)$. The **contraction** of the set $A \subset V$ of vertices consists in replacing A by a single vertex a and replacing each arc into (resp. out of) A by an arc with same index into (resp. out of) a

Theorem 550

*G minimally connected, $A \subset V$ generating a strongly connected subgraph of G .
Then the contraction of A gives a minimally connected graph*

Theorem 551

G a minimally connected graph, G' be the minimally connected graph obtained by the contraction of an elementary circuit of G . Then

$$\nu(G) = \nu(G') + 1$$

Theorem 552

G minimally connected of order $n \geq 2 \implies G$ has ≥ 2 anti-nodes

Theorem 553

$G = (V, U)$. Then the graph C' obtained by contracting each strongly connected component of G contains no circuits

Arborescences

Definition 554 (Root)

Vertex $a \in V$ in $G = (V, U)$ is a **root** if all vertices of G can be reached by paths *starting* from a

Not all graphs have roots

Definition 555 (Quasi-strong connectedness)

G is **quasi-strongly connected** if $\forall x, y \in V$, exists $z \in V$ (denoted $z(x, y)$ to emphasize dependence on x, y) from which there is a path to x and a path to y

Strongly connected $\implies$ quasi-strongly connected (take $z(x, y) = x$); converse not true

Quasi-strongly connected $\implies$ connected

Arborescence

Definition 556 (Arborescence)

An **arborescence** is a tree that has a root

Lemma 557

$G = (V, U)$ has a root $\iff G$ quasi-strongly connected

Theorem 558

H graph of order $n > 1$. TFAE (and all characterise an arborescence)

- 1. H quasi-strongly connected without cycles*
- 2. H quasi-strongly connected with $n - 1$ arcs*
- 3. H tree having a root a*
- 4. $\exists a \in V$ s.t. all other vertices are connected with a by 1 and only 1 path from a*
- 5. H quasi-strongly connected and loses quasi-strong connectedness if any arc is removed*
- 6. H quasi-strongly connected and $\exists a \in V$ s.t.*

$$d_H^-(a) = 0$$

$$d_H^-(x) = 1 \quad \forall x \neq a$$

- 7. H has no cycles and $\exists a \in V$ s.t.*

$$d_H^-(a) = 0$$

$$d_H^-(x) = 1 \quad \forall x \neq a$$

Theorem 559

G has a partial graph that is an arborescence $\iff G$ quasi-strongly connected

Theorem 560

$G = (V, E)$ simple connected graph and $x_1 \in V$. It is possible to direct all edges of E so that the resulting graph $G_0 = (V, U)$ has a spanning tree H s.t.

- 1. H is an arborescence with root x_1*
- 2. The cycles associated with H are circuits*
- 3. The only elementary circuits of G_0 are the cycles associated with H*

Counting trees

Proposition 561

*X a set with n distinct objects, $n_1, \dots, n_p$ nonnegative integers s.t.
 $n_1 + \dots + n_p = n$. The number of ways to place the n objects into p boxes
 $X_1, \dots, X_p$ containing $n_1, \dots, n_p$ objects respectively is*

$$\binom{n}{n_1, \dots, n_p} = \frac{n!}{n_1! \cdots n_p!}$$

Proposition 562 (Multinomial formula)

Let $a_1, \dots, a_p \in \mathbb{R}$ be p real numbers, then

$$(a_1 + \dots + a_p)^n = \sum_{n_1, \dots, n_p \geq 0} \binom{n}{n_1, \dots, n_p} (a_1)^{n_1} \cdots (a_p)^{n_p}$$

Theorem 563

Denote $T(n; d_1, \dots, d_n)$ the number of distinct trees H with vertices $x_1, \dots, x_n$ and with degrees $d_H(x_1) = d_1, \dots, d_H(x_n) = d_n$. Then

$$T(n; d_1, \dots, d_n) = \binom{n-2}{d_1-1, \dots, d_n-1}$$

Theorem 564

The number of different trees with vertices $x_1, \dots, x_n$ is n^{n-2}

There is a whole industry of similar results (as well as for arborescences), but we will stop here. The main point is that we are talking about a large number of possibilities..

Why use graphs/networks?

Binary relations

Undirected graphs

Directed graphs

`FIGS/random_bg_64.pdf`

Matrices associated to a graph/digraph

Trees

Why use graphs/networks?

Binary relations

Trees

Definition 565 (Forest, trees and branches)

- ▶ A connected graph with no cycle is a **tree**
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- ▶ A graph (connected or not) without any cycle is a **forest**. Each component is a tree

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Is the “Karate graph” a tree?

```
is_acyclic(G_Z)
```

```
## [1] FALSE
```

```
is_tree(G_Z)
```

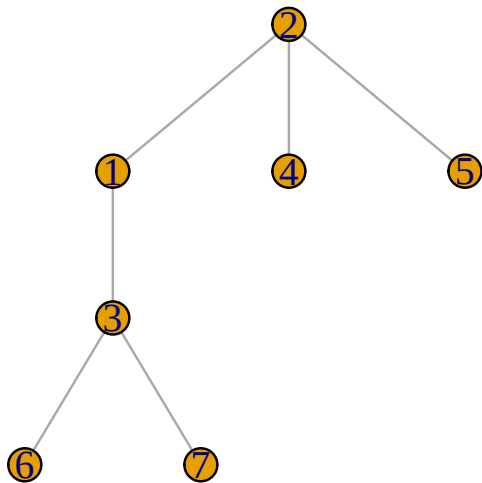
```
## [1] FALSE
```

So we need a friend to play with!

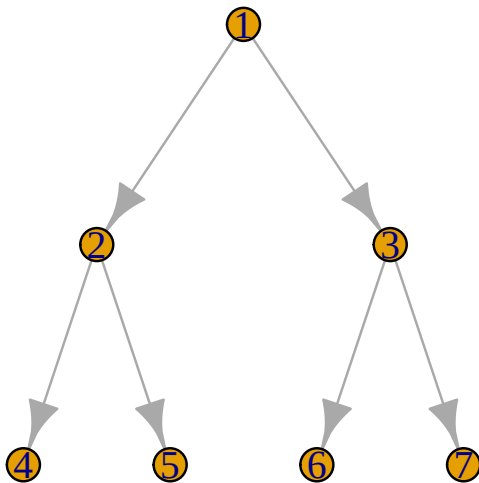
```
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```

```
G_td <- make_tree(7, 2)
```

An undirected tree



A (out) directed tree



Property 566

- ▶ *Every edge of a tree is a bridge*
- ▶ *Given two vertices u and v of a tree, there is an unique path linking u to v*
- ▶ *A tree with p vertices and q edges satisfies $q = p - 1$. Thus, a tree is minimally connected*

(First property: the deletion of any edge of a tree disconnects it)

Every edge of a tree is a bridge

```
E(G_tu)

## + 6/6 edges from 23c1b35:
## [1] 1--2 1--3 2--4 2--5 3--6 3--7

bridges(G_tu)

## + 6/6 edges from 23c1b35:
## [1] 2--4 2--5 1--2 3--6 3--7 1--3

all(sort(E(G_tu)) == sort(bridges(G_tu)))

## [1] TRUE
```

Spanning tree

Definition 567 (Spanning tree)

A **spanning tree** of a connected graph G is a subgraph of G that contains all the vertices of G and is a tree.

A graph may have many spanning trees

Minimal spanning tree

Definition 568 (Value of a spanning tree)

The **value of a spanning tree** T of order p is

$$\sum_{i=1}^{p-1} f(e_i)$$

where f is the function that maps the edge set into $\mathbb{R}$

Definition 569 (Minimal spanning tree)

Let G be an undirected network, and let T be a **minimal spanning tree** of G . Then T is a spanning tree whose the value is minimum

Algorithm to find a minimal spanning tree

Let $G = (V(G), E(G))$ be an undirected network and T be a minimal spanning tree

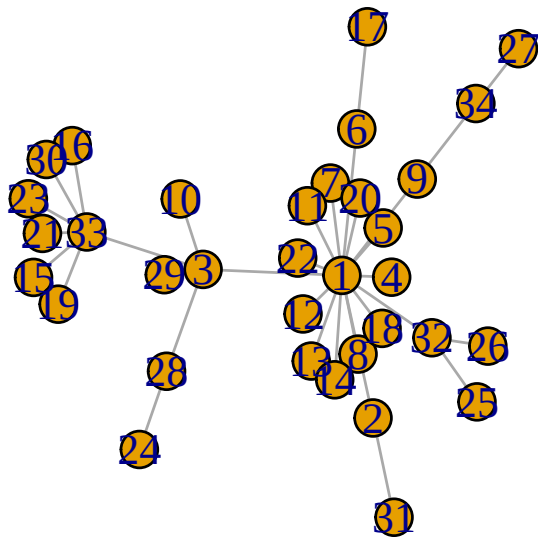
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3. For each edge e in sorted order if the endpoints of e are disconnected in T add e to T

Finding a minimal spanning tree of the Karate graph

The function `mst` finds minimal spanning trees, using distances if no edge weights are provided

```
G_mst = mst(G_Z)
```


A minimal spanning tree of the Karate graph



Minimal connector problem

- ▶ Model: a graph G such that edges represent all possible connections, and each edge has a positive value which represents its cost; an undirected network G
- ▶ Solution: a minimal spanning tree T of G
 - ▶ a spanning tree of G is a subgraph of G that contains all the vertices of G and is a tree.
 - ▶ the cost of the spanning tree is the sum of values of the edges of T
 - ▶ a spanning tree such that no other spanning tree has a smaller cost is a minimal spanning tree.

Theorem 570 (Characterisation of trees)

$H = (V, U)$ a graph of order $|V| = n > 2$. The following are equivalent and all characterise a tree :

- 1. H connected and has no cycles*
- 2. H has $n - 1$ arcs and no cycles*
- 3. H connected and has exactly $n - 1$ arcs*
- 4. H has no cycles, and if an arc is added to H , exactly one cycle is created*
- 5. H connected, and if any arc is removed, the remaining graph is not connected*
- 6. Every pair of vertices of H is connected by one and only one chain*

Definition 571 (Pendant vertex)

A vertex is **pendant** if it is adjacent to exactly one other vertex

Theorem 572

A tree of order $n \geq 2$ has at least two pendant vertices

Theorem 573

A graph $G = (V, U)$ has a partial graph that is a tree $\iff G$ connected

(A partial graph is a graph generated by a subset of the arcs)

Spanning tree

Can build a spanning tree as follows:

- ▶ Consider any arc u_0
- ▶ Find arc u_1 that does not form a cycle with u_0
- ▶ Find arc u_2 that does not form a cycle with $\{u_0, u_1\}$
- ▶ Continue
- ▶ When you cannot continue anymore, you have a spanning tree

Definition 574 (Minimally connected graph)

G is **minimally connected** if it is strongly connected and removal of any arc destroys strong-connectedness

A minimally connected graph is 1-graph without loops

Definition 575 (Contraction)

$G = (V, U)$. The **contraction** of the set $A \subset V$ of vertices consists in replacing A by a single vertex a and replacing each arc into (resp. out of) A by an arc with same index into (resp. out of) a

Theorem 576

*G minimally connected, $A \subset V$ generating a strongly connected subgraph of G .
Then the contraction of A gives a minimally connected graph*

Arborescences

Definition 577 (Root)

Vertex $a \in V$ in $G = (V, U)$ is a **root** if all vertices of G can be reached by paths *starting* from a

Not all graphs have roots

Definition 578 (Quasi-strong connectedness)

G is **quasi-strongly connected** if $\forall x, y \in V$, exists $z \in V$ (denoted $z(x, y)$ to emphasize dependence on x, y) from which there is a path to x and a path to y

Strongly connected $\implies$ quasi-strongly connected (take $z(x, y) = x$); converse not true

Quasi-strongly connected $\implies$ connected

Arborescence

Definition 579 (Arborescence)

An **arborescence** is a tree that has a root

Lemma 580

$G = (V, U)$ has a root $\iff G$ quasi-strongly connected

Theorem 581

H graph of order $n > 1$. TFAE (and all characterise an arborescence)

- 1. H quasi-strongly connected without cycles*
- 2. H quasi-strongly connected with $n - 1$ arcs*
- 3. H tree having a root a*
- 4. $\exists a \in V$ s.t. all other vertices are connected with a by 1 and only 1 path from a*
- 5. H quasi-strongly connected and loses quasi-strong connectedness if any arc is removed*
- 6. H quasi-strongly connected and $\exists a \in V$ s.t.*

$$d_H^-(a) = 0 \quad \text{and} \quad d_H^-(x) = 1, \quad \forall x \neq a$$

- 7. H has no cycles and $\exists a \in V$ s.t.*

$$d_H^-(a) = 0 \quad \text{and} \quad d_H^-(x) = 1, \quad \forall x \neq a$$

Theorem 582

G has a partial graph that is an arborescence $\iff G$ quasi-strongly connected

Theorem 583

$G = (V, E)$ simple connected graph and $x_1 \in V$. It is possible to direct all edges of E so that the resulting graph $G_0 = (V, U)$ has a spanning tree H s.t.

- 1. H is an arborescence with root x_1*
- 2. The cycles associated with H are circuits*
- 3. The only elementary circuits of G_0 are the cycles associated with H*

Counting trees

Proposition 584

*X a set with n distinct objects, $n_1, \dots, n_p$ nonnegative integers s.t.
 $n_1 + \dots + n_p = n$. The number of ways to place the n objects into p boxes
 $X_1, \dots, X_p$ containing $n_1, \dots, n_p$ objects respectively is*

$$\binom{n}{n_1, \dots, n_p} = \frac{n!}{n_1! \cdots n_p!}$$

Proposition 585 (Multinomial formula)

Let $a_1, \dots, a_p \in \mathbb{R}$ be p real numbers, then

$$(a_1 + \dots + a_p)^n = \sum_{n_1, \dots, n_p \geq 0} \binom{n}{n_1, \dots, n_p} (a_1)^{n_1} \cdots (a_p)^{n_p}$$

Theorem 586

Denote $T(n; d_1, \dots, d_n)$ the number of distinct trees H with vertices $x_1, \dots, x_n$ and with degrees $d_H(x_1) = d_1, \dots, d_H(x_n) = d_n$. Then

$$T(n; d_1, \dots, d_n) = \binom{n-2}{d_1-1, \dots, d_n-1}$$

Theorem 587

The number of different trees with vertices $x_1, \dots, x_n$ is n^{n-2}

There is a whole industry of similar results (as well as for arborescences), but we will stop here. The main point is that we are talking about a large number of possibilities..